A first-mode law for the separation between the Kneser continuation and the regular iteration on $(1, \eta)$

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Abstract

For $1 < b < \eta = \mathrm{e}^{1/\mathrm{e}}$, we investigate the separation between the regular super-exponential R_b and the boundary value K_b of a Kneser-type continuation in the base. With both solutions normalised by $F(0) = 1$, write $K_b = R_b \circ (\mathrm{id} + P_b)$, where P_b is 1-periodic. The measured separation is dominated by its first Fourier mode:

$$D_b(z) := K_b(z) - R_b(z) \approx c_1(b) R_b'(z)(\mathrm{e}^{2\pi \mathrm{i} z} - 1).$$

Across the sampled bases, $|\hat{c}_1| = |c_1/\Lambda|$ varies by less than 5%, where $\Lambda = \exp(4\pi^2/\log\lambda)$ and λ is the attracting multiplier. The elementary parameter relation $\log b = \lambda \mathrm{e}^{-\lambda}$ proves that this candidate scale has saddle-node exponent $\log\Lambda \sim -C_\eta/\sqrt{\eta - b}$, with $C_\eta = 20.3508008\ldots$; its relation to the separation remains conjectural. A separate computation of the parabolic horn map for $u \mapsto \mathrm{e}^u - 1$ gives $|A_1| = 0.08905843641221$ in a fixed Fatou-coordinate normalisation. The measured $|\hat{c}_1|$ increases toward that value. Matching the direction of the transition suggests the inverse horn map as the limiting object: its coefficients satisfy $B_1 = -A_1$ and $B_2 = -A_2 + 2\pi\mathrm{i}A_1^2$. At six sampled bases, $|c_2/c_1^2|$ increases toward the candidate target $|B_2/B_1^2| = 1.57442268552971\ldots$. All ratios c_n/c_1^n are invariant under time translations; a distinct, larger formal conjugacy relation singles out the quadratic coefficient and iterative residue of the zero-constant-mode cylinder germ. We then show that the Kneser continuation can be eliminated altogether: a welding identity expresses every c_n, up to an additive $O(\Lambda)$, through the Fourier modes t_n of the transition map $T_b = R_b^{-1} \circ S_b$ between the regular solutions at the two real fixed points, $c_n/\Lambda^n = t_n \mathrm{e}^{-2\pi \mathrm{i} n t_0} + O(\Lambda)$. This reproduces the ladder values, arguments included, to their full precision, and reduces the horn-map conjectures to a confluence statement for the saddle-node unfolding. We prove that confluence statement: the Kneser-free invariants converge to the inverse horn-map coefficients for every n. A unisingular parabolic-renormalisation theorem proves $B_1 \neq 0$, giving the limiting phase and normalised ratios. On the Kneser side, we prove the decay estimate for the sewing solution with explicit constants. That solution is characterised uniquely by its two regular representations and the height of its seam. The base-continued family has it as boundary value whenever its θ-representations stay uniformly bounded along the ladder. The sewn family varies holomorphically near every real $b \in (1, \eta)$; near η its imaginary part at real heights has an explicit leading Λ-profile and is not identically zero on any interval about 0. Computing T_b directly up to $\eta - b = 3 \cdot 10^{-5}$ confirms the limits for $n = 1, 2, 3$ with deficits $\propto |\log\lambda|^2$. Numerical consistency checks of the separation coefficients are reported separately from exact identities; their quoted decimal comparisons have no certified error bounds. An interval-certified quasiconformal argument extends the exterior Kneser family across a non-cusp Shell–Thron boundary subarc on a common complex neighbourhood of the full height segment $[0, 1]$. Identification with the inner sewn family remains open.

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1 Introduction

Tetration is the solution of the Abel-type functional equation

$$F(z+1) = b^{F(z)}, \qquad F(0) = 1. \tag{1}$$

For $b > \eta := \mathrm{e}^{1/\mathrm{e}}$ the two fixed points of $w \mapsto b^w$ are a complex conjugate pair, and Kneser's construction [10] produces the solution that is real-analytic on $(-2, \infty)$ and tends to the two fixed points as $\operatorname{Im} z \to \pm\infty$. Its conformal uniformisation is uniquely characterised by the initial-region criterion of [27]; high-precision algorithms for it are well developed [11, 12].

For $1 < b < \eta$ the situation is different. The map $E_b(w) = b^w$ has two *real* fixed points

$$L = \mathrm{e}^{\lambda} \ \ (\text{attracting},\ 0 < \lambda < 1), \qquad L_2 = \mathrm{e}^{\lambda_2} \ \ (\text{repelling},\ \lambda_2 > 1), \tag{2}$$

and the classical Koenigs–Schröder construction at the attracting point yields a real-analytic solution R_b of (1) by elementary means. This is the solution usually called "the" tetration in that range; it is one of the four regular super-exponentials studied for $b = \sqrt{2}$ in [12], and it is the reason why the interior of the Shell–Thron region is explicitly excluded from [20].

There is, however, a second natural solution. The two-fixed-point construction is holomorphic in the base across the Shell–Thron boundary, so one may take

$$K_b := \lim_{\varepsilon \to 0^+} \kappa_{b+\mathrm{i}\varepsilon}, \tag{3}$$

the boundary value on $(1, \eta)$ of the Kneser-type family continued from the upper half of the region. Paulsen [21] proved that K_b is *not* the regular solution: there is no open subinterval of $(1, \eta)$ on which $K_b = R_b$. His proof is an analytic contradiction argument and gives no rate; the only quantitative datum in that paper is a single number for $b = \sqrt{2}$, obtained from a 120-digit contour computation,

$$\operatorname{Im} \kappa_{\sqrt{2}^+}(1/2) \approx -1.18899697185401045226976795872715599664 \cdot 10^{-48}, \tag{4}$$

reported there as evidence that the limit is not real, and not compared with any scale.

This note is about the size and the shape of $D_b := K_b - R_b$. The statements we can prove are elementary (Section 2 and Proposition 2); the substance is numerical, and we have tried to be explicit throughout about which is which (Section 8).

Summary of findings.

1. The measured separation is dominated by one complex coefficient (Section 3). With $P_b(0) = 0$, the leading profile is $c_1(\mathrm{e}^{2\pi \mathrm{i} z} - 1)$; omitting the constant term creates a z-dependent error.

2. The normalised modulus $|\hat{c}_1| = |c_1/\Lambda|$ stays near 0.0850–0.0889 across the reported data (Section 4). This comparison fixes the imaginary origin: the modulus is invariant under real, but not arbitrary complex, time translations.

3. The resolved low-order spectrum is compatible with $c_n = \kappa_n c_1^n$ for bounded low-order κ_n (Section 5). It does not determine the large-n growth or the nearest singularity. A uniform strip-bound proof remains open (Section 5.1).

4. The scale Λ, independently of its conjectured relation to D_b, has the proved saddle-node exponent of Proposition 6. The measured $|\hat{c}_1|$ increases toward the first parabolic horn coefficient in the chosen normalisation (Section 6.1).

5. The reversed transition gives the second-mode target $|B_2| = 0.012487\ldots$, and the six measured ratios $|\kappa_2|/|\kappa_2^\infty|$ rise from 0.873 to 0.979 (Section 6.2). This is evidence for a limit, not its proof.

6. Translation invariance permits comparisons at every order. For the distinct problem of classifying a simple parabolic cylinder germ under tangent-to-identity formal conjugacy, the parameters are its quadratic coefficient a_2 and residue ρ, with $\kappa_2 = 2\pi\mathrm{i}(\frac{1}{2} - \rho)$ (Section 6.4). This classification does not prohibit testing higher Fourier coefficients.

7. The separation is a transition-map invariant (Section 7): $\hat{c}_n = t_n \mathrm{e}^{-2\pi i n t_0} + O(\Lambda)$ for the Kneser-free map $T_b = R_b^{-1} \circ S_b$ (Theorem 13); the conjectures follow for every n from a confluence statement (Theorem 24), and the limits are confirmed numerically to $\eta - b = 3 \cdot 10^{-5}$.

2 Setup

Throughout, $1 < b < \eta$ and $E(w) = b^w$.

Lemma 1 (Exact parametrisation of the family)**.** *Let L be a fixed point of E and $\lambda = E'(L) = L \log b$ its multiplier. Then*

$$L = \mathrm{e}^{\lambda}, \qquad \log b = \lambda\, \mathrm{e}^{-\lambda}. \tag{5}$$

The map $\lambda \mapsto \lambda \mathrm{e}^{-\lambda}$ increases on $(0,1)$ and decreases on $(1,\infty)$ with maximum $\mathrm{e}^{-1} = \log \eta$ at $\lambda = 1$; consequently $\lambda \in (0,1)$ parametrises $(1,\eta)$ through the attracting fixed point, $\lambda_2 \in (1,\infty)$ parametrises it through the repelling one, and the two branches meet at the parabolic base $b = \eta$.

Proof. $L = b^L$ gives $\log L = L \log b = \lambda$, hence $L = \mathrm{e}^{\lambda}$ and $\log b = \lambda/L = \lambda \mathrm{e}^{-\lambda}$. The monotonicity statements are those of $\lambda \mathrm{e}^{-\lambda}$. □

We write λ for the multiplier at the attracting fixed point and set

$$h := \frac{2\pi}{|\log \lambda|}, \qquad \Lambda := \exp\Big(\frac{4\pi^2}{\log \lambda}\Big) = \mathrm{e}^{-2\pi h}. \tag{6}$$

The quantity h is the modulus of the imaginary period of the regular solution: if σ denotes the Koenigs coordinate at L, normalised by $\sigma(E(w)) = \lambda\sigma(w)$, then $R(z) = \sigma^{-1}(c\lambda^z)$ for a suitable constant c, and λ^z has period $2\pi\mathrm{i}/\log\lambda$, of modulus h. Note in particular that, on the real segment, R is *exactly periodic* in the imaginary direction and therefore has no limit as $\operatorname{Im} z \to \pm\infty$; the two-sided limit characterisation of the Kneser solution degenerates precisely on $(1,\eta)$, which is the structural reason the interval is delicate.

3 The first-mode law

Proposition 2. *Let R and K be holomorphic solutions of (1), both normalised by $F(0) = 1$. Assume a branch of R^{-1} is defined on the required images of K and is compatible with the shift: $R^{-1}(E(w)) = R^{-1}(w) + 1$ wherever used. Put $P = R^{-1} \circ K - \mathrm{id}$. Then $P(z+1) = P(z)$ on the common domain and $P(0) = 0$, with the inverse branch fixed by $R^{-1}(1) = 0$.*

If $P = \sum_{n\geq 0} c_n \mathrm{e}^{2\pi i n z}$ converges locally uniformly near 0 and the points under consideration, then $c_0 = -\sum_{n\geq 1} c_n$. Define $Q(z) = \sum_{n\geq 2} c_n(\mathrm{e}^{2\pi i n z} - 1)$. Taylor expansion gives

$$D(z) = c_1 R'(z)(\mathrm{e}^{2\pi \mathrm{i} z} - 1) + R'(z)Q(z) + O(P(z)^2), \tag{7}$$

locally where the second derivative of R is bounded along the Taylor segments. Omitting Q at quadratic accuracy additionally requires $R'Q = O(P^2)$.

Proof. The functional equation and the compatible inverse branch imply $(R^{-1} \circ K)(z+1) = (R^{-1} \circ K)(z) + 1$. The normalisation gives $P(0) = 0$; evaluating the convergent series at zero yields the identity for c_0. Thus $P = c_1(\mathrm{e}^{2\pi \mathrm{i} z} - 1) + Q$; substitute this into $R(z+P(z)) - R(z) = R'(z)P(z) + O(P(z)^2)$. □

The numerical question is whether the tail Q and the Taylor remainder are small enough for the leading profile to dominate. We test this through

$$q(z) := \frac{D(z)}{R'(z)(\mathrm{e}^{2\pi \mathrm{i} z} - 1)}, \tag{8}$$

away from zeros of the denominator. First-mode dominance predicts an approximately constant q. Table 1 shows $|q|$ at four values of z, including $z = \frac{1}{2} - \frac{\mathrm{i}}{2}$, where the first positive mode is amplified by $\mathrm{e}^{\pi} \approx 23.14$ and the first negative mode is suppressed by the same factor.

Table 1: The first-mode law (7), tested through (8). The last column is $\max|q| / \min|q|$ over the four z. For comparison, the naive normalisation $D/(R'\mathrm{e}^{2\pi \mathrm{i} z})$, which omits the constant mode, has spread 1.917 at every point of the data set.

base	$z = \frac{1}{2}$	$z = -\frac{1}{2}$	$z = \frac{1}{4}$	$z = \frac{1}{2} - \frac{\mathrm{i}}{2}$	spread
$1.1 + 0.0005\mathrm{i}$	$1.987\,636 \times 10^{-9}$	$1.987\,636 \times 10^{-9}$	$1.987\,636 \times 10^{-9}$	$1.987\,637 \times 10^{-9}$	1.0000
$1.05 + 0.003\mathrm{i}$	$1.422\,053 \times 10^{-7}$	$1.422\,053 \times 10^{-7}$	$1.422\,053 \times 10^{-7}$	$1.422\,061 \times 10^{-7}$	1.0000

4 The normalised first coefficient $\hat{c}_1 = c_1/\Lambda$

Since K is defined as a boundary value (3), we compute c_1 at $b + \mathrm{i}y$ for a ladder of $y \downarrow 0$ and extrapolate. The relevant normalisation is by the scale Λ of (6):

$$\hat{c}_1(b) := \lim_{y \to 0^+} \frac{c_1(b + \mathrm{i}y)}{\Lambda(b + \mathrm{i}y)}. \tag{9}$$

Remark 3 (Translation invariance)**.** A simultaneous shift $K_\delta(z) = K(z+\delta)$, $R_\delta(z) = R(z+\delta)$ gives $P_\delta(z) = P(z+\delta)$ and $c_n \mapsto c_n \mathrm{e}^{2\pi \mathrm{i} n \delta}$. Thus every $\kappa_n = c_n / c_1^n$, for $c_1 \neq 0$, is invariant under complex translations. In contrast, $|c_1| \mapsto |c_1| \mathrm{e}^{-2\pi \operatorname{Im} \delta}$: $|\hat{c}_1|$ is invariant only under real shifts. Its comparison with a parabolic horn coefficient requires a fixed imaginary normalisation. Arguments of coefficients are meaningful once an origin has been specified. Independent source and target translations additionally change the constant mode, while leaving the ratios of positive modes invariant.

Table 2 collects the values. The first six rows are computed here; the seventh is Paulsen's number (4) divided by $2R'_{\sqrt{2}}(1/2)\Lambda(\sqrt{2})$; the last two use values of K published from `fatou.gp` runs, i.e. a completely independent implementation in PARI/GP, at two complex bases in the interior of the Shell–Thron region.

Remark 4 (Where the model stops)**.** The published `fatou.gp` set contains a third interior base, $b = 2 + \mathrm{i}$, with $|\lambda| = 0.98767$ and hence $|\Lambda| = 0.806$. There the "exponentially small" quantity is not small at all, the truncation to one mode is not justified, and indeed the same computation returns $|\hat{c}_1| = 0.1065$. We regard this as a useful marker of the boundary of validity rather than as a counterexample.

Table 2: The near-invariant $|\hat{c}_1|$. $|D|$ ranges over forty-five orders of magnitude; $|\hat{c}_1|$ varies by 4.6%. The column $|D|$ is the value of $|D(\frac{1}{2})|$ at the lowest rung of the ladder actually computed, not the limit. The last column compares with the horn-map coefficient $|A_1| = 0.08905843641221$ of the parabolic limit computed in Section 6.1. Rows marked † are complex bases, where $\hat{c}_1$ is a sample of the same function off the real segment and the last column is not meaningful.

b	λ (or $\lvert\lambda\rvert$)	Λ	$\lvert D\rvert$	$\lvert\hat{c}_1\rvert$	$\lvert\hat{c}_1\rvert/\lvert A_1\rvert$	source
1.05	0.051362	1.68×10^{-6}	1.01×10^{-8}	0.085017	0.95462	this work, 20 digits
1.10	0.105964	2.30×10^{-8}	3.24×10^{-10}	0.086427	0.97045	this work, 20 digits
1.15	0.164802	3.10×10^{-10}	7.08×10^{-12}	0.087205	0.97919	this work, 20 digits
1.20	0.229312	2.28×10^{-12}	7.30×10^{-14}	0.087732	0.98511	this work, 20 digits
1.25	0.301735	4.91×10^{-15}	2.04×10^{-16}	0.088125	0.98952	this work, 24 digits
1.30	0.385935	9.82×10^{-19}	1.16×10^{-19}	0.088434	0.99299	this work, 32 digits
$\sqrt{2}$	0.693147	1.66×10^{-47}	1.19×10^{-48}	0.088941	0.99868	[21], 120 digits
$0.8 + 0.4\mathrm{i}^{\dagger}$	0.387654	1.05×10^{-3}	9.86×10^{-5}	0.085592	—	`fatou.gp`, 80 digits
$1 + \mathrm{i}^{\dagger}$	0.710560	2.13×10^{-2}	4.41×10^{-3}	0.087272	—	`fatou.gp`, 80 digits

Internal consistency. At $b = 1.10$ the value $|\hat{c}_1| = 0.08642738$ is obtained from six settings (working precision 16 and 20 digits, θ sample height 0.10 and 0.15) and, separately, from a finer ladder reaching $y = 5 \times 10^{-4}$, which gives 0.08642720; the absolute difference is $1.8 \cdot 10^{-7}$ (about 2.1 parts per million). At $b = 1.20$ and $b = 1.25$ the same quantity computed at working precisions 20, 24 and 32 digits agrees to all seven digits printed, with signal-to-noise ratios up to $2 \cdot 10^{21}$.

5 The Fourier spectrum of P

Rather than fitting, one can measure P directly: for z on a horizontal line $\operatorname{Im} z = y_0$ compute $P(z) = R^{-1}(K(z)) - z$, choosing the branch of the logarithm in R^{-1} nearest to z, and transform. Lowering y_0 amplifies the mode n by $\mathrm{e}^{-2\pi n y_0}$ and so makes higher modes observable; raising y_0 does the same for negative modes.

Table 3: Spectrum of P at $b = 1.1 + 5 \times 10^{-4}\mathrm{i}$ ($\Lambda = 2.3001 \times 10^{-8}$, $h = 2.7992$). The values of $|c_0|, |c_1|, |c_2|$ are stable across the three sampling lines; $|c_3|$ is not resolved. The mode-to-mode ratio is 2.86×10^{-9}, an order of magnitude below Λ.

y_0	$\lvert c_0\rvert$	$\lvert c_1\rvert$	$\lvert c_2\rvert$	$\lvert c_3\rvert$
0.0	1.9876×10^{-9}	1.9876×10^{-9}	5.6899×10^{-18}	noise
-0.5	1.9876×10^{-9}	1.9876×10^{-9}	5.6899×10^{-18}	1.77×10^{-26}
-1.0	1.9876×10^{-9}	1.9876×10^{-9}	5.6899×10^{-18}	1.32×10^{-26}

Three facts come out of Table 3 and its analogues at $b = 1.05, 1.15$.

Numerical observation 5.

1. $c_0/c_1 = -1$ to ten significant digits, confirming the normalisation structure of Proposition 2.

2. Sampling on $\operatorname{Im} z = +0.5$ and $+1.0$ gives $|c_{-1}|$ equal to 4.0×10^{-23} and 1.3×10^{-23} respectively. A genuine coefficient does not depend on the sampling line; the measured values decrease as the line is raised, i.e. they are the numerical floor. Hence $|c_{-1}|/|c_1| \lesssim$

$5\cdot10^{-15}$: the hypothesis that P has no negative modes, which expresses the boundedness of K in the upper half-plane, is verified over fifteen orders of magnitude.

3. Writing $\kappa_n = c_n/c_1^n$, the measured κ_2 is $O(1)$:

$$|\kappa_2| = 1.36643,\ 1.44023,\ 1.47531 \quad \text{and} \quad \arg\kappa_2 = 1.41804,\ 1.47802,\ 1.51415$$

at $b = 1.05, 1.10, 1.15$. By Remark 3 these are translation invariants; they need not survive arbitrary nonlinear changes of time coordinate.

5.1 What the spectrum does not prove

A periodic function bounded by M on a horizontal line at depth d satisfies $|c_n| \le M\mathrm{e}^{-2\pi nd}$ for positive modes. A bound at $d = h = 2\pi/|\log\lambda|$ with M uniform as $b \to \eta^-$ would therefore establish $|c_1| = O(\Lambda)$. It would not, on its own, establish a nonzero limit of c_1/Λ.

The few resolved coefficients do not rule out this approach. For example, if the positive-mode series is written as $P - c_0 = \sum_{n\ge1}\kappa_n(c_1\xi)^n$, $\xi = \mathrm{e}^{2\pi\mathrm{i}z}$, then its radius in ξ is determined by $L_\kappa = \limsup_{n\to\infty}|\kappa_n|^{1/n}$. Only under the additional, unproved condition $L_\kappa = 1$ would its singularity depth be

$$d = \frac{-\log|c_1|}{2\pi} = h + \frac{\log(1/|\hat c_1|)}{2\pi}. \tag{10}$$

For general finite positive L_κ, subtract $\log L_\kappa/(2\pi)$ from the right-hand side. The final equality in (10) is an algebraic identity from $c_1 = \Lambda\hat c_1$; computing its two sides is not an independent measurement of a singularity. Moreover, a singularity deeper than h would be compatible with analyticity on the desired strip. The unresolved issue is a uniform bound obtained independently of the unknown coefficient.

In a domain where the Koenigs coordinate σ and a logarithm branch exist, $p(z) = \lambda^{-z}\sigma(K(z))$ is periodic and $P = \log(p/c)/\log\lambda$. Zeros of p can create logarithmic singularities, but other obstructions to continuation also require control. The present low-order spectrum neither locates all such obstructions nor settles the origin of the scale Λ.

6 The parabolic end

By Lemma 1, $\log b = \lambda\mathrm{e}^{-\lambda}$. Writing $\lambda = 1-\varepsilon$ and expanding,

$$\log b = \mathrm{e}^{-1}\Big(1 - \frac{\varepsilon^2}{2} - \frac{\varepsilon^3}{3} + O(\varepsilon^4)\Big), \qquad \text{so} \qquad \varepsilon = \sqrt{2\mathrm{e}\,(\log\eta - \log b)}\,(1+O(\varepsilon)),$$

and $|\log\lambda| = \varepsilon + \varepsilon^2/2 + \cdots$. (Numerically, the ratio of $|\log\lambda|$ to $\sqrt{2\mathrm{e}(\log\eta - \log b)}$ is 1.0386, 1.0108, 1.0033, 1.0003 for $\eta - b = 10^{-2}, 10^{-3}, 10^{-4}, 10^{-6}$.) Substituting into (6) and using $\log\eta - \log b \sim (\eta - b)/\eta$:

Proposition 6. *As $b \to \eta^-$,*

$$\Lambda(b) = \exp\Big(-\frac{C_\eta}{\sqrt{\eta - b}}\,(1+o(1))\Big), \qquad C_\eta = \frac{4\pi^2}{\sqrt{2\mathrm{e}^{\,1-1/\mathrm{e}}}} = 20.3508008\ldots \tag{11}$$

This is the classical shape $\exp(-C/\sqrt{\varepsilon})$ of exponentially small splitting at a saddle-node bifurcation. It is the right shape for structural reasons: at $b = \eta$ the two real fixed points (2) merge into the parabolic point $w = \mathrm{e}$ of $\mathrm{e}^{w/\mathrm{e}}$, and the two regular super-exponentials of the parabolic base [13] are the attracting- and repelling-petal Fatou parametrisations. The decay of Λ motivates a vanishing-separation conjecture. It does not establish it: control of $\hat c_1$, higher modes, and R_b' is also needed to transfer the scale to D_b.

For unfoldings of a parabolic germ into a pair of fixed points, the analytic classification is by an "unfolded" Écalle–Voronin modulus which converges, as the parameter tends to the parabolic value, to the modulus of the limiting germ [18, 24]; exponentially small differences between the two natural parametrisations are measured by the horn maps of that limit [7, 28, 15, 26, 19]. Our data are consistent with such a statement, and we can now test it directly.

6.1 The horn map of the parabolic limit

At $b = \eta$ put $w = \mathrm{e}(1 + u)$; the map becomes the parabolic germ

$$f(u) = \mathrm{e}^u - 1 = u + \tfrac{u^2}{2} + \tfrac{u^3}{6} + \cdots, \tag{12}$$

with attracting petal on $\operatorname{Re} u < 0$ and repelling petal on $\operatorname{Re} u > 0$, and with the (divergent) formal Abel function

$$\alpha(u) = -\frac{2}{u} + \frac{\log(-u)}{3} + \sum_{k\geq 1} c_k u^k, \qquad \alpha(f(u)) = \alpha(u) + 1. \tag{13}$$

Both Fatou coordinates are normalised by the *same* α:

$$\Phi_{\mathrm{att}}(u) = \lim_{n\to\infty} \big[\alpha(f^n(u)) - n\big], \qquad \Phi_{\mathrm{rep}}(u) = \lim_{n\to\infty} \big[\alpha(f^{-n}(u)) + n\big],$$

the branch of $\log(-u)$ being continued along the orbit. On the upper gate the transition $h = \Phi_{\mathrm{rep}} \circ \Phi_{\mathrm{att}}^{-1}$ commutes with $z \mapsto z + 1$ and tends to the identity as $\operatorname{Im} z \to +\infty$ (both coordinates being asymptotic to the same α), so

$$h(z) - z = \sum_{n\geq 1} A_n \, \mathrm{e}^{2\pi i n z}, \tag{14}$$

which is the lifted Écalle–Voronin horn map in the chosen normalisation. The modulus $|A_1|$ is unchanged under simultaneous real translations, but changes under imaginary translations; the specified α fixes that freedom. Computing $h(z) - z$ at N equally spaced points of a horizontal line and transforming gives

$$\boxed{|A_1| = 0.08905843641221} \qquad |A_2| = 0.037349258777, \qquad |A_3| = 0.02246384. \tag{15}$$

These values are unchanged, to all digits shown, when the working precision runs over 30–50 digits, the number of terms of (13) over 20–60, the matching radius over 0.005–0.02, the transform size over $N = 16, 24, 32$, and the sampling height over $\operatorname{Im} z = 1, 1.5, 2$; the constant term A_0 comes out as zero to 10^{-47} or better, as it must.

Comparing with Table 2: the hyperbolic-side quantity $|\hat{c}_1(b)|$ increases toward $|A_1|$ on the sampled bases, the ratio being 0.95462 at $b = 1.05$ and 0.99868 at $b = \sqrt{2}$. The value at $\sqrt{2}$ is not ours: it is Paulsen's 120-digit contour computation (4), so the 0.13% agreement there is between two computations sharing neither code, method, nor author.

Conjecture 7. $\lim_{b\to\eta^-} |\hat{c}_1(b)| = |A_1|$, the first horn-map coefficient (15) of the parabolic germ (12).

This is the expected shape of an unfolded Écalle–Voronin statement [18, 24]: the exponentially small splitting of the two natural parametrisations of the perturbed map, measured in the units forced by the multiplier, converges to the modulus of the parabolic limit. The numerical evidence concerns $n = 1$; the limit and the required normalisation match have not been established.

6.2 The second invariant: the reversed transition map

The obvious extension of Conjecture 7, namely $c_n/\Lambda^n \to A_n$, is inconsistent with the presently measured $n = 2$ values by a factor of about three: $|\hat{c}_2| = |\kappa_2|\,|\hat{c}_1|^2$ measures 0.00993, 0.01076, 0.01125 at $b = 1.05, 1.10, 1.15$ against $|A_2| = 0.037349$. The reason is a direction, not a rescaling, and reversing it yields a substantially closer numerical target. Neither comparison alone establishes a limiting relation.

The hyperbolic-side object is $P = R^{-1} \circ K - \mathrm{id}$, the attracting (regular) time as a function of the repelling (Kneser) time, i.e. the transition map $\alpha_{\mathrm{att}} \circ \alpha_{\mathrm{rep}}^{-1}$; the h of (14) is $\Phi_{\mathrm{rep}} \circ \Phi_{\mathrm{att}}^{-1}$, the other way round. Subject to establishing this identification of the Kneser time with the repelling time, the candidate limiting object is h^{-1}, and inverting a 1-periodic map tangent to the identity mixes the modes: writing $h^{-1}(z) - z = \sum_{n\geq 1} B_n \mathrm{e}^{2\pi \mathrm{i} n z}$,

$$B_1 = -A_1, \qquad B_2 = -A_2 + 2\pi\mathrm{i}A_1^2, \qquad B_3 = -\Big(A_1\big(2\pi\mathrm{i}B_2 + \tfrac{(2\pi\mathrm{i})^2}{2}B_1^2\big) + 4\pi\mathrm{i}A_2B_1 + A_3\Big), \tag{16}$$

so that $n = 1$ is unaffected in modulus — which is why Conjecture 7 was insensitive to the direction — while at $n = 2$ the term $2\pi\mathrm{i}A_1^2$, carrying the factor 2π, changes the order of magnitude. Computing $\Phi_{\mathrm{att}} \circ \Phi_{\mathrm{rep}}^{-1}$ *directly* by the algorithm of Section 6.1 run backwards, rather than through (16), gives the same numbers, with the constant term B_0 vanishing to 10^{-59}:

$$|B_1| = 0.089058436412213331563\ (= |A_1|), \qquad |B_2| = 0.012487384111564700811, \tag{17}$$

$$\begin{aligned} \kappa_2^\infty := \frac{B_2}{B_1^2} &= -0.02532049309975755123 + 1.57421906523195162 67\mathrm{i}, \\ |\kappa_2^\infty| &= 1.5744226855297069\ldots \end{aligned} \tag{18}$$

unchanged in all digits shown over working precisions 40–60 digits, transform sizes $N = 16, 24, 32$ and sampling heights $\operatorname{Im} z = 1, 1.5, 2$.

Proposition 8. *The first inverse upper horn coefficient satisfies $B_1 \neq 0$.*

Proof. For $f(u) = \mathrm{e}^u - 1$, the negative real ray belongs to its immediate parabolic basin A, since $u < f(u) < 0$ there. The only finite singular value of f is -1, an asymptotic value along that ray; f has no critical points. By [4, Lemma 87], the restriction $f|_A : A \to A$ has no other singular values. Thus f belongs to the one-petal class S_∞ of [4, Definition 8]. By [4, Theorem 9 and Corollary 11], its upper parabolic renormalisation has exactly one attracting petal at 0. This renormalisation is the exponential projection of the upper inverse horn map, up to nonzero linear scalings of source and target [4, Appendix A]. In our normalisation that projection is $H(\zeta) = \zeta + 2\pi\mathrm{i}B_1\zeta^2 + O(\zeta^3)$. After normalising the multiplier to one, such linear scalings preserve whether the quadratic coefficient vanishes. One attracting petal makes it nonzero, whence $B_1 \neq 0$. □

The hyperbolic-side κ_2, extrapolated to $y = 0$ along the ladders, is then

b	κ_2	$\lvert\kappa_2\rvert/\lvert\kappa_2^\infty\rvert$	$\lvert\hat{c}_2\rvert/\lvert B_2\rvert$
1.05	$1.374049\,\mathrm{e}^{+1.413420\mathrm{i}}$	0.87273	0.79532
1.10	$1.440851\,\mathrm{e}^{+1.477690\mathrm{i}}$	0.91516	0.86188
1.15	$1.479111\,\mathrm{e}^{+1.511255\mathrm{i}}$	0.93946	0.90077
1.20	$1.505578\,\mathrm{e}^{+1.533318\mathrm{i}}$	0.95627	0.92800
1.25	$1.525582\,\mathrm{e}^{+1.549420\mathrm{i}}$	0.96898	0.94875
1.30	$1.541516\,\mathrm{e}^{+1.561918\mathrm{i}}$	0.97910	0.96538

monotonically increasing, with the deficit $1-|\hat{c}_2|/|B_2|$ a varying multiple 4.5–4.9 of the $n=1$ deficit on the same bases, and with $\arg\kappa_2$ approaching $\arg\kappa_2^\infty = 1.586879$ from below $(1.4134, 1.4777, 1.5113, 1.5333, 1.5494, 1.5619)$. The last three rows are runs made after the truncation of Remark 11 was removed; there the two sampling lines agree to 10^{-6} at $b=1.25$ and 10^{-14} at $b=1.30$. For $b=1.05$ and $b=1.10$ two mutually independent ladders agree on κ_2 to four and six digits.

Conjecture 9. $\lim_{b\to\eta^-} \kappa_2(b) = \kappa_2^\infty$ of (18), the second coefficient of the reversed horn map of the parabolic germ (12), normalised by the first.

6.3 How fast the limit is approached

In the fixed imaginary normalisation, write $\delta_1(b) = 1-|\hat{c}_1(b)|/|B_1|$ for the first-mode modulus deficit. Over nine bases — eight computed here and $b=\sqrt{2}$ from (4) — δ_1 runs from $4.53\cdot10^{-2}$ at $b=1.05$ to $1.22\cdot10^{-3}$ at $b=\sqrt{2}$, and the natural variable for it is the multiplier scale $\varepsilon := |\log\lambda|$:

b	ε	$\eta-b$	δ_1	δ_1/ε^2
1.05	2.96886	0.39467	0.0453065	0.005140
1.10	2.24465	0.34467	0.0295431	0.005863
1.15	1.80301	0.29467	0.0208085	0.006401
1.20	1.47267	0.24467	0.0148953	0.006868
1.25	1.19820	0.19467	0.0104955	0.007310
1.30	0.95209	0.14467	0.0070327	0.007758
1.35	0.71389	0.09467	0.0042041	0.008249
1.40	0.45443	0.04467	0.0018301	0.008862
$\sqrt{2}$	0.36651	0.03045	0.0012202	0.009084

The historical exploratory fits to these nine values report a pure-power exponent $p=1.737$ (log-RMS 0.040). For models $\log\delta_1 = \log C + p\log\varepsilon + \sum_{j=1}^{m} a_j\varepsilon^j$, free fits with $m=1,2,3$ give $p = 1.928, 1.974, 1.99893$ and log-RMS $0.0032, 0.00058, 0.00017$. These sample residuals suggest, but do not identify, the candidate asymptotic law

$$\delta_1(b) = C\varepsilon^2(1+O(\varepsilon)), \qquad \varepsilon = |\log\lambda|, \qquad C \approx 0.01013. \tag{19}$$

The previously printed uncertainty ±0.00002 was neither a confidence interval nor a certified error bound and is withdrawn. The reported fixed $p=2$ fit used three correction terms, whereas the reported $p=7/4$ fit used one. Their residuals are not a fair exponent comparison. Matched correction orders, sensitivity to omitted points, and data closer to $\varepsilon=0$ are required before excluding alternatives. Adding correction parameters also reduces residuals without necessarily improving extrapolation.

The parameter relation gives $\eta-b \sim \varepsilon^2/(2\mathrm{e}^{1-1/\mathrm{e}})$ and $1-\lambda\sim\varepsilon$. If the normalised deficit were differentiable in the unfolding parameter with a nonzero linear term, a quadratic correction in ε would follow. Those hypotheses have not been proved. The saddle-node geometry alone does not establish regularity of the normalised observable. The measured range reaches only $\varepsilon\approx0.367$ and finite-range fits change substantially with the choice of variable.

Paulsen publishes only $\operatorname{Im} K_{\sqrt{2}}(1/2)$. Using $\operatorname{Im} D(1/2) \approx -2|c_1|R'(1/2)\sin(\arg c_1)$ to recover a modulus requires a phase estimate. The corrected deficit 0.0012202 uses a phase extrapolated from our nearest two bases; it is not an independent modulus measurement. Omitting this correction gives 0.0013210. The one-correction exponents for the corrected,

uncorrected, and omitted external point are $1.928, 1.853, 1.919$, respectively. These are sensitivity checks, not proof of an exponent. The reproducible `deficit_audit.py` audit separates eight internal points, nine points with the phase correction, and nine without it. Their cubic-correction free exponents are approximately 1.99053, 1.99889, and 1.40683. More seriously, the two archived ladders at $b = 1.35$ differ in $|\hat{c}_1|$ by about $8.1 \cdot 10^{-5}$, about 22% of the deficit when expressed in deficit units. Substituting archived alternative readings substantially changes the fits. This observed spread is a sensitivity warning, not a probability model or a certified uncertainty interval.

Writing $\delta_\kappa = 1 - |\kappa_2|/|\kappa_2^\infty|$, the measured ratios are

$$\delta_\kappa/\delta_1 = 2.8090,\ 2.8717,\ 2.9093,\ 2.9356,\ 2.9557,\ 2.9718 \quad (b = 1.05, \dots, 1.30).$$

Aitken extrapolation gives 3.04, with no error bound or established convergence model. The exact identity $\delta_2 = 1 - (1 - \delta_\kappa)(1 - \delta_1)^2$ implies a difference of 2 between the ratios only asymptotically, if δ_κ/δ_1 remains bounded and both deficits vanish.

An earlier explanation posited the common relative change $a_2 \mapsto a_2(1+s)$, $a_3 \mapsto a_3(1+s)$. Its algebra is

$$1 - \frac{\kappa_2(s)}{\kappa_2^\infty} = P\frac{s}{1+s}, \qquad P = \frac{1-\rho}{1/2-\rho} = 2.99514 - 0.03209\mathrm{i},$$
$$\delta_1 = 1 - |1+s| = -\operatorname{Re} s + O(|s|^2).$$

Consequently, for $q = (1 - \kappa_2/\kappa_2^\infty)/\delta_1$, $\operatorname{Re}(q/P)$ should approach -1, provided $|s|^2 = o(\delta_1)$. The previous version mistakenly used $+1$. The reported values 1.0224, 1.0082, 1.0021, 0.9985, 0.9960, 0.9942 do not support that model. Moreover, the modulus-deficit ratio depends on $\operatorname{Re}(Ps)/(-\operatorname{Re} s)$, not generally on $|P|$. The proposed target $|P| = 2.995307$ and the phase-limit estimate derived from this model are therefore withdrawn. Direct phase measurements remain usable, but a controlled limiting phase alignment remains open.

Remark 10. The fits above are superseded by (28): in the unfolding parameter $p = |\log\lambda| \log\lambda_2$, the deficit is analytic with $\delta_1 = 0.0101990063\, p + O(p^2)$.

6.4 Translation gauge and formal conjugacy

Standard Fatou coordinates on petals with the prescribed asymptotic form are unique up to additive constants, and their transition maps change by source and target translations [3, Sections 2.2–2.3]. All κ_n are invariant under these translations (Remark 3). Arbitrary periodic changes of Abel coordinate need not preserve those asymptotic conditions; separate changes on the two sides do not in general act by a simultaneous conjugacy.

A different question arises after setting the transition's constant mode to zero. Its cylinder germ is then

$$F(\zeta) = \zeta \exp\Big(2\pi\mathrm{i} \sum_{n\ge1} B_n \zeta^n\Big) = \zeta + a_2\zeta^2 + a_3\zeta^3 + \cdots, \quad a_2 = 2\pi\mathrm{i}B_1, \quad a_3 = 2\pi\mathrm{i}B_2 + \tfrac{(2\pi\mathrm{i})^2}{2}B_1^2. \tag{20}$$

The multiplier is $F'(0) = 1$; a_2 is the quadratic coefficient. By Proposition 8, this is a simple parabolic germ, and formal conjugacy tangent to the identity preserves a_2 and the iterative residue $\rho = 1 - a_3/a_2^2$. Allowing a nonzero linear change also permits normalising a_2 to 1; see [3, Definition 1 and Appendix]. Substitution gives the exact identity

$$\kappa_2 = \frac{B_2}{B_1^2} = 2\pi\mathrm{i}\Big(\frac{1}{2} - \rho\Big), \qquad \rho = 0.249455254258832 - 0.004029881638351\mathrm{i}. \tag{21}$$

If the constant mode c_0 is retained, the multiplier instead equals $\mathrm{e}^{2\pi \mathrm{i} c_0}$; the tangent-to-identity formula requires its removal. Subtracting c_0 in the output time coordinate does so without changing positive Fourier coefficients.

The gauge-check script conjugates h^{-1} simultaneously by $\psi(z) = z + \sum_{n\geq 1} s_n \mathrm{e}^{2\pi \mathrm{i} n z}$, which descends to a tangent-to-identity cylinder change. With the three tested coefficient lists, $|c_1|$ and κ_2 remain unchanged to twelve printed digits, while κ_3 becomes $22.9 - 2.7\mathrm{i}$, $-26.2 + 81.9\mathrm{i}$, and $121.2 - 63.8\mathrm{i}$. This checks invariance under that larger group; its rounded inputs do not certify the underlying horn coefficients to twelve digits.

It does not follow that higher Fourier comparisons are meaningless. Once both sides use compatible normalised coordinates, $\kappa_n(b) \to B_n/B_1^n$ remains a testable proposal for every n. The present $n \geq 3$ readings lack reliable numerical resolution; that is why they are not used as convergence evidence. Formal coordinate changes also need not converge, so finite formal classification does not eliminate the analytic information carried by a horn map.

Remark 11 (Precision and storage)**.** An earlier harness stored the $O(1)$ Taylor coefficients of K with 24 significant digits. This produced a floor near 10^{-25} in reconstructed spectra even when the working precision was increased. The previously reported $|\hat{c}_2| = 0.01203$ at $b = 1.20$ was not a reliable measurement. The table in Section 6.2 uses the corrected storage runs for $b = 1.15, 1.20, 1.25, 1.30$; the former warning that these reruns were pending is obsolete. Agreement across sampling lines is a consistency test, not an error bound. The first-mode values extracted from separately stored D avoid this particular coefficient-storage floor, without thereby becoming certified estimates.

7 The separation is a transition-map invariant

Everything so far compares K_b, a boundary value of a family built by the θ-mapping, with R_b. This section removes K_b from the problem. The separation coefficients turn out to be determined, up to a relative error of order Λ, by the transition map between the two *regular* solutions at the two real fixed points — an object built from two convergent Schröder series in a fraction of a second, at any $b < \eta$.

7.1 The transition map

Let τ be the Koenigs coordinate at the repelling point L_2, $\tau(E(w)) = \lambda_2 \tau(w)$, $\tau'(L_2) = 1$, and put

$$S(z) = \tau^{-1}(-\lambda_2^z),$$

continued to an entire function by $S(z+1) = E(S(z))$. It maps $\mathbb{R}$ decreasingly onto the segment (L, L_2) and has period $2\pi\mathrm{i}/\log\lambda_2$. Let $R^{-1}(w) = \log(\sigma(w)/\sigma(1))/\log\lambda$ be the attracting Abel coordinate, $R^{-1}(1) = 0$, which is multivalued with period $\mathrm{i}h = 2\pi\mathrm{i}/|\log\lambda|$. On the horizontal strip where S takes values in the immediate basin of L and a continuous branch is chosen, put

$$T(z) := R^{-1}(S(z)), \qquad T(z+1) = T(z) + 1, \qquad T(z) - z = \sum_{n\in\mathbb{Z}} t_n \mathrm{e}^{2\pi \mathrm{i} n z}.$$

Lemma 12 (Real structure)**.** *For real $b \in (1, \eta)$, $\operatorname{Im} T(x) \equiv h/2 \pmod{h}$ for every real x. Consequently, on the branch with $\operatorname{Im} t_0 = h/2$, $T(x) - x - \mathrm{i}h/2$ is real, $t_{-n} = \overline{t_n}$, and the line $\operatorname{Im} w = -h/2$ of attracting time is the image of $\mathbb{R}$ under the branch $T - \mathrm{i}h$.*

Proof. $S(x) \in (L, L_2)$, where $\sigma > 0$, while $1 < L$ gives $\sigma(1) < 0$. Hence $\sigma(S(x))/\sigma(1)$ is negative and $\operatorname{Im}\log(\sigma(S(x))/\sigma(1)) \in \pi + 2\pi\mathbb{Z}$; divide by $\log\lambda = -2\pi/h$. □

Translations of S ($S \mapsto S(\cdot+\delta)$, $\delta \in \mathbb{C}$) send $t_0 \mapsto t_0+\delta$, $t_n \mapsto t_n \mathrm{e}^{2\pi \mathrm{i} n\delta}$, so

$$\tau_n(b) := t_n \, \mathrm{e}^{-2\pi \mathrm{i} n t_0} \qquad (\operatorname{Im} t_0 = h/2) \tag{22}$$

does not depend on the normalisation of S; it depends on that of R only through the real translation fixed by $R(0)=1$. The condition $\operatorname{Im} t_0 = h/2$ refers to the real normalisation of S and fixes the branch of R^{-1}; a branch shift by $\mathrm{i}kh$ would multiply τ_n by Λ^{-nk}. The branch is fixed once in the real normalisation and then carried along.

7.2 The welding identity

Theorem 13. *Let K be a solution of (1), holomorphic on the sets below where it is represented, and suppose there are 1-periodic holomorphic maps $\varphi_\pm = \mathrm{id} + (\text{periodic})$ with*

1. *$K = R \circ \varphi_+$ on an upper half-plane U_+, where $P = \varphi_+ - \mathrm{id} = \sum_{n\geq 0} c_n \mathrm{e}^{2\pi \mathrm{i} n z}$ converges on U_+;*
2. *$K = S \circ \varphi_-$ on a lower half-plane U_-, where φ_- is univalent, $G := \varphi_-^{-1} - \mathrm{id}$ is bounded on $V_- = \varphi_-(U_-)$, which contains a lower half-plane, and $G(w) \to g_0$ as $\operatorname{Im} w \to -\infty$;*
3. *$U_+ \cap U_-$ contains a horizontal line ℓ with $\varphi_-(\ell)$ inside the strip of T.*

Then, with the branch of T fixed by continuity from $\varphi_+ \circ \varphi_-^{-1}$, for every $n \geq 1$

$$t_n = \sum_{m \geq n} c_m \big[\mathrm{e}^{2\pi \mathrm{i} m G}\big]_{n-m}, \qquad t_0 = g_0 + \sum_{m \geq 0} c_m \big[\mathrm{e}^{2\pi \mathrm{i} m G}\big]_{-m}, \tag{23}$$

where $[F]_k$ is the k-th Fourier coefficient; in particular $[\mathrm{e}^{2\pi \mathrm{i} m G}]_0 = \mathrm{e}^{2\pi \mathrm{i} m g_0}$ and $[\mathrm{e}^{2\pi \mathrm{i} m G}]_k = 0$ for $k > 0$.

Proof. On ℓ, $R \circ \varphi_+ = S \circ \varphi_- = R \circ T \circ \varphi_-$, so $\varphi_+ - T \circ \varphi_-$ takes values in the discrete period lattice of R and is constant; absorb it into the branch of T. Hence $T = \varphi_+ \circ \varphi_-^{-1}$ on the curve $\varphi_-(\ell)$, i.e. $T(w) - w = G(w) + P(w + G(w))$ there. The Fourier coefficients are computed by Cauchy's theorem on this curve. The region below it lies in V_-, and $\sum c_m \mathrm{e}^{2\pi \mathrm{i} m \varphi_-^{-1}(w)}$ converges uniformly on the curve $\varphi_-(\ell)$ itself, because $\ell \subset U_+$. Since G is bounded and holomorphic on V_-, with limit g_0 at $-\mathrm{i}\infty$, so is $\mathrm{e}^{2\pi \mathrm{i} m G}$, whose Fourier series therefore contains only modes $k \leq 0$ and has constant term $\mathrm{e}^{2\pi \mathrm{i} m g_0}$. Expanding $P(w+G) = \sum_m c_m \mathrm{e}^{2\pi \mathrm{i} m w} \mathrm{e}^{2\pi \mathrm{i} m G(w)}$ and collecting modes gives (23); the Fourier coefficients of $T - \mathrm{id}$ do not depend on the line inside its strip of holomorphy. □

Corollary 14. *Under the hypotheses of Theorem 13, for every $n \geq 1$,*

$$t_n \, \mathrm{e}^{-2\pi \mathrm{i} n g_0} = c_n + \mathrm{e}^{-2\pi \mathrm{i} n g_0} \sum_{m > n} c_m [\mathrm{e}^{2\pi \mathrm{i} m G}]_{n-m}.$$

If $c_n \neq 0$, this may be written as

$$t_n \, \mathrm{e}^{-2\pi \mathrm{i} n g_0} = c_n (1 + \rho_n'), \qquad |\rho_n'| \leq \sum_{m > n} \frac{|c_m|}{|c_n|} \, \big|[\mathrm{e}^{2\pi \mathrm{i} m G}]_{n-m}\big| \, \big|\mathrm{e}^{-2\pi \mathrm{i} n g_0}\big|,$$

and $g_0 = t_0 - c_0 - \sum_{m \geq 1} c_m [\mathrm{e}^{2\pi \mathrm{i} m G}]_{-m}$. Heuristically, if the c_m decay like Λ^m and G is bounded on a line at height comparable to h above the gluing line, both corrections are $O(\Lambda)$. The precise statement for the sewing solution is Theorem 32(4). On the branch of Lemma 12, $g_0 = t_0 - \mathrm{i}h + O(\Lambda)$, so that for the sewing solution, by Theorem 32(4),

$$\hat{c}_n := \frac{c_n}{\Lambda^n} = \tau_n(b) + O(\Lambda). \tag{24}$$

Since $B_1 \neq 0$, also $c_1 \neq 0$ for b sufficiently close to η and $\kappa_n := c_n / c_1^n = t_n / t_1^n + O(\Lambda)$ for each fixed n.

The identity $\mathrm{e}^{-2\pi \mathrm{i} n(t_0 - \mathrm{i}h)} = \Lambda^n \mathrm{e}^{-2\pi \mathrm{i} n t_0}$ is where the scale Λ enters: *the Λ law is the statement that the gluing line sits at attracting time $-h/2$, half a period below the real axis, while the gate carrying the positive modes of T sits a further half period below it.* Nothing in the argument uses Kneser's construction.

7.3 Welding construction

Proposition 15. *For $b \in (1, \eta)$ glue the half-cylinders $C_- = \{\operatorname{Im} w' \le 0\}/\mathbb{Z}$ (repelling time) and $C_+ = \{\operatorname{Im} w \ge -h/2\}/\mathbb{Z}$ (attracting time) along their boundaries by $w = T(w') - \mathrm{i}h$. The result is conformally $\mathbb{C}^*$; its universal-cover coordinate z defines a solution*

$$K^W(z) = R(w(z)) \text{ on } C_+, \qquad K^W(z) = S(w'(z)) \text{ on } C_-,$$

unique up to a translation in z, which satisfies the hypotheses of Theorem 13. It is holomorphic except where $w(z)$ meets the singular set $\bigcup_k ((-\infty, -2] + \mathrm{i}kh)$ of R.

Unlike S, which is entire (the Poincaré function of a repelling point), R is not: its inverse Koenigs map extends only through inverse branches of E. So K^W carries, besides the cut near $(-\infty, -2]$, the copies near $(-\infty, -2] + \mathrm{i}kh$, $k \ge 1$, lying in its R-represented region. The same holds for the Kneser-type solutions at complex bases inside the Shell–Thron region, whose upper representation is through the attracting regular superfunction. An earlier draft wrote "entire" here, which is wrong.

Proof. By Lemma 12 the gluing map is a real-analytic diffeomorphism between the boundary circles that extends holomorphically to a neighbourhood, so the glued surface X is a Riemann surface; each end is a punctured disc, hence $X \cup \{\text{two points}\}$ is a simply connected compact Riemann surface and $X \cong \mathbb{C}^*$. The definitions agree on the seam because $R \circ (T - \mathrm{i}h) = S$. The generator of $\pi_1(X)$ lifts to $z \mapsto z + 1$ and acts as $w \mapsto w + 1$, $w' \mapsto w' + 1$, whence $K^W(z+1) = E(K^W(z))$. Near each puncture the change of coordinate is a holomorphic map of punctured discs fixing the puncture, which is the statement that $w - z$ (resp. $w' - z$) has only non-negative (resp. non-positive) Fourier modes. Uniqueness is the uniqueness of the uniformisation. □

Proposition 16 (Local parameter dependence of the sewn family)**.** *For every $b_0 \in (1, \eta)$, the normalised geometric sewing of Proposition 15 extends to a holomorphic family over a complex disc Δ_{b_0} about b_0. On every common regular domain of the universal-cover coordinate, $K_b^W(z)$ is jointly holomorphic in (b, z). These local families agree on overlaps and therefore define the inner sewn family on an open complex neighbourhood of the entire real interval $(1, \eta)$. No uniform width of that neighbourhood is asserted.*

Proof. Fix b_0. Both fixed points and their Koenigs coordinates vary holomorphically near b_0. The compact set $S_{b_0}([0,1]) \subset (L, L_2)$ lies in the attracting basin and avoids its fixed point. Thus the chosen branch of $T_b = R_b^{-1} \circ S_b$ is jointly holomorphic on a parameter disc times a narrow strip about $[0,1]$; the relation $T_b(w'+1) = T_b(w') + 1$ extends it to the whole strip. The real circle map is orientation preserving: $\tau_b'(S_b(x)) > 0$ by inverse iteration to L_2, $\sigma_b'(S_b(x)) > 0$ by forward iteration to L, and hence $S_b'(x) < 0$ and $(R_b^{-1})'(S_b(x)) = \sigma_b'(S_b(x))/(\sigma_b(S_b(x)) \log \lambda_b) < 0$. Consequently $T_{b_0}'(x) > 0$. Compactness of the circle and local injectivity give a smaller strip on which T_{b_0} is injective modulo 1; shrinking the parameter disc preserves this property and makes the image of the real circle an analytic Jordan curve in the cylinder.

Put $U_b = T_b + \varpi_b$, where $\varpi_b = 2\pi\mathrm{i}/\log \lambda_b$; at b_0 this is $T_{b_0} - \mathrm{i}h$. Use U_b to identify a thin open annular collar in the repelling cylinder with its image in the attracting cylinder. More explicitly, for a sufficiently small $\delta > 0$ take the repelling chart $\{\operatorname{Im} w' < \delta/2\}/\mathbb{Z}$ and

the attracting chart above the curve $U_b(\{\operatorname{Im} w' = -\delta/2\})$; identify them on $|\operatorname{Im} w'| < \delta/2$. After shrinking the parameter disc, these charts and the transition $(b, w') \mapsto (b, U_b(w'))$ form a Hausdorff holomorphic family of cylinders. At $b = b_0$ it is the analytic-collar version of the boundary welding in Proposition 15. Adjoin the two ends using the parameter independent local coordinates $\mathrm{e}^{2\pi \mathrm{i} w}$ and $\mathrm{e}^{-2\pi \mathrm{i} w'}$. This yields a proper holomorphic submersion whose fibres are compact Riemann surfaces of genus zero. The two added points and the upper-chart point $w = 0$ are three disjoint holomorphic sections (the last remains above the seam after shrinking the disc).

Local holomorphic triviality of a proper family with mutually biholomorphic compact fibres [8], followed by the unique fibrewise Möbius map sending these three sections to $0, \infty, 1$, gives a jointly holomorphic coordinate q_b on the compactified fibres. Choose the local lift $z = (2\pi\mathrm{i})^{-1} \log q_b$ for which the marked upper point $w = 0$ has $z = 0$. On the lift of the overlap, $R_b(U_b(w')) = S_b(w')$ because ϖ_b is a period of R_b. The two regular superfunctions therefore define K_b^W, jointly holomorphic wherever their represented branches are regular, and $K_b^W(0) = R_b(0) = 1$. On the real parameter slice this is exactly the geometric welding. Its marked coordinate makes the local families unique, so the identity theorem patches them on overlapping discs. □

Corollary 17 (Non-reality at fractional real heights)**.** *For every real $b < \eta$ sufficiently close to η, and every $\delta > 0$ sufficiently small, some real x with $0 < |x| < \delta$ satisfies $\operatorname{Im} K_b^W(x) \neq 0$. In particular the sewn solution is not real-valued on any real interval about $z = 0$.*

Proof. Suppose that $K_b^W(x)$ were real for all real x in a neighbourhood of 0. Shrink that neighbourhood so its values lie in a real interval about 1 on which the attracting Koenigs coordinate σ_b is holomorphic and real, with $\sigma_b(K_b^W(x))/\sigma_b(1) > 0$. The upper representation of Theorem 32 gives

$$\frac{\sigma_b(K_b^W(x))}{\sigma_b(1)} = \lambda_b^{\,x + P_b(x)}, \qquad P_b(0) = 0.$$

Since $\log \lambda_b$ is real and nonzero, continuity at $x = 0$ makes $\operatorname{Im} P_b(x) = 0$ on this interval. The function P_b is holomorphic and periodic on a strip containing the whole real line, so reflection and the identity theorem make it real on that line. But its Fourier expansion has only non-negative modes. A real-valued periodic function has conjugate positive and negative coefficients; therefore all its nonconstant coefficients vanish. This contradicts $c_1(b) \neq 0$ for b sufficiently close to η, established in Corollary 14 using Proposition 8. □

This analytic non-reality statement concerns the sewn family near η. It is consistent with the tiny non-real part observed numerically by Paulsen at $b = \sqrt{2}$ [21, §6], but does not identify his base-continued family with K^W.

Corollary 18 (Complex cusp profile and its imaginary trace)**.** *Let $A = B_1 \mathrm{e}^{2\pi \mathrm{i} a} \neq 0$ and put $f = F_\eta^{\mathrm{att}}$. On some fixed complex disk about $z = 0$, as $b \to \eta^-$ through real bases,*

$$\frac{K_b^W(z) - R_b(z)}{\Lambda(b)} \longrightarrow f'(z) A(\mathrm{e}^{2\pi \mathrm{i} z} - 1)$$

locally uniformly, hence with all z-derivatives. In particular, on a fixed real interval about 0,

$$\frac{\operatorname{Im} K_b^W(x)}{\Lambda(b)} \longrightarrow f'(x) \operatorname{Im}\Big[A(\mathrm{e}^{2\pi \mathrm{i} x} - 1)\Big]$$

uniformly on compact subintervals. The right-hand side is not identically zero on any interval about 0.

Proof. By Theorem 32(4), with $B = \mathrm{e}^{2\pi(Y_0+D+2)}$, there is a constant C independent of b and n such that $|c_n(b)| \le C(\Lambda B)^n$ for all $n \ge 1$ and b sufficiently close to η. Here we used $|\tau_n| \le M\mathrm{e}^{2\pi n(Y_0+1)}$ and absorbed the factor Λ in the error bound into C. Since $P_b(0) = 0$, its constant mode is $-\sum_{n\ge1} c_n$. Choose a disk in the common upper representation supplied by Corollary 34. The Fourier tail is geometric on every smaller disk once b is close enough to η, and there

$$P_b(z) = c_1(b)(\mathrm{e}^{2\pi\mathrm{i}z} - 1) + O(\Lambda^2)$$

locally uniformly. Theorem 32(4) and Corollary 29 give $c_1/\Lambda \to A$. On a fixed disk about 0, $R_b \to F_\eta^{\mathrm{att}}$ with its derivatives by Corollary 34 and its proof. Taylor's formula gives $K_b^W(z) = R_b(z) + R_b'(z)P_b(z) + O(\Lambda^2)$ on a smaller disk. This proves the complex limit, with derivative convergence by Cauchy's formula. For real x, $R_b(x)$ and $f'(x)$ are real, giving the imaginary trace. If its last factor vanished on an interval, both the first and second derivatives at $x = 0$ would vanish, forcing $\operatorname{Re} A = \operatorname{Im} A = 0$, contrary to $A \ne 0$. □

Corollary 19 (Obstruction at the real cusp)**.** *For every sufficiently small complex neighbourhood V of 0, one can choose a real $x_* \in V \setminus \{0\}$ and constants $c, \delta > 0$ such that*

$$|\operatorname{Im} K_b^W(x_*)| \ge c\Lambda(b) \qquad (\eta - \delta < b < \eta,\ b \in \mathbb{R}).$$

There is therefore no jointly holomorphic family $F(b, z)$ on a product of a disc about $b = \eta$ and V which is real at real heights for real $b > \eta$ and agrees with K_b^W for all real $b < \eta$ sufficiently close to η. In particular, direct holomorphic continuation of a real-normalised Kneser family through the real cusp cannot identify it with the inner sewn branch.

Proof. The limiting profile in Corollary 18 is not identically zero on any real interval about 0. Choose x_* where it is nonzero. Since $\Lambda(b) > 0$ for real $b < \eta$, the uniform limit gives the stated lower bound after reducing δ. For fixed real x_*, Schwarz reflection in the parameter says that a holomorphic function $b \mapsto F(b, x_*)$ which is real on a real interval above η is also real on the real interval below η in the same disc. This contradicts the lower bound if $F = K^W$ there. □

Corollary 20 (Conjugate inner branches and their jump)**.** *For real $b \in (1, \eta)$ put $K_b^{W,\#}(z) = \overline{K_b^W(\bar z)}$ on the reflected regular domain. This reflected family is locally jointly holomorphic in (b, z) near every such real base. Both branches obey the tetration equation, are normalised at $z = 0$, agree at every nonnegative integer height where defined, and have a simple zero at $z = -1$ with the same logarithmic monodromy $2\pi\mathrm{i}/\log b$ at $z = -2$. For x_* as in Corollary 19,*

$$|K_b^W(x_*) - K_b^{W,\#}(x_*)| \ge 2c\Lambda(b) \qquad (\eta - \delta < b < \eta,\ b \in \mathbb{R}).$$

More precisely, on a fixed real interval about 0 the quotient of their difference by $\Lambda(b)$ tends locally uniformly to $2\mathrm{i}(F_\eta^{\mathrm{att}})'(x)\operatorname{Im}[A(\mathrm{e}^{2\pi\mathrm{i}x} - 1)]$.

Proof. For complex b near a real base, define the reflected family by $K_b^{W,\#}(z) = \overline{K_{\bar b}^W(\bar z)}$. Proposition 16 makes it jointly holomorphic. Reflection preserves the equation for real b and therefore, by the identity theorem, on its local complex parameter disc. The integer values follow from the equation and normalisation. The zero at -1 and its simplicity follow from Proposition 84, as does the monodromy. On real heights the difference is exactly $2\mathrm{i}\operatorname{Im} K_b^W(x)$, so the lower bound and the profile follow from Corollaries 19 and 18. □

Proposition 21 (The cusp splitting is beyond all algebraic orders)**.** *On a sufficiently small real interval J about 0, put*

$$M(x) = (F_\eta^{\mathrm{att}})'(x)\operatorname{Im}[A(\mathrm{e}^{2\pi\mathrm{i}x} - 1)].$$

The interval J may be chosen so that $M(x) \neq 0$ for every $x \in J \setminus \{0\}$. For every such x and every integer $q \geq 1$, the germ along $t > 0$ of

$$t \longmapsto K^W_{\eta - t^q}(x)$$

has no meromorphic extension through $t = 0$. In particular, no finite ramification of the base parameter makes this value meromorphic at the real cusp; it has no convergent Puiseux–Laurent germ there. If it admits a single-valued holomorphic continuation on a full punctured t-disc, the singularity at $t = 0$ is essential.

Proof. The profile M is real analytic, vanishes at 0, and, by Corollary 18, is not identically zero on any interval about 0. Thus 0 is an isolated zero, and we shrink J accordingly. Suppose such a meromorphic germ $G(t)$ existed. Its reflection $G^{\#}(t) = \overline{G(\bar{t})}$ would also be meromorphic, so $D(t) = G(t) - G^{\#}(t)$ would be meromorphic near 0. For positive real t, the imaginary profile and the saddle-node scale of Λ give

$$D(t) = 2\mathrm{i}\,\mathrm{Im}\,K^W_{\eta - t^q}(x) = O\Big(\exp(-c_0 t^{-q/2})\Big) = o(t^N) \qquad \text{for every } N \geq 0$$

with some $c_0 > 0$. A meromorphic Laurent germ with this decay on the positive axis is identically zero. This contradicts the nonzero limit $D(t)/\Lambda(\eta - t^q) \to 2\mathrm{i}M(x)$. □

Corollary 22 (A two-jet witness at the normalisation height)**.** *Put $f = F^{\mathrm{att}}_{\eta}$ and $A = B_1 \mathrm{e}^{2\pi \mathrm{i} a} \neq 0$. As $b \to \eta^-$ through real bases,*

$$\frac{1}{\Lambda(b)} \begin{pmatrix} \mathrm{Im}[(K^W_b)'(0)] \\ \mathrm{Im}[(K^W_b)''(0)] \end{pmatrix} \longrightarrow \begin{pmatrix} 2\pi f'(0)\,\mathrm{Re}\,A \\ 4\pi f''(0)\,\mathrm{Re}\,A - 4\pi^2 f'(0)\,\mathrm{Im}\,A \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Consequently at least one of the first two z-derivatives at 0, with the same choice of derivative for every $q \geq 1$, has no meromorphic base germ through $t = 0$ after $b = \eta - t^q$. The corresponding derivative of the reflected inner branch differs from that of K^W_b by at least $c\Lambda(b)$ in modulus for some $c > 0$ and all sufficiently close real $b < \eta$.

Proof. The complex limit of Corollary 18 and Cauchy's formula give convergence of the first two derivatives. The derivatives of R_b at 0 are real for real b, giving the displayed vector. Since $f'(0) \neq 0$ by Corollary 34, vanishing of its first entry forces $\mathrm{Re}\,A = 0$, and then vanishing of its second forces $\mathrm{Im}\,A = 0$, a contradiction. For a nonzero entry, the reflection argument in the proof of Proposition 21 applies to that derivative. Reflection doubles its imaginary part, which also gives the lower bound. □

Remark 23 (Conjugate bypasses of the cusp)**.** Suppose the real-normalised Kneser germ admits analytic continuation along two conjugate parameter paths bypassing η and reaching a common real interval below η. If the continuation along one path agrees there with K^W, uniqueness of analytic continuation and Schwarz reflection make the continuation along the reflected path $\overline{K^W_b(\bar{z})}$. At the fixed real height x_* of Corollary 19, the two values differ in modulus by at least $2c\Lambda(b)$. The existence of such continuations is not asserted here.

Two remarks place Proposition 15 relative to K_b. For non-real b with $\mathrm{Im}\,b > 0$ the same sewing (along the curve $T_b(\mathbb{R})$) yields a solution tending to L and L_2 as $\mathrm{Im}\,z \to \pm\infty$. The geometric sewn family depends holomorphically on b near every real point of $(1, \eta)$ (Proposition 16). These two limits alone do not characterise the solution (Remark 43); identifying it with the base-continued K_b requires the seam condition of Theorem 38. The numerical comparison in Table 4 shows that τ_1 and t_2/t_1^2 reproduce the ladder values of $\hat{c}_1$ and κ_2 — moduli *and* arguments — to the precision of the ladder extrapolation, e.g. eight digits at $b = 1.10$.

Table 4: Kneser ladders (Sections 4, 6.2, extrapolated to $\operatorname{Im} b = 0$) against the Kneser-free invariants $\tau_1 = t_1 \mathrm{e}^{-2\pi \mathrm{i} t_0}$ and t_2/t_1^2 (script `transition_map.py`).

b	$\hat{c}_1$ (ladder)	τ_1	κ_2 (ladder)	t_2/t_1^2
1.05	$0.0850235\,\mathrm{e}^{1.534136\mathrm{i}}$	$0.0850226\,\mathrm{e}^{1.534165\mathrm{i}}$	$1.374049\,\mathrm{e}^{1.413420\mathrm{i}}$	$1.374336\,\mathrm{e}^{1.413645\mathrm{i}}$
1.10	$0.08642720\,\mathrm{e}^{1.55109214\mathrm{i}}$	$0.08642720\,\mathrm{e}^{1.55109214\mathrm{i}}$	$1.440851\,\mathrm{e}^{1.477690\mathrm{i}}$	$1.440851\,\mathrm{e}^{1.477690\mathrm{i}}$
1.15	$0.0872053\,\mathrm{e}^{1.561012\mathrm{i}}$	$0.0872052\,\mathrm{e}^{1.561013\mathrm{i}}$	$1.479111\,\mathrm{e}^{1.511255\mathrm{i}}$	$1.479107\,\mathrm{e}^{1.511258\mathrm{i}}$
1.30	0.088434	$0.0884321\,\mathrm{e}^{1.577580\mathrm{i}}$	$1.541516\,\mathrm{e}^{1.561918\mathrm{i}}$	$1.541515\,\mathrm{e}^{1.561919\mathrm{i}}$

7.4 Reduction of the conjectures to confluence

Write $a := \Phi_{\mathrm{att}}(1/\mathrm{e} - 1) = 3.0292972144180360989\ldots$ for the α-normalised attracting Fatou coordinate of the base point $w = 1$ at $b = \eta$ (real, since the orbit of $u = 1/\mathrm{e} - 1$ stays on the negative axis).

Theorem 24. *Assume the following confluence statement* (C)*: there are constants $s(b) \in \mathbb{C}$ such that, as $b \to \eta^-$,*

$$R_b^{-1}(\mathrm{e}(1+u)) \to \Phi_{\mathrm{att}}(u) - a, \qquad S_b^{-1}(\mathrm{e}(1+u)) + s(b) \to \Phi_{\mathrm{rep}}(u),$$

uniformly on compact subsets of the upper gate, with the branches continued as in Section 6.1. Then for every $n \geq 1$

$$\tau_n(b) \longrightarrow B_n\,\mathrm{e}^{2\pi \mathrm{i} n a}.$$

Consequently, also $t_n/t_1^n \to B_n/B_1^n$ for every $n \geq 1$. Together with Corollary 14, and subject to the identification of K_b discussed in Sections 7.7–7.9, this gives $\hat{c}_n \to B_n \mathrm{e}^{2\pi \mathrm{i} n a}$ for all n. The limiting phase $\arg \hat{c}_1 \to \arg B_1 + 2\pi a = 1.5865330757\ldots \pmod{2\pi}$ and the ratio limit of Conjecture 9 follow for the sewing solution.

Proof. Replace S_b by $S_b(\cdot - s(b))$, which leaves τ_n unchanged. By (C), $T_b \to \Phi_{\mathrm{att}} \circ \Phi_{\mathrm{rep}}^{-1} - a = h^{-1} - a$ uniformly on $[0,1] + \mathrm{i}Y_g$ for fixed Y_g in the gate, so the Fourier coefficients on that line converge: $t_n \to B_n$ ($n \geq 1$) and $t_0 \to -a$, the latter because $B_0 = 0$ in the α normalisation. Hence $\tau_n = t_n \mathrm{e}^{-2\pi \mathrm{i} n t_0} \to B_n \mathrm{e}^{2\pi \mathrm{i} n a}$. □

Statement (C) is a confluence theorem for the saddle-node unfolding $u \mapsto \mathrm{e}^{\lambda \mathrm{e}^{-\lambda} \mathrm{e}(1+u)}/\mathrm{e} - 1$ in the direction where the two fixed points are real (the "Glutsyuk" direction): the Koenigs coordinates at the two hyperbolic points converge, on the gate, to the sectorial Fatou coordinates of the parabolic germ. Results of this type are due to Glutsyuk [9] and, for the full unfolded modulus, Mardešić–Roussarie–Rousseau [18] and Christopher–Rousseau [5]. Rather than matching their normalisations, we prove (C) directly for this family in Section 7.5, so Theorem 24 holds unconditionally.

7.5 Proof of the confluence statement (C)

We prove (C) directly for the family at hand; no normalisation from the literature is needed. In u-coordinates ($w = \mathrm{e}(1+u)$, $\mu = \mathrm{e}\log b$)

$$f_b(u) = \mathrm{e}^{\mu - 1 + \mu u} - 1, \qquad f_\eta(u) = \mathrm{e}^u - 1.$$

For $b < \eta$ close to η the map has, in a fixed disc D_r, exactly the two fixed points $u_1 = \mathrm{e}^{\lambda - 1} - 1 < 0 < u_2 = \mathrm{e}^{\lambda_2 - 1} - 1$ (Rouché), and

$$f_b(u) - u = (u - u_1)(u - u_2)\, k_b(u), \qquad k_b \to k_\eta, \quad k_\eta(u) = \tfrac{1}{2} + \tfrac{u}{6} + \cdots, \tag{25}$$

uniformly on D_r, with k_b zero-free there. Put $A_j = 1/\log \lambda_j$, so $A_1 < 0 < A_2$.

Lemma 25 (Residues). *As $b \to \eta^-$: $A_1(u_1-u_2) \to 2$, $A_1+A_2 \to \nu := 1-4k'_\eta(0) = \frac{1}{3}$, hence $A_1u_1 + A_2u_2 \to 2$ and $A_ju_j^2 \to 0$.*

Proof. $\lambda_1 = 1+(u_1-u_2)k_b(u_1)$ and $\lambda_2 = 1+(u_2-u_1)k_b(u_2)$. With $d = u_1-u_2 \to 0$ and $1/\log(1+x) = 1/x + \frac{1}{2} + O(x)$, $A_1+A_2 = \frac{k_b(u_2)-k_b(u_1)}{d\,k_b(u_1)k_b(u_2)} + 1 + O(d) \to 1-4k'_\eta(0)$, and $A_1 d \to 1/k_\eta(0) = 2$. □

Numerically $A_1+A_2 = 0.3333331469$ and $A_1(u_1-u_2) = 1.99676$ at $\lambda = 0.99$. Define the *model time*

$$\Psi_b(u) = A_1\log(u-u_1) + A_2\log(u-u_2),$$

which is exactly the time of the vector field $(u-u_1)(u-u_2)/(\dots)$ whose time-one map has multipliers λ_1, λ_2, and let $\alpha_2(u) = -2/u + \frac{1}{3}\log(-u)$. By Lemma 25, for u, p in a simply connected region at positive distance from 0, with logarithms continued along the same path,

$$\Psi_b(u) - \Psi_b(p) \longrightarrow \alpha_2(u) - \alpha_2(p). \tag{26}$$

Lemma 26 (Uniform one-step error). *$F_b := \Psi_b \circ f_b - \Psi_b - 1$ is single-valued and holomorphic on a fixed disc $D_{r'}$, $F_b \to F_\eta := \alpha_2 \circ f_\eta - \alpha_2 - 1$ uniformly there, and*

$$|F_b(u)| \le M\,|u-u_1|\,|u-u_2| \qquad (u \in D_{r'/2})$$

with M independent of b.

Proof. By (25), $(f(u)-u_1)/(u-u_1) = 1+(u-u_2)k(u)$ and $(f(u)-u_2)/(u-u_2) = 1+(u-u_1)k(u)$, so

$$F = A_1\log\frac{1+(u-u_2)k}{1+(u-u_1)k} + (A_1+A_2)\log(1+(u-u_1)k) - 1,$$

holomorphic for $|u| < r'$ small. The first logarithm is $O(|u_1-u_2|)$ uniformly, and $A_1(u_1-u_2)$, A_1+A_2 are bounded (Lemma 25); so $|F| \le M_0$ on $|u| = r'$ uniformly, and the displayed formula converges. At $u = u_1$ the first term is $A_1\log\lambda_1 = 1$ and the second vanishes, so $F(u_1) = 0$; likewise $F(u_2) = 0$. The maximum principle applied to $F/((u-u_1)(u-u_2))$ on $D_{r'}$ gives the bound, and letting $b \to \eta$ gives $F_\eta = O(u^2)$. □

Lemma 27 (Uniform petals and summability). *Put $Z_b(u) = A_1\log\frac{u-u_1}{u-u_2}$ and $\Omega_{R,b} = \{u : \operatorname{Re} Z_b(u) > R\}$ (a disc of Apollonius about u_1). There is R_0 such that for $R \ge R_0$ and all b near η: $\Omega_{R,b} \subset D_{r'/2}$, $f_b(\Omega_{R,b}) \subset \Omega_{R,b}$, $\operatorname{Re} Z_b(f_b^k u) \ge \operatorname{Re} Z_b(u) + \frac{3}{4}k$, and*

$$\sum_{k\ge N} |f_b^k u - u_1|\,|f_b^k u - u_2| \le \frac{C}{R+\frac{3}{4}N} \qquad (u \in \Omega_{R,b}),$$

with C independent of b and u. As $b \to \eta$, $\Omega_{R,b}$ converges on compacta to the petal $\{\operatorname{Re}(-2/u) > R\}$.

Proof. $Z(f(u)) - Z(u) = A_1\log\frac{1+(u-u_2)k}{1+(u-u_1)k} = 1+O(|u|+|u_1|+|u_2|)$ uniformly, by the expansion in the proof of Lemma 26; on $\Omega_{R,b} \subset D_{C/R+|u_1|}$ this is within $\frac{1}{4}$ of 1 once R is large, which gives invariance and the growth of $\operatorname{Re} Z$. With $q = (u-u_1)/(u-u_2)$, $|q| = \mathrm{e}^{-\sigma}$, $\sigma = |\log\lambda_1|\operatorname{Re} Z$, one has $u-u_2 = (u_2-u_1)/(q-1)$, whence

$$|u-u_1||u-u_2| = \frac{|q|\,|u_2-u_1|^2}{|1-q|^2} \le \Big(\frac{|u_2-u_1|}{|\log\lambda_1|}\Big)^2 \frac{(1+\sigma)^2\mathrm{e}^{-\sigma}}{(\operatorname{Re} Z)^2},$$

using $1-\mathrm{e}^{-\sigma} \ge \sigma/(1+\sigma)$. The first factor is bounded (Lemma 25) and $(1+\sigma)^2\mathrm{e}^{-\sigma} \le 4$; summing $(R+\frac{3}{4}k)^{-2}$ over $k \ge N$ gives the bound. The last assertion follows from $\operatorname{Re} Z_b(u) \to \operatorname{Re}(-2/u)$, i.e. $A_1(u_2-u_1) \to -2$. □

Theorem 28 (Confluence)**.** *Statement* (C) *holds. More precisely, with branches continued along paths in the respective parabolic basins:*

1. $R_b^{-1}(\mathrm{e}(1+u)) \to \Phi_{\mathrm{att}}(u) - a$ *uniformly on compact subsets of the parabolic basin of* f_η*;*
2. *for a fixed* p' *in the gate (below), putting* $s(b) = \Phi_{\mathrm{rep}}(p') - \mathcal{S}_b(p')$*,* $\mathcal{S}_b(u) + s(b) \to \Phi_{\mathrm{rep}}(u)$ *uniformly on compact subsets of the repelling basin (backward orbits tending to* 0*). Here* $\mathcal{S}_b$ *is the real-normalised repelling time, i.e. the branch of* S_b^{-1} *that is real on the segment* (u_1, u_2) *and is continued through the band of Lemma 30, then along backward-invariant paths.*

The gate *of height* Y_* *is* $\mathcal{G}(Y_*) = \Phi_{\mathrm{rep}}^{-1}(\{Y_* \le \operatorname{Im} z \le Y_* + 1,\ -1 \le \operatorname{Re} z \le 2\})$*. For* Y_* *large it is a compact subset of both parabolic basins of* f_η*. For* b *near* η *it lies in the band* $\mathcal{B}_b(Y_* - 2)$*, because* $Z_b \to -2/u$ *uniformly on it (Lemma 25) and* $\operatorname{Im}(-2/u) = \operatorname{Im}\alpha(u) + O(1) = \operatorname{Im}\Phi_{\mathrm{rep}}(u) + O(1)$ *there.*

Proof. (1) On $\Omega_{R,b}$ put $\mathcal{A}_b(u) = \lim_n[\Psi_b(f_b^n u) - n] = \Psi_b(u) + \sum_{k\ge 0} F_b(f_b^k u)$; the series converges absolutely by Lemmas 26 and 27. Since $f^n u \to u_1$ and the Koenigs map satisfies $\sigma(f^n u) = \lambda_1^n \sigma(u)$, $\Psi_b(f^n u) - n = A_1 \log \sigma(u) + A_2 \log(u_1 - u_2) + o(1)$, because $A_1 \log \lambda_1 = 1$. Thus $\mathcal{A}_b$ is the Koenigs Abel coordinate up to an additive constant, and $R_b^{-1}(\mathrm{e}(1+u)) = \mathcal{A}_b(u) - \mathcal{A}_b(u^*)$, $u^* = 1/\mathrm{e} - 1$. Let K be a compact subset of the petal $P_R = \{\operatorname{Re}(-2/u) > R\}$ and $U \supset K \cup \{f_\eta^{N_0} u^*\}$ a simply connected region in P_R at positive distance from 0. For b near η, $U \subset \Omega_{R,b}$ and $u_1, u_2 \notin U$. For $u, p \in U$,

$$\mathcal{A}_b(u) - \mathcal{A}_b(p) = \Psi_b(u) - \Psi_b(p) + \sum_{k\ge 0} [F_b(f_b^k u) - F_b(f_b^k p)].$$

The first term converges by (26). In the sum, each term converges because $f_b^k \to f_\eta^k$ and $F_b \to F_\eta$ uniformly. The tails beyond N are $\le 2MC/(R + \frac{3}{4}N)$ uniformly in b (Lemma 27). Hence the difference converges uniformly to $\alpha_2(u) - \alpha_2(p) + \sum_k [F_\eta(f_\eta^k u) - F_\eta(f_\eta^k p)] = \Phi_{\mathrm{att}}(u) - \Phi_{\mathrm{att}}(p)$. The last identity holds because $\Phi_{\mathrm{att}} = \lim[\alpha(f^n u) - n]$ and $\alpha - \alpha_2 = O(u) \to 0$ along orbits. A general compact K of the basin satisfies $f_\eta^N(K) \Subset P_R$ for some N. Then $f_b^N(K) \Subset P_R$ for b near η, and $\mathcal{A}_b = \mathcal{A}_b \circ f_b^N - N$. Taking $p = f_b^{N_0} u^*$, which tends to $f_\eta^{N_0} u^* \in U$, and using uniform convergence on a neighbourhood gives (1). (On $\Omega_{R,b}$ the function $\mathcal{A}_b$ is multivalued, since this punctured Apollonius disc surrounds u_1. It is single-valued on U, which avoids u_1. On K, $\sigma_b \ne 0$ because $f_b^N(K)$ avoids u_1. On non-immediate components of the basin the branch is defined by $\mathcal{A}_b \circ f_b^N - N$. The Koenigs map is normalised by $\sigma_b'(u_1) = 1$.)

(2) Apply (1)'s argument to $G_b(v) = -f_b^{-1}(-v)$, the principal inverse branch reflected. Then $G_\eta(v) = -\log(1 - v) = v + \frac{v^2}{2} + \frac{v^3}{3} + \cdots$, which is of the same prepared form with $k_\eta'(0) = \frac{1}{3}$ and residue $-\frac{1}{3}$. The attracting fixed point of G_b is $-u_2$, with multiplier $1/\lambda_2$. Its Koenigs Abel coordinate is $-S_b^{-1}(\mathrm{e}(1-v))$ up to a constant, and its parabolic Fatou coordinate is $-\Phi_{\mathrm{rep}}(-v)$ up to a constant (since $\alpha_f(-v) = -\alpha_G(v) + \mathrm{const}$). The free constant is fixed by p'. □

With Theorem 28, Theorem 24 becomes unconditional. Four points need checking; each is settled here. The proof of Lemma 30 uses only Lemmas 25–27, so there is no circularity.

(a) Inversion on the gate line. Fix $Y_* \ge Y_B + 6$ and the line $\ell = [0,1] + \mathrm{i}Y_*$. By Rouché, the solution $v_b(z)$ of $\mathcal{S}_b(v) + s(b) = z$ near $\Phi_{\mathrm{rep}}^{-1}(z)$ exists and converges uniformly on ℓ. Moreover $S_b(z - s(b)) = \mathrm{e}(1 + v_b(z))$, because $\mathcal{S}_b$ is a branch of S_b^{-1}.

(b) The gate line lies in the strip of holomorphy of T*.* The points $v_b(z)$, $z \in \ell$, lie within $o(1)$ of the gate, hence in the band $\mathcal{B}_b(Y_B + 3)$. By Lemma 30(ii) their S-times $z - s(b)$ satisfy $|\operatorname{Im}(z - s(b))| \le h/2 - Y_B - 3 + \pi/3 + C/Y_B + C\varepsilon < h/2 - Y_B - 2$. So the shifted line $\ell - s(b)$ lies in the strip Σ of Lemma 30(iii), where T is holomorphic. Therefore the Fourier coefficients

of $\tilde{T} = T(\cdot - s)$ on ℓ are $t_n e^{-2\pi i n s}$ and $\tilde{t}_0 = t_0 - s$, with the t_n those of T on $\mathbb{R}$. No prior knowledge of $\operatorname{Im} s$ is used.

(c) The branch of R_b^{-1}. Continue R_b^{-1} from u^* along the negative axis to $f_\eta^{N_0} u^*$, where it is real. Then go inside $P_R \setminus D_\delta(0)$ through the upper half-plane to the gate, and then inside the band down to the segment (u_1, u_2). This path passes above u_1 and does not wind around u_2. Along it $R_b^{-1} = \Psi_b + \sum_k F_b \circ f_b^k + c_R$, so the change of $\operatorname{Im} R_b^{-1}$ is $A_1 \Delta \arg(u - u_1) + A_2 \Delta \arg(u - u_2) + O(C/Y + \varepsilon) = h/2 + 0 + \text{small}$. Lemma 12 forces the exact value, so on the segment $\operatorname{Im} R_b^{-1} = +h/2$. The branch in Theorem 28(1) is therefore the one with $\operatorname{Im} t_0 = h/2$ used in (22).

(d) $B_0 = 0$. At $b = \eta$, on the gate, $\Phi_{\text{att}} - \alpha_2 = \sum_{k \geq 0} F_\eta(f^k u)$ and $\Phi_{\text{rep}} - \alpha_2 = -\sum_{k \geq 1} F_\eta(f^{-k} u)$, with the same principal $\log(-u)$ in α_2. That logarithm stays continuous along both orbits, since $-f^{\pm k} u$ remains in the lower half-plane. By Lemma 26 and $f^{\pm k} u \approx -2/(x \pm k + iY)$, both sums are $O(1/Y)$. Hence $\Phi_{\text{att}} \circ \Phi_{\text{rep}}^{-1}(z) - z \to 0$ as $\operatorname{Im} z \to +\infty$, so its constant Fourier mode vanishes: $B_0 = 0$. Numerically $|B_0| < 10^{-59}$.

By (a)–(d) and Theorem 28, $\tilde{T} \to h^{-1} - a$ uniformly on ℓ, where h^{-1} denotes the inverse horn map (not the period h). So $\tilde{t}_n \to B_n$ and $\tilde{t}_0 \to -a$, and $\tau_n = \tilde{t}_n e^{-2\pi i n \tilde{t}_0}$ gives:

Corollary 29. *For every $n \geq 1$, $\tau_n(b) \to B_n e^{2\pi i n a}$ as $b \to \eta^-$. Since $B_1 \neq 0$, also $t_n / t_1^n \to B_n / B_1^n$ for every $n \geq 1$.*

A direct pointwise check of Theorem 28 on two gate points (`confluence_check.py`) gives errors 0.086, 0.021, 0.0050, 0.00079 (attracting side) at $\lambda = 0.8, 0.9, 0.95, 0.98$, decreasing like ε^2.

7.6 The Kneser side I: proof of the decay estimate (W2)

Throughout, $b < \eta$ is real and close to η, $\varepsilon = |\log \lambda|$, and S has the real normalisation of Section 7, so $T(z) - z = \sum t_n e^{2\pi i n z}$ with $\operatorname{Im} t_0 = h/2$ and $t_{-n} = \overline{t_n}$ (Lemma 12).

Lemma 30 (The band)**.** *Let $Z_b(u) = A_1 \log \frac{u - u_1}{u - u_2}$ with the branch $\operatorname{Im} Z_b \in (0, h)$ on $\mathbb{C} \setminus ((-\infty, u_1] \cup [u_2, \infty))$. This branch has $\operatorname{Im} Z_b = h/2$ on (u_1, u_2) and $\operatorname{Im} Z_b < h/2$ on $\operatorname{Im} u > 0$. For $Y > 0$ let*

$$\mathcal{B}_b(Y) = \{u : \ Y < \operatorname{Im} Z_b(u) < h - Y\},$$

a lens bounded by two circular arcs through u_1, u_2. There is an absolute Y_B such that for $Y \geq Y_B$ and all b near η:

(i) every $u \in \mathcal{B}_b(Y)$ has its forward orbit in $D_{r'}$ tending to u_1 and its principal backward orbit in $D_{r'}$ tending to u_2, with $\sum_{k \in \mathbb{Z}} |F_b(f_b^k u)| \leq C/Y + C\varepsilon$;

(ii) the real-normalised times $\mathcal{R}_b = \Psi_b + \sum_{k \geq 0} F_b \circ f_b^k + c_R$ and $\mathcal{S}_b = \Psi_b - \sum_{k \geq 1} F_b \circ f_b^{-k} + c_S$ are holomorphic and injective on $\mathcal{B}_b(Y)$. They are branches of R_b^{-1} and S_b^{-1}, with $\mathcal{S}_b$ real on (u_1, u_2) and $\mathcal{R}_b$ as in item (c) above, and

$$\left|\operatorname{Im} \mathcal{S}_b(u) - (\operatorname{Im} Z_b(u) - h/2)\right| \leq \tfrac{\pi}{3} + C/Y + C\varepsilon;$$

(iii) S_b maps the strip $\Sigma = \{|\operatorname{Im} w'| < h/2 - Y - 2\}$ into $\mathcal{B}_b(Y)$ with $\mathcal{S}_b \circ S_b = \mathrm{id}$. Consequently $T = \mathcal{R}_b \circ S_b$ is holomorphic on Σ, and

$$T(w') - w' = (c_R - c_S) + \sum_{k \in \mathbb{Z}} F_b(f_b^k(S_b(w'))).$$

Proof. (i) Put $q = e^{Z/A_1} = (u - u_1)/(u - u_2)$ and $\zeta = Z/A_1 = -\varepsilon Z$, so $\operatorname{Im} \zeta \in (-2\pi + \varepsilon Y, -\varepsilon Y)$ on the band. On $-2\pi \leq \operatorname{Im} \zeta \leq 0$ one has $|e^\zeta - 1| \geq c \min(1, \delta(\zeta)) \max(1, |e^\zeta|)$ with $\delta(\zeta) =$

$\min(|\zeta|, |\zeta + 2\pi\mathrm{i}|)$ and an absolute $c > 0$; the only zeros are 0 and $-2\pi\mathrm{i}$. On the band $\delta(\zeta) = \varepsilon \min(|Z|, |Z - \mathrm{i}h|) > \varepsilon Y$. Since $u - u_2 = (u_2 - u_1)/(q-1)$ and $|u_2 - u_1|/\varepsilon \to 2$ (Lemma 25), this gives

$$|u-u_2| \le C(\varepsilon + Y^{-1}), \qquad |u-u_1||u-u_2| = \frac{|q|\,|u_2-u_1|^2}{|q-1|^2} \le \begin{cases} C\,\min(|Z|, |Z-\mathrm{i}h|)^{-2}, & \delta(\zeta) \le 1, \\ C\varepsilon^2 \mathrm{e}^{-\varepsilon|\operatorname{Re} Z|}, & \delta(\zeta) > 1, \end{cases} \tag{27}$$

using $|q|/\max(1,|q|)^2 = \mathrm{e}^{-|\operatorname{Re}\zeta|}$. So $\mathcal{B}_b(Y) \subset D_{C/Y+|u_1|+|u_2|} \subset D_{r'/2}$ for Y large. By the proof of Lemma 27, $Z \circ f - Z = 1 + O(|u| + \varepsilon)$, so $\operatorname{Re} Z$ increases by at least $\frac{3}{4}$ per step while the orbit is in $D_{r'/2}$.

The imaginary part is controlled through $\Psi_b = Z_b + (A_1 + A_2)\log(u - u_2)$, with step error $|F_b| \le M|u-u_1||u-u_2|$ (Lemma 26). Summing (27) along $\operatorname{Re} Z_k \ge \operatorname{Re} Z_0 + \frac{3}{4}k$ bounds $\sum_k |F_b(u_k)|$ by $C'/Y + C'\varepsilon$. Points with $\delta(\zeta_k) \le 1$ contribute $\sum_k (x_k^2 + Y^2)^{-1} \le C/Y$, and the others $\varepsilon^2 \sum \mathrm{e}^{-\varepsilon|x_k|} \le C\varepsilon$. Now $\operatorname{Im} Z - \operatorname{Im}\Psi = -(A_1+A_2)\arg(u-u_2)$ with $\arg(u-u_2) \in (0, 2\pi)$ on the lens. So $\operatorname{Im} Z$ drifts by at most $C'/Y + C'\varepsilon + \frac{2\pi}{3} + o(1)$ along the orbit, which is $< Y/2$ for $Y \ge Y_B \ge 8$. By induction the orbit stays in $\mathcal{B}_b(Y/2) \subset D_{r'/2}$, and $\operatorname{Re} Z_k \to +\infty$, i.e. $u_k \to u_1$. The continued values of Z along the orbit have $\operatorname{Im} \in (Y/2, h - Y/2) \subset (0, h)$, so they agree with the fixed slit-plane branch. The backward orbit is the forward orbit of the reflected inverse family $G_b(v) = -f_b^{-1}(-v)$, to which the same argument applies (its prepared form is checked in the proof of Theorem 28(2)). Its period is $h_2 = h + 2\pi/3 + o(1)$, so $\mathcal{B}_b(Y)$ is contained in the corresponding band of G_b at height $Y(1 + O(\varepsilon)) - 1$. This is harmless for $Y \ge Y_B$.

(ii) As in the proof of Theorem 28(1), $\Psi_b(f^n u) - n \to A_1 \log \sigma_b(u) + \text{const}$ along forward orbits. So $\mathcal{R}_b$ is a branch of the attracting Abel coordinate, and $\mathcal{S}_b$ likewise of the repelling one. Choose the constants c_R (by item (c) above) and c_S so that $\mathcal{S}_b$ is real on the segment. Both F_b and f_b are real, so the sums are real there. On the segment $\operatorname{Im} Z = h/2$ and $\arg(u - u_2) = \pi$, which fixes $\operatorname{Im} c_S$. The displayed estimate then follows from $\operatorname{Im}\Psi = \operatorname{Im} Z + (A_1 + A_2)\arg(u - u_2)$, $A_1 + A_2 \to \frac{1}{3}$ and (i). Injectivity: if $\mathcal{S}_b(u) = \mathcal{S}_b(u')$ then the principal backward orbits satisfy $\tau_b(u_{-n}) = \tau_b(u'_{-n})$ near u_2, where τ_b is injective, so $u_{-n} = u'_{-n}$. Moreover f_b is injective on $D_{r'}$ (its non-injectivity has period $2\pi\mathrm{i}/\log b$ in w, far larger than the disc), so $u = u'$. The same argument with σ_b and forward orbits proves injectivity of $\mathcal{R}_b$.

(iii) Fix $x \in \mathbb{R}$. $S_b(x) \in (u_1, u_2) \subset \mathcal{B}_b(Y)$ and $\mathcal{S}_b(S_b(x)) = x$, since both sides are the real-normalised repelling time. Move y' from 0 towards $\pm(h/2 - Y - 2)$. As long as $S_b(x + \mathrm{i}y')$ stays in the lens, continuation gives $\mathcal{S}_b(S_b(x + \mathrm{i}y')) = x + \mathrm{i}y'$. It cannot leave through the corners u_1, u_2, where $|\operatorname{Re}\mathcal{S}_b| \to \infty$. It cannot leave through the arcs either: there $\operatorname{Im} Z \in \{Y, h - Y\}$, so by (ii) $|\operatorname{Im}\mathcal{S}_b| \ge h/2 - Y - \pi/3 - o(1) > |y'|$. Hence $S_b(\Sigma) \subset \mathcal{B}_b(Y)$, $T = \mathcal{R}_b \circ S_b$ on Σ, and $T(w') - w' = \mathcal{R}_b(u) - \mathcal{S}_b(u)$ with $u = S_b(w')$ is the displayed bi-infinite sum. □

Lemma 31 (Uniform mode bound)**.** *Put $Y_0 := Y_B + 3$. There is M, which can be taken ≤ 1, and $b_1 < \eta$ such that for $b \in (b_1, \eta)$ and all $n \ne 0$,*

$$|t_n| \le M\mathrm{e}^{-2\pi|n|d}, \qquad d := \tfrac{h}{2} - Y_0 - 1.$$

Proof. Take $Y = Y_B + 1$ in Lemma 30(iii), so that the estimate (ii) is available on the closure of the lens. Σ has half-width $h/2 - Y_B - 3 = d + 1$, so the lines $\operatorname{Im} w' = \pm d$ lie in Σ. On them, by (iii) and (i), $T - \mathrm{id} - t_0 = \sum_k F_b(f_b^k S_b)$ − mean has modulus at most $2(C/Y_B + C\varepsilon)$. Let M be the supremum of this over $b \in (b_1, \eta)$. Cauchy's estimate on these lines gives the bound for $n > 0$ and $n < 0$. This does not use Theorem 28. □

For $H > 0$ let $\mathfrak{A}_H$ be the space of 1-periodic $g = \sum g_n \mathrm{e}^{2\pi\mathrm{i}nz}$ with $\|g\|_H = \sum |g_n| \mathrm{e}^{2\pi|n|H} < \infty$. It is a Banach algebra (the weight is submultiplicative), its elements are holomorphic and bounded by $\|g\|_H$ on $|\operatorname{Im} z| \le H$, and the projections $\Pi_{<0}, \Pi_{\ge 0}$ onto negative and non-negative modes have norm 1. Also $\|\mathrm{e}^g\|_H \le \mathrm{e}^{\|g\|_H}$ and $\|\mathrm{e}^{g_1} - \mathrm{e}^{g_2}\|_H \le \|g_1 - g_2\|_H \, \mathrm{e}^{\max \|g_j\|_H}$.

Theorem 32 (W2). *Let M, d be as in Lemma 31, $\rho = \frac{1}{4}$, and choose $D \geq 1$ with $4\pi M \mathrm{e}^{-2\pi D} \leq \rho(1-\mathrm{e}^{-2\pi D})^2$. Put $H = d - D - \rho$ and $\tau = T - \mathrm{id} - t_0$. Assume $H > 1$, which holds for b near η since $d \sim \pi/\varepsilon$, and that b is so close to η that $C_1 \Lambda \mathrm{e}^{4\pi(Y_0+D+1)} < \frac{1}{2}$, where $C_1 = 4M + 2\rho$. Then:*

1. *The map $\mathcal{T}(Q) = -\Pi_{<0}[\tau \circ (\mathrm{id} + Q)]$ is a contraction, with constant $\leq \frac{1}{2}$, of the ball $\{Q \in \Pi_{<0}\mathfrak{A}_H : \|Q\|_H \leq \rho\}$ into itself. Its fixed point Q^* and $P' := \Pi_{\geq 0}[\tau \circ (\mathrm{id} + Q^*)]$ satisfy*
$$T \circ (\mathrm{id} + Q^*) = \mathrm{id} + t_0 + P' \qquad \text{on } |\operatorname{Im} z| < H.$$

2. *$\varphi_- = \mathrm{id} + Q^*$ is univalent on $\{\operatorname{Im} z < H - 1\}$ and $\varphi_+ = \mathrm{id} + t_0 + P'$ on $\{\operatorname{Im} z > -(H-1)\}$. The function equal to $R \circ (\varphi_+ - \mathrm{i}h)$ above and $S \circ \varphi_-$ below is a solution of (1); it is the sewing solution K_b^W of Proposition 15.*

3. *The equation $\varphi_+(z_0) = \mathrm{i}h$ has a unique solution with $|z_0 - (\mathrm{i}h - t_0)| < 1$, and $|z_0 - (\mathrm{i}h - t_0)| \leq C_1 \Lambda \mathrm{e}^{4\pi(Y_0+D+1)}$. After the translation $z \mapsto z + z_0$ one has $K_b^W(0) = 1$ and $P = R^{-1} \circ K_b^W - \mathrm{id}$ with $P(0) = 0$.*

4. *For every $n \geq 1$ the Fourier coefficients of this P satisfy*
$$\left| \frac{c_n}{\Lambda^n} - \tau_n(b) \right| \leq C_n \Lambda, \qquad C_n = C\, M(M+1)\, \mathrm{e}^{2\pi(n+2)(Y_0+D+2)},$$
with an absolute constant C.

Proof. (1) Write $\tau \circ (\mathrm{id} + Q) = \sum_{m \neq 0} t_m \mathrm{e}^{2\pi \mathrm{i} m z} \mathrm{e}^{2\pi \mathrm{i} m Q}$. Then
$$\|\tau \circ (\mathrm{id} + Q)\|_H \leq \sum_{m \neq 0} |t_m| \mathrm{e}^{2\pi|m|(H+\rho)} \leq 2M \sum_{m \geq 1} \mathrm{e}^{-2\pi m D} \leq \rho$$
by Lemma 31 and the choice of D. Similarly
$$\|\tau \circ (\mathrm{id} + Q_1) - \tau \circ (\mathrm{id} + Q_2)\|_H \leq \sum_{m \neq 0} |t_m| \mathrm{e}^{2\pi|m|(H+\rho)} 2\pi|m|\, \|Q_1 - Q_2\|_H \leq \frac{4\pi M \mathrm{e}^{-2\pi D}}{(1-\mathrm{e}^{-2\pi D})^2} \|Q_1 - Q_2\|_H,$$
which is at most $\frac{1}{2}\|Q_1 - Q_2\|_H$. Since $|\operatorname{Im}(z + Q)| \leq H + \rho < d$ on the strip, the Fourier series of T converges at $z + Q(z)$ and equals T there (Lemma 30). The identity follows by splitting $\tau \circ (\mathrm{id} + Q^*) = P' + \Pi_{<0}[\ldots] = P' - Q^*$.

(2) The series of Q^* has only negative modes and converges on $\operatorname{Im} z \leq H$, so it converges on the whole lower half-plane $\operatorname{Im} z < H$. There $|Q^{*\prime}| \leq 2\pi \mathrm{e}^{-2\pi}\rho/(1-\mathrm{e}^{-2\pi})^2 < 1$ on $\operatorname{Im} z \leq H - 1$, so $\operatorname{Re} \varphi_-' > 0$ and φ_- is univalent (Noshiro–Warschawski). The same holds for φ_+ above. On the strip, $R \circ (\varphi_+ - \mathrm{i}h) = R \circ (T \circ \varphi_- - \mathrm{i}h) = R \circ T \circ \varphi_- = S \circ \varphi_-$, because $\mathrm{i}h$ is a period of R and $R \circ T = S$. The functional equation holds because $\varphi_\pm$ commute with $z \mapsto z + 1$. The resulting function realises the sewing of Proposition 15: the images of $\varphi_\pm$ are the two half-cylinders and the seam in the z-plane is $\Gamma = \varphi_-^{-1}(\mathbb{R})$, a curve in $|\operatorname{Im} z| \leq \rho$. Since $\varphi_\pm - \mathrm{id}$ is bounded, the argument principle shows that the restrictions of $\varphi_\pm$ to the two sides of Γ map onto the two half-cylinders. By uniqueness of the uniformisation it is K_b^W up to translation. (The identity $R \circ T = S$ is used on the band, where orbits stay in $D_{r'}$ and $\mathcal{R}_b$ inverts R; Lemma 30.)

(3) For $\operatorname{Im} z \geq h/2 - 1$ we have $|P'(z) - p_0'| \leq \sum_{n \geq 1} |p_n'| \mathrm{e}^{-2\pi n(h/2-1)}$ and $|p_n'| \leq \rho\, \mathrm{e}^{-2\pi n H}$. Moreover $p_0' = \sum_{m \geq 1} t_m [\mathrm{e}^{2\pi \mathrm{i} m Q^*}]_{-m}$, and $|[\mathrm{e}^{2\pi \mathrm{i} m Q^*}]_{-m}| \leq \mathrm{e}^{2\pi m \rho} \mathrm{e}^{-2\pi m H}$, so $|p_0'| \leq 2M \mathrm{e}^{-2\pi(d+H-\rho)} \leq 2M\Lambda \mathrm{e}^{2\pi(2Y_0+D+2+2\rho)}$, and $\sum_{n \geq 1} |p_n'| \mathrm{e}^{-2\pi n(h/2-1)} \leq 2\rho\Lambda \mathrm{e}^{2\pi(Y_0+D+2+\rho)}$. Both are $\leq C_1 \Lambda \mathrm{e}^{4\pi(Y_0+D+1)}/2$. Since $\operatorname{Im}(\mathrm{i}h - t_0) = h/2$ and $|P''| \leq \rho$ there, Rouché's theorem on the disc of radius 1 gives the

unique root and the bound, using the smallness hypothesis. Then $K_b^W(z_0) = R(\varphi_+(z_0) - \mathrm{i}h) = R(0) = 1$, and after translation $P(z) = \varphi_+(z + z_0) - \mathrm{i}h - z$, so $P(0) = 0$.

(4) For $n \geq 1$ the n-th mode of P is $c_n = p'_n \mathrm{e}^{2\pi \mathrm{i} n z_0}$. Because Q^* has no constant or positive modes, $[\mathrm{e}^{2\pi \mathrm{i} m Q^*}]_0 = 1$ and $[\mathrm{e}^{2\pi \mathrm{i} m Q^*}]_k = 0$ for $k > 0$. Hence $p'_n = t_n + \sum_{m>n} t_m [\mathrm{e}^{2\pi \mathrm{i} m Q^*}]_{n-m}$, and

$$|p'_n - t_n| \leq \sum_{m>n} M \mathrm{e}^{-2\pi m d} \mathrm{e}^{2\pi m \rho} \mathrm{e}^{-2\pi(m-n)H} \leq 2M \mathrm{e}^{-2\pi n(d-\rho)} \mathrm{e}^{-2\pi(d+H-\rho)}.$$

With $z_0 = \mathrm{i}h - t_0 + \zeta$, $|\zeta| \leq C_1 \Lambda \mathrm{e}^{4\pi(Y_0+D+1)}$ by (3), we have $\mathrm{e}^{2\pi \mathrm{i} n z_0} = \Lambda^n \mathrm{e}^{-2\pi \mathrm{i} n t_0} \mathrm{e}^{2\pi \mathrm{i} n \zeta}$ and $|t_n \mathrm{e}^{-2\pi \mathrm{i} n t_0}| = |\tau_n|$, so $|t_n| = |\tau_n| \mathrm{e}^{-\pi n h}$. Dividing,

$$\frac{c_n}{\Lambda^n} - \tau_n = \tau_n (\mathrm{e}^{2\pi \mathrm{i} n \zeta} - 1) + (p'_n - t_n) \mathrm{e}^{-2\pi \mathrm{i} n t_0} \mathrm{e}^{2\pi \mathrm{i} n \zeta}.$$

In the first term $|\tau_n| = |t_n| \mathrm{e}^{\pi n h} \leq M \mathrm{e}^{2\pi n (Y_0+1)}$ and $|\mathrm{e}^{2\pi \mathrm{i} n \zeta} - 1| \leq 4\pi n |\zeta|$ when $2\pi n |\zeta| \leq 1$; in the second, $|\mathrm{e}^{2\pi \mathrm{i} n \zeta}| \leq \mathrm{e}$ in that case. If instead $2\pi n |\zeta| > 1$, then $\Lambda \geq (2\pi n C_1)^{-1} \mathrm{e}^{-4\pi(Y_0+D+1)}$, and the trivial bound $|c_n| \Lambda^{-n} + |\tau_n| \leq C' M \mathrm{e}^{2\pi n (Y_0+D+2)}$ is already $\leq C_n \Lambda$ (this uses $D \geq 1$). In the second term, $|(p'_n - t_n) \mathrm{e}^{-2\pi \mathrm{i} n t_0}| \leq 2M \mathrm{e}^{2\pi n (Y_0+1+\rho)} \Lambda \mathrm{e}^{2\pi(2Y_0+2+D+2\rho)}$, by $d = h/2 - Y_0 - 1$ and $H = d - D - \rho$. Adding the two gives the stated bound, since $n(Y_0+1+\rho) + 2Y_0 + 2D + 2.5 \leq (n+2)(Y_0 + D + 2)$. □

Corollary 33. *For the sewing solution, $\hat{c}_n(K_b^W) \to B_n \mathrm{e}^{2\pi \mathrm{i} n a}$ for every $n \geq 1$ as $b \to \eta^-$. Moreover, $\kappa_n(K_b^W) \to B_n / B_1^n$ and $\lim \arg \hat{c}_1 = \arg B_1 + 2\pi a$.*

Proof. Theorem 32(4) and Corollary 29: C_n does not depend on b and $\Lambda \to 0$. Proposition 8 makes c_1 nonzero for b near η, justifying division and the limiting phase. □

Corollary 34 (Local parabolic limit of the sewn solution)**.** *Put $u^* = 1/\mathrm{e} - 1$ and $a = \Phi_{\mathrm{att}}(u^*)$. On some fixed disk about $z = 0$, the function*

$$F_\eta^{\mathrm{att}}(z) = \mathrm{e}(1 + \Phi_{\mathrm{att}}^{-1}(z + a))$$

is holomorphic, satisfies $F_\eta^{\mathrm{att}}(0) = 1$, and has $(F_\eta^{\mathrm{att}})'(0) \neq 0$. As $b \to \eta^-$ through real bases, $K_b^W \to F_\eta^{\mathrm{att}}$ locally uniformly on that disk, with all z-derivatives. Consequently there are $r > 0$ and $b_2 < \eta$ such that, for $b_2 < b < \eta$, the only preimage of 1 under K_b^W in $B(0, r)$ is 0; both K_b^W and $1/K_b^W$ are uniformly bounded there.

Proof. Theorem 28(1) gives locally uniform convergence of $A_b(u) := R_b^{-1}(\mathrm{e}(1+u))$ to $A_\eta(u) := \Phi_{\mathrm{att}}(u) - a$ near u^*, with $A_b(u^*) = 0$. The derivative $A'_\eta(u^*)$ is nonzero: the series in the proof of Theorem 28, Lemmas 26 and 27, and Cauchy's estimate give $\Phi'_{\mathrm{att}}(u) = 2/u^2 + O(1/u)$ on a narrower attracting petal. The Abel identity and $f'_\eta(u) \neq 0$ propagate nonvanishing back along the orbit of u^*. Uniform convergence of these local inverses therefore gives $R_b \to F_\eta^{\mathrm{att}}$ on a fixed disk about 0.

In Theorem 32, write $P_b(z) = \zeta_b + P'_b(z + z_0(b))$, where $\zeta_b = z_0(b) + t_0(b) - \mathrm{i}h(b) \to 0$. The proof of that theorem gives $p'_0(b) \to 0$ and $|p'_n(b)| \leq \rho \mathrm{e}^{-2\pi n H(b)}$ for $n \geq 1$, while $H(b) \to \infty$ and $\operatorname{Im} z_0(b) = h(b)/2 + o(1)$. Hence $P_b \to 0$ uniformly on every fixed bounded z-disk on which the upper representation is defined. That representation contains a fixed disk about 0 for all sufficiently large b, and $K_b^W(z) = R_b(z + P_b(z))$ there. This proves the asserted convergence; Cauchy's formula gives convergence of derivatives. Choose a small closed disk on which F_η^{att} is nonzero and $F_\eta^{\mathrm{att}} - 1$ has only its simple zero at 0. Rouché's theorem and $K_b^W(0) = 1$ give the final assertions. □

The construction is directly computable (`weld_solve.py`). At $b = 1.10$ the iteration contracts by $\sim 10^{-4}$ per step and gives $\hat{c}_1 = 0.0864272029\,\mathrm{e}^{1.5510921403\mathrm{i}}$, against $\tau_1 = 0.0864272022\,\mathrm{e}^{1.5510921408\mathrm{i}}$.

The relative difference is $8.8\cdot10^{-9}=0.38\Lambda$, and at $b=1.30$ it is $3.9\cdot10^{-19}=0.40\Lambda$. At these bases h is too small for the hypotheses of Theorem 32, so the agreement is not an instance of the theorem. It does match the first-order expansion of the correction, $c_1/(\Lambda\tau_1)-1\approx\Lambda\big[-4\pi^2|\tau_1|^2-2\pi\mathrm{i}\tau_1-4\pi\mathrm{i}\tau_2\bar\tau_1/\tau_1\big]$, whose modulus is 0.3834Λ and 0.3984Λ respectively (derived independently by a referee).

7.7 The Kneser side II: identification (W1)

It remains to identify K_b^W with the boundary value K_b of (3). We prove the local part and isolate the one global input.

Theorem 35 (W1, local)**.** *Let $b_0\in(b_1,\eta)$. There is a disc $\Delta\ni b_0$ in the base plane such that:*

1. *For $b'\in\Delta$, with $L(b'),L_2(b')$ the continued fixed points, $R_{b'},S_{b'}$ the regular superfunctions, $\varpi(b')=2\pi\mathrm{i}/\log\lambda(b')$ the period of $R_{b'}$, and $\hat T_{b'}=R_{b'}^{-1}\circ S_{b'}+\varpi/2$, the contraction of Theorem 32 works with ρ and D unchanged, $2M$ in place of M, and $H-\frac12$ in place of H. The solution is $K_{b'}^W=R_{b'}\circ(\varphi_++\varpi)$ above, where $\varphi_+=T_{b'}\circ\varphi_-$, and $S_{b'}\circ\varphi_-$ below. It is normalised by the root of $\varphi_+(z_0)=-\varpi$ in the disc of radius 1 about $-\varpi-t_0$. (For real b', $\varpi=-\mathrm{i}h$.) Its fixed point, the resulting solution $K_{b'}^W$, and the normalising root $z_0(b')$ are holomorphic in b'.*

2. *For $b'\in\Delta$ with $\operatorname{Im}b'>0$ one has $\arg\lambda>0$, $\arg\lambda_2<0$, and $K_{b'}^W(z)\to L(b')$ as $\operatorname{Im}z\to+\infty$, $K_{b'}^W(z)\to L_2(b')$ as $\operatorname{Im}z\to-\infty$, uniformly for $\operatorname{Re}z$ in compacts. In the upper and lower half-planes (of the translated coordinate) it has the representations $R_{b'}\circ(\mathrm{id}+\theta_{\mathrm{up}})$ and $S_{b'}\circ(\mathrm{id}+\theta_{\mathrm{dn}})$. These are the defining representations of the complex-base construction in `_cbuild.py`.*

3. *(Local uniqueness.) If a solution F with $F(0)=1$ admits such representations and, after the translation that puts its seam representation in the form $T\circ(\mathrm{id}+Q)=\mathrm{id}+t_0+P'$, has $\|Q\|_H\le\rho$ and its normalising root in that disc, then $F=K_{b'}^W$. (Without the root condition the statement is false: the translates of $K_{b'}^W$ by the other roots c_j, $j\ge1$, have the same Q.)*

Proof. (1) By Lemma 30(iii), T is holomorphic on the strip of half-width $d+1$, so there is slack 1 in d; we use half of it. The Koenigs maps at $L(b')$, $L_2(b')$ and the principal logarithms defining R^{-1} depend holomorphically on b'. To choose one parameter disc for the unbounded strip, first work on the compact fundamental rectangle $0\le\operatorname{Re}z\le1$, $|\operatorname{Im}z|\le d(b_0)+\frac12$. Its image under S_{b_0} is compactly contained in the common domain of the two chosen Abel charts; their local parameter continuations therefore give a single disc Δ on this rectangle. The identity $T_{b'}(z+1)=T_{b'}(z)+1$ extends the result to the whole strip. Thus $\hat T_{b'}-\mathrm{id}$ is jointly holomorphic on $\Delta\times\{|\operatorname{Im}z|<d(b_0)+\frac12\}$. Shrinking Δ again, its modes satisfy $|t_n(b')|\le2M\mathrm{e}^{-2\pi|n|(d(b_0)-1/2)}$ (Cauchy), and the estimates of Theorem 32(1) hold with $2M$ in place of M: the original choice of D gives a Lipschitz constant at most $2\rho=\frac12$ and a ball-image norm at most $\rho(1-\mathrm{e}^{-2\pi D})/\pi<\rho$. The iterates $\mathcal T^k(0)$ are holomorphic in b' and converge uniformly, so Q^* is holomorphic in b'. So are P' and, by the implicit function theorem (Rouché), z_0.

(2) From $\lambda\mathrm{e}^{-\lambda}=\log b'$, $\partial\lambda/\partial b=\mathrm{e}^\lambda/(b(1-\lambda))$ is positive for $\lambda<1$ and negative for $\lambda_2>1$. Hence $\operatorname{Im}b'>0$ gives $\operatorname{Im}\lambda>0>\operatorname{Im}\lambda_2$ for Δ small. Above, $K^W=R(z+c+o(1))$, and $R(w)=\sigma^{-1}(c'\lambda^w)\to L$ when $\operatorname{Re}(w\log\lambda)\to-\infty$. This is the case as $\operatorname{Im}w\to+\infty$, since $\operatorname{Re}(w\log\lambda)=\operatorname{Re}w\log|\lambda|-\operatorname{Im}w\arg\lambda$. Below, the same argument with $\arg\lambda_2<0$ applies. The representations are those of Theorem 32(2), transported by $z\mapsto z+z_0$.

(3) The fixed point of a contraction is unique in its ball. P' and z_0 are then determined, the latter by $F(0)=1$ together with the root condition. □

Corollary 36. *Suppose that, for all b' in a nonempty open subset of $\Delta \cap \{\operatorname{Im} b' > 0\}$, the Kneser-type solution $\kappa_{b'}$ continued from $b > \eta$ satisfies the hypothesis of Theorem 35(3), and that κ is jointly holomorphic in (b', z) off a closed singular set whose complement is connected in z. Then $\kappa_{b'} = K^W_{b'}$ on $\Delta \cap \{\operatorname{Im} b' > 0\}$. The boundary value K_b exists on $\Delta \cap \mathbb{R}$ and equals K^W_b. The upper family continues analytically across $\Delta \cap \mathbb{R}$. The conclusions of Corollary 33 then hold for K_b, including the first-mode and ratio limits of Conjectures 7 and 9.*

Proof. On the open set, Theorem 35(3) gives $\kappa = K^W$. Both families are holomorphic in (b', z) for z in a lower half-plane, where $K^W = S \circ \varphi_-$ has no cuts. So they agree on $\Delta \cap \{\operatorname{Im} b' > 0\}$ by the identity theorem, and then everywhere by continuation in z. The rest is Theorem 35(1) and Corollary 33. □

The hypothesis of Corollary 36 is an inequality on an open set of bases. It concerns only the lower correction of κ near the seam. The ladders are consistent with it: they reproduce τ_1 and t_2/t_1^2 to their precision (Table 4), and the explicit sewing solution matches the ladder value of $\hat{c}_1$ at $b = 1.10$ to eight digits. This check is numerical and not certified. A proof free of numerics would need a global uniqueness theorem for Kneser-type solutions at complex bases inside the Shell–Thron region, one that selects the no-winding solution: the class defined only by the limits at $\pm i\infty$ also contains solutions of the form $R \circ (z + 2\pi i m z/\log\lambda + \dots)$. The base continuation from $b > \eta$ passes the Shell–Thron boundary, where $|\Lambda| = 1$ and the present contraction does not apply. This is the one input we do not supply.

7.8 A characterisation of the sewing solution, and boundary values

Corollary 36 needs a quantitative input about κ at one base. Here we replace it with a qualitative characterisation. From it follows a boundary-value theorem whose only hypothesis is a uniform bound on the two θ-representations along a ladder $b + iy$, $y \downarrow 0$.

Definition 37. For real $b \in (1, \eta)$ and $Y > 0$ let $\mathcal{C}_b(Y)$ be the set of solutions F of (1) with $F(0) = 1$ such that ("solution" means holomorphic off a closed singular set and satisfying the equation where defined)

(a) $F = R \circ (\mathrm{id} + P)$ on $\{\operatorname{Im} z > -2Y\}$, with P holomorphic, 1-periodic and bounded there, and $P(0) = 0$ (the branch $R^{-1}(1) = 0$);

(b) $F = S \circ (\mathrm{id} + Q)$ on $\{\operatorname{Im} z < -Y\}$, with Q holomorphic, 1-periodic and bounded there, and $|\operatorname{Im} Q - h/2| \le 1$ on the strip $O = \{-2Y < \operatorname{Im} z < -Y\}$.

Condition (b) says that the real line of repelling time, i.e. the segment (L, L_2), sits half a period of attracting time below the real axis of F. Heuristically, some such condition is necessary. For the sewing solution the roots of $\varphi^W_+(c) \in ih\mathbb{Z}$ near the axis are $c_j = c_0 + ijh + O(\Lambda)$. The translates $K^W(\cdot + c_j)$ keep an R-representation above and an S-representation on some lower half-plane, and give $|c_1| \asymp \Lambda^{1+j}$. Condition (b) excludes $j \neq 0$, as the proof shows.

Theorem 38 (Characterisation)**.** *There are $b_1 < \eta$ and Y_B such that for $b \in (b_1, \eta)$ and $Y_0 + D + 3 < Y < (h - Y_0 - D - 3)/2$ (with Y_0, D as in Theorem 32),*

$$\mathcal{C}_b(Y) = \{K^W_b\},$$

with K^W_b normalised as in Theorem 32(3). In particular $\hat{c}_n \to B_n e^{2\pi i n a}$ for every n holds for the unique element of $\mathcal{C}_b(Y)$.

Proof. $K_b^W \in \mathcal{C}_b(Y)$. In the normalised coordinate, $\varphi_-(z) = z + z_0 + Q^*(z+z_0)$ is defined for $\operatorname{Im}(z+z_0) < H$, i.e. for $\operatorname{Im} z < -(Y_0 + 1 + D + \rho) + O(\Lambda)$, which contains $\{\operatorname{Im} z < -Y\}$. There $|Q - z_0| \le \rho$, so $|\operatorname{Im} Q - h/2| \le \rho + O(\Lambda) < 1$. The R-representation holds for $\operatorname{Im}(z+z_0) > -(H-1)$, i.e. above $-(h - Y_0 - D - 2 - \rho)$, which contains $\{\operatorname{Im} z > -2Y\}$. P and Q are bounded there.

Uniqueness. Let $F \in \mathcal{C}_b(Y)$, $\varphi_+ = \mathrm{id} + P$ and $\varphi_- = \mathrm{id} + Q$. Work with the strip $\Sigma_q = \{|\operatorname{Im} w'| < d - D\}$ of S-time. By Lemma 31 and the choice of D, on $\overline{\Sigma_q}$

$$|T - \mathrm{id} - t_0| \le \frac{2M\mathrm{e}^{-2\pi D}}{1 - \mathrm{e}^{-2\pi D}} \le \frac{\rho}{2\pi}, \qquad |T' - 1| \le \rho < 1,$$

so T is univalent on the convex set Σ_q (Noshiro–Warschawski), continuous up to the boundary, and within $\rho/2\pi$ of a translation.

(i) *The overlap.* Let $O = \{-2Y < \operatorname{Im} z < -Y\}$. For $z \in O$, $\operatorname{Im}\varphi_-(z) \in (h/2 - 2Y - 1,\, h/2 - Y + 1) \subset (-(d-D), d-D)$, because $2Y < h - Y_0 - D - 2$ and $Y > Y_0 + D + 2$. Since $\sigma \circ R(w) = \sigma(1)\lambda^w$ wherever R is defined, $R \circ \varphi_+ = S \circ \varphi_- = R \circ T \circ \varphi_-$ gives $\lambda^{\varphi_+} = \lambda^{T \circ \varphi_-}$. Hence $\varphi_+ = T \circ \varphi_- - \mathrm{i}h + k\mathrm{i}h$ on O for a constant $k \in \mathbb{Z}$. No injectivity is needed for this step.

(ii) *The lifted sewn surface.* Let $V_- = \{\operatorname{Im} w' < d - D\}$ (repelling time) and let V_+ be the region above the curve $T(\{\operatorname{Im} w' = -(d-D)\}) - \mathrm{i}h$ (attracting time). Let $\tilde{X}$ be $V_- \sqcup V_+$ modulo $w = T(w') - \mathrm{i}h$ for $w' \in \Sigma_q$. Since T is a homeomorphism of $\overline{\Sigma_q}$ onto its image, the identification graph is closed and $\tilde{X}$ is a Hausdorff Riemann surface. Translation by 1 acts freely on it, and $X = \tilde{X}/\mathbb{Z} \cong \mathbb{C}^*$, both ends being punctured discs. The function $\tilde{K}$, equal to R on V_+ and S on V_-, is well defined, because $R(T(w') - \mathrm{i}h) = S(w')$ on the band. The sewing solution is $\tilde{K} \circ \tilde{\Phi}^W$, where $\tilde{\Phi}^W$ is the lift of its uniformisation; sewing along the real line or along any other essential curve in Σ_q gives the same $\tilde{X}$.

Define $\tilde{\Phi} : \mathbb{C} \to \tilde{X}$ by $\varphi_+ - k\mathrm{i}h$ on $\{\operatorname{Im} z > -2Y\}$ and φ_- on $\{\operatorname{Im} z < -Y\}$. These agree on O by (i). They land in $V_\mp$ as follows. $\operatorname{Im} Q$ is bounded and harmonic on a half-cylinder, hence extends harmonically over the puncture. So on $\{\operatorname{Im} z \le -Y - \delta\}$ it is bounded by its values on the line $\operatorname{Im} z = -Y - \delta \subset O$. This gives $\operatorname{Im}\varphi_- \le -Y + h/2 + 1 < d - D$. Likewise, on a line in O near $\operatorname{Im} z = -2Y$, (i) gives $\operatorname{Im}(\varphi_+ - k\mathrm{i}h) \ge \operatorname{Im} z - 1 - \rho/2\pi$. The minimum principle then gives $\operatorname{Im}(\varphi_+ - k\mathrm{i}h) \ge -2Y - 1 - \rho/2\pi$ above that line. This exceeds the height $-h/2 - d + D + \rho/2\pi$ of the lower boundary of V_+, because $2Y < h - Y_0 - D - 3$.

(iii) $\tilde{\Phi}$ *is a translate of* $\tilde{\Phi}^W$. $\tilde{\Phi}$ commutes with $z \mapsto z + 1$ and descends to a holomorphic $\Phi : \mathbb{C}/\mathbb{Z} \to X \cong \mathbb{C}^*$. Φ is proper, since $\varphi_\pm - \mathrm{id}$ are bounded, and has local degree one at each end, since $P \to p_0$ and $Q \to q_0$. A proper holomorphic map of $\mathbb{C}^*$ that fixes the ends with local degree one is linear. Hence $\tilde{\Phi} = \tilde{\Phi}^W(\cdot + c)$ for some $c \in \mathbb{C}$, and $F = \tilde{K} \circ \tilde{\Phi} = K_b^W(\cdot + c)$ in the coordinate of Theorem 32(1)–(2). Univalence of $\varphi_\pm$ is a conclusion, not an assumption.

(iv) *The root.* Take a line $\ell \subset O$ with $\operatorname{Im} z = -Y''$ and Y'' close to $2Y$. There $\operatorname{Im}\varphi_- \le -2Y + h/2 + 1 + o(1) < H - \rho$, so $\varphi_-(\ell)$ lies in $\varphi_-^W(\{\mathrm{Im} < H\})$. The inclusion $\varphi_-^W(\{\mathrm{Im} < H\}) \supset \{\mathrm{Im} < H - \rho\}$ holds because $|Q^*| \le \rho$: for w in the smaller set, $\varphi_-^W - w$ has winding number 1 around 0 on the boundary of a large fundamental rectangle. By injectivity of $\tilde{\Phi}^W$, the points $z + c$ with $z \in \ell$ lie in $\{\mathrm{Im} < H\}$. There $Q(z) = c + Q^*(z+c)$, so $|\operatorname{Im} c - h/2| \le 1 + \rho$. On the upper chart, $\varphi_+(z) - k\mathrm{i}h = \varphi_+^W(z + c) - \mathrm{i}h$, and $P(0) = 0$ gives $\varphi_+^W(c) = \mathrm{i}h(1-k)$. Since $|P'| \le \rho$ where φ_+^W is defined, $|c - (\mathrm{i}h(1-k) - t_0)| \le \rho$. Comparing imaginary parts with the bound on $\operatorname{Im} c$ gives $|k|h \le 1 + 2\rho$, so $k = 0$. Then c lies in the Rouché disc of Theorem 32(3), so $c = z_0$ and $F = K_b^W$. The last sentence of the theorem is Corollary 33. □

Theorem 39 (Boundary values)**.** *Let $b \in (b_1, \eta)$ and Y be as in Theorem 38. Suppose that for every sequence $y_k \downarrow 0$ the Kneser-type solutions $\kappa_k = \kappa_{b+\mathrm{i}y_k}$ admit representations (a) and (b) with $R_{b+\mathrm{i}y_k}$, $S_{b+\mathrm{i}y_k}$ in place of R_b, S_b. Suppose also that P_k, Q_k are bounded uniformly in k, and that $|\operatorname{Im} Q_k - h(b_k)/2| \le \frac{1}{2}$ on O, with $b_k = b + \mathrm{i}y_k$. Then $\kappa_k \to K_b^W$ locally uniformly off*

the (moving) singular sets, the boundary value (3) *exists, and its separation coefficients satisfy* $|\hat{c}_n(b) - \tau_n(b)| \le C_n\Lambda$. *Consequently, if the hypothesis holds for all* b *near* η, $\hat{c}_n \to B_n e^{2\pi i n a}$ *for all* n *and Conjectures 7 and 9 hold for* K_b.

Proof. $R_{b'} \to R_b$ and $S_{b'} \to S_b$ locally uniformly (Koenigs maps depend holomorphically on the base), and so do the constants h, λ. By Montel's theorem a subsequence of (P_k, Q_k) converges locally uniformly to (P, Q), bounded and periodic, with $P(0) = 0$ and $|\operatorname{Im} Q - h/2| \le \frac{1}{2}$. The limit $F = R_b \circ (\mathrm{id} + P) = S_b \circ (\mathrm{id} + Q)$ solves (1) with $F(0) = 1$, so $F \in \mathcal{C}_b(Y)$ and $F = K_b^W$ by Theorem 38. Every subsequence has the same limit. The Fourier coefficients of P_k converge to those of P, and Theorem 32(4) applies. □

The hypothesis of Theorem 39 is the form in which the complex-base construction computes κ: `_cbuild.py` solves precisely for bounded $\theta_{\mathrm{up}}, \theta_{\mathrm{dn}}$ with these two representations. Numerically it holds along every ladder we ran. The ladder values of c_1 are of size Λ, not $O(1)$ or Λ^2. This is consistent with the height condition, but it pins the offset only in combination with the value of $|\tau_1|$. What is not proved is that the hypothesis holds for the family continued from $b > \eta$ through the Shell–Thron boundary. That is a statement about Paulsen's family, not about the separation.

7.9 Uniform bounds come for free

The hypothesis of Theorem 39 asks for bounds that are uniform along a ladder. We show that uniformity is automatic. It suffices that the continued family lie in the characterisation class on some open set of bases, with no quantitative condition.

Let $N = \bigcup_{b_0 \in (b_1, \eta)} \Delta(b_0)$ be the union of the discs of Theorem 35, and $N^+ = N \cap \{\operatorname{Im} b' > 0\}$. Both are connected: the discs are centred on an interval, and consecutive upper half-discs overlap. For complex b', Definition 37 is read with $R_{b'}, S_{b'}$, the period $\varpi(b') = 2\pi i / \log \lambda(b')$ in place of $-ih$, and $h(b') = |\operatorname{Im} \varpi(b')|$.

Theorem 40 (Uniformity)**.** *Shrinking the discs if necessary:*

1. *The sewing solutions* $K_{b'}^W$ *of the discs agree on overlaps and form one family, holomorphic in* (b', z) *on* $N \times \mathbb{C}$ *minus the singular set. Its representation functions* $P_{b'}, Q_{b'}$ *are bounded uniformly on* $\{\operatorname{Im} z > -2Y\}$ *and* $\{\operatorname{Im} z < -Y\}$ *respectively, for* b' *in any compact subset of* N*, including compact pieces of the real segment.*

2. *For every* $b' \in N$, $\mathcal{C}_{b'}(Y) = \{K_{b'}^W\}$.

3. *Let* κ *be any family of solutions, jointly holomorphic in* (b', z) *off a closed singular set, whose regular set in* z *is connected for each* b' *and which contains* $N^+ \times W$ *for some open* $W \subset \{\operatorname{Im} z < -Y\}$*, with* $\kappa_{b'} \in \mathcal{C}_{b'}(Y)$ *for all* b' *in a nonempty open subset of* N^+*. Then* $\kappa = K^W$ *on* N^+*. Its* θ*-representations are bounded uniformly along every ladder* $b + iy$, $y \downarrow 0$, $b \in (b_1, \eta)$*. So the hypothesis of Theorem 39 holds, and with it the conclusions there, including Conjectures 7 and 9 for* κ.

Proof. (1) On $\Delta(b_0) \cap \Delta(b_0')$ let $H \le H'$ be the two strip widths. Since $\|\cdot\|_H \le \|\cdot\|_{H'}$, the fixed point of the H'-contraction lies in the H-ball. It is a fixed point there, so it equals the H-fixed point by uniqueness. The normalising roots coincide by Rouché. The singular set $\{(b', z) : \varphi_+(z) + \varpi(b') \in \text{cuts of } R_{b'}\}$ is closed in $N \times \mathbb{C}$, because the cuts move continuously with b'. Uniform bounds on compacta follow from continuity of the fixed point in the base and from the explicit bounds $\|Q^*\|_H \le \rho$, $\|P'\|_H \le \rho$ of Theorem 32, which hold throughout each disc by Theorem 35(1). The continuation of the representations to the half-planes of Definition 37 is as in the proof of Theorem 38: through the band, which persists for b' near the real axis because $T_{b'}$ depends holomorphically on b'.

(2) The proof of Theorem 38 runs verbatim with the following per-base data. The admissible range is $Y_0 + D + 3 < Y < (h(b') - Y_0 - D - 3)/2$ with $H - \frac{1}{2}$ in place of H; it contains a fixed Y for all b' in each small disc. The strip Σ_q, on which $|\hat{T}_{b'} - \mathrm{id} - \hat{t}_0| \le \rho/2\pi$ and $|\hat{T}'_{b'} - 1| \le \rho$, persists by continuity in b'. So does the identity $R_{b'} \circ T_{b'} = S_{b'}$ on it, by analytic continuation in b' of an identity valid at real b'. The identity $\sigma \circ R(w) = \sigma(1)\lambda^w$ holds for all b'. The sewn surface is again $\mathbb{C}^*$. Proper maps of $\mathbb{C}^*$ fixing the ends with local degree one are linear. Finally, the normalising roots $c_j = c_0 - j\varpi + O(|\Lambda|)$ have imaginary parts separated by $h(b')$. None of these steps uses reality of the base. The one place it entered, $\operatorname{Im} t_0 = h/2$, is replaced by $\operatorname{Im} \hat{t}_0 \approx 0$ for the shifted map $\hat{T}_{b'}$, which holds after shrinking the discs.

(3) By (2), $\kappa_{b'} = K^W_{b'}$ on a nonempty open subset of N^+. For $z \in W$, where $K^W_{b'} = S_{b'} \circ \varphi_-$ has no cuts, both sides are holomorphic in b' on the connected set N^+. So they agree on $N^+ \times W$, and then for all z off the singular set by continuation in z (the complement of the cuts is connected). Uniform boundedness is then (1). □

Corollary 41 (A bounded complex cusp corridor)**.** *There are $b_3 < \eta$, $r > 0$, and a connected open set $N_\eta \subset \mathbb{C}$ containing (b_3, η), with $\eta \in \partial N_\eta$, such that K^W is jointly holomorphic on $N_\eta \times B(0,r)$ and $K^W_b \to F^{\mathrm{att}}_\eta$ locally uniformly there as $b \to \eta$ within N_η, together with all z-derivatives. Throughout N_η, the only preimage of 1 in $B(0,r)$ is 0, and K^W_b and $1/K^W_b$ are uniformly bounded on that disk. No lower bound on the width of N_η is asserted.*

Proof. Choose r inside the disk of Corollary 34 so that F^{att}_η has no zero on $\overline{B(0,r)}$ and $F^{\mathrm{att}}_\eta - 1$ has just its simple zero at 0. The minimum of $|F^{\mathrm{att}}_\eta|$ on the closed disk and of $|F^{\mathrm{att}}_\eta - 1|$ on its boundary is positive; let one third of the smaller minimum be m. By that corollary, for every real $x \in (b_3, \eta)$, with b_3 sufficiently close to η, $\|K^W_x - F^{\mathrm{att}}_\eta\|_{\overline{B(0,r)}} < m$. Theorem 35 gives a base disk U_x centred at x on which K^W_b is jointly holomorphic near the closed z-disk. Shrink it so that $|b - x| < (\eta - x)/4$ and

$$\|K^W_b - K^W_x\|_{\overline{B(0,r)}} < \min(m, \eta - x) \qquad (b \in U_x).$$

Two overlapping disks centred on the real line have an overlapping real interval. Both families equal the uniquely normalised real solution of Theorem 32 there, so the identity theorem patches them. Their union $N_\eta = \bigcup_x U_x$ is connected and excludes η. The preceding bounds give zero-freeness and, by Rouché's theorem, exactly one preimage of 1 in $B(0,r)$, necessarily 0. They also bound the functions and their reciprocals uniformly. If $b \to \eta$ inside N_η and $b \in U_x$, then $|b - x| < (\eta - x)/4$ forces $x \to \eta$; the last displayed bound and Corollary 34 give the stated convergence. Cauchy's formula gives convergence of derivatives on smaller disks. □

Remark 42. The open-set hypothesis in (3) is implied by membership at a single base b'_* with strict inequality $|\operatorname{Im} Q - h/2| < 1$, provided the limits $\kappa_{b'}(z) \to L(b')$, $L_2(b')$ as $\operatorname{Im} z \to \pm\infty$ are locally uniform in b'. Indeed, at a complex base these limits give R- and S-representations near $\pm\mathrm{i}\infty$ with bounded periodic corrections, provided there is no winding. In detail, $H(q) = \lambda^P$, with $q = \mathrm{e}^{2\pi\mathrm{i}z}$, satisfies $|H| = o(|q|^{-\arg\lambda/2\pi})$ with $\arg\lambda < 2\pi$, so it extends holomorphically to $q = 0$. A zero of H at $q = 0$ is exactly winding. We take $\operatorname{Im} z$ large enough to avoid the other zeros of H (points where $\kappa = L$). The winding number is an integer that varies continuously, so it persists. The S-side is the same; τ^{-1} has no critical points. On the remaining strip, which is compact modulo 1, the representations of $\kappa_{b'_*}$ continue to nearby b' because $\kappa_{b'} \to \kappa_{b'_*}$ uniformly there, and the height condition is open. Local uniformity of the defining limits in the base is exactly the kind of statement a construction of the family, such as Paulsen's [21], provides.

Consequently, nothing quantitative about κ remains to be checked. It is enough that the base-continued family has its two regular representations, with the seam half a period below

the real axis, on some open set of bases near (b_1, η). That is how `_cbuild.py` computes it, and the ladders are consistent with it. A proof requires the precise definition and uniqueness theorem of [21], which we have not been able to consult.

7.10 Numerics up to $\eta - b = 3 \cdot 10^{-5}$

Because T needs no ladder, the invariants can be computed essentially at the parabolic value. The modes are measured on the line $\operatorname{Im} z = -h_2/2 + Y_0$, $h_2 = 2\pi/\log\lambda_2$, close to the gate carrying the positive modes, so that the precision needed is independent of h; results at $Y_0 = 1, 1.5, 2, 2.5, 3$ agree to all printed digits, and at $b = 1.3$ they agree with sampling on $\mathbb{R}$ at 40 digits.

λ	$\eta - b$	$\lvert\tau_1\rvert/\lvert B_1\rvert$	δ_1/ε^2	$\arg\tau_1$	$\lvert t_2/t_1^2\rvert$	$\lvert t_3/t_1^3 - B_3/B_1^3\rvert$
0.5	9.04×10^{-2}	0.99601457	0.008295	1.581415	1.555683	0.148
0.8	1.21×10^{-2}	0.99952725	0.009494	1.585920	1.572187	0.0177
0.9	2.84×10^{-3}	0.99989062	0.009853	1.586391	1.573905	0.00410
0.95	6.87×10^{-4}	0.99997362	0.010028	1.586499	1.574298	0.000989
0.98	1.08×10^{-4}	0.99999587	0.010131	1.586528	1.574403	0.000155
0.99	2.67×10^{-5}	0.99999897	0.010165	1.586532	1.574418	0.0000385
limit	0	1	≈ 0.01020	1.586533	1.574423	0

All three invariants approach their parabolic targets. The third mode converges to $B_3/B_1^3 = -2.254799 - 0.024409\mathrm{i}$, so it is a meaningful and verified comparison.

The deficit law. The natural variable for the approach is not ε but the product of the two logarithmic multipliers,

$$p := |\log\lambda|\,\log\lambda_2 = \varepsilon^2\bigl(1 - \tfrac{\varepsilon}{3} + O(\varepsilon^2)\bigr).$$

This is the analytic parameter $\hat\epsilon$ of the Mardešić–Roussarie–Rousseau normal form, up to a factor analytic and nonzero at 0: the multipliers there are $\exp(\mu_{0,\infty})$ with $\mu_0\mu_\infty = -4\hat\epsilon/(1 - a^2\hat\epsilon)$. Twelve bases with $1 - \lambda = k/200$, $k = 1, \dots, 12$, fitted by polynomials of degree $2, 3, 4$ in p, give coefficients that agree to eleven digits:

$$\begin{aligned} \log\frac{\tau_1(b)}{B_1 \mathrm{e}^{2\pi\mathrm{i}a}} &= \kappa^{(1)}p + O(p^2), \quad \kappa^{(1)} = -0.0101990063459 - 0.0132509561825\mathrm{i}, \\ \log\frac{t_2/t_1^2}{B_2/B_1^2} &= \kappa^{(2)}p + O(p^2), \quad \kappa^{(2)} = -0.0306585435 - 0.0358138682\mathrm{i}. \end{aligned} \tag{28}$$

The coefficients of p^2 are about $4 \cdot 10^{-4}$ in modulus. The constant terms vanish to 10^{-16}, and in ε there is no linear term. Consequently

$$\delta_1 = C\,p + O(p^2), \qquad C = -\operatorname{Re}\kappa^{(1)} = 0.0101990063,$$

and $\delta_\kappa/\delta_1 \to \operatorname{Re}\kappa^{(2)}/\operatorname{Re}\kappa^{(1)} = 3.00603$. This settles Section 6.3. The deficit is analytic of first order in the unfolding parameter. The exponents 1.74 and 2 and the constant 0.01013 reported there came from fitting a finite range in variables that differ from p at relative order ε. Candidate closed forms such as $1/(10\pi^2)$ are excluded. By Corollary 14, the Kneser-side $\hat c_n$ inherits (28) up to $O(\Lambda)$, which is invisible at every order in p. The first-order coefficients of $\log(\tau_n/(B_n \mathrm{e}^{2\pi\mathrm{i}na}))$ for $n = 1, 2, 3$ are

$$-0.0101990063 - 0.0132509562\mathrm{i}, \quad -0.0510565562 - 0.0623157805\mathrm{i}, \quad -0.1400443481 - 0.1692244639\mathrm{i}$$

(script `deficit_modes.py`). They are not linear in n, so the first-order effect of the unfolding is not a translation of Fatou coordinates. A derivation of $\kappa^{(1)}$ is left open, and it is not a routine perturbation computation. Write the attracting coordinate as $\Psi_p + \sum_k F_p \circ f_p^k$. The model time is $\Psi_p = \nu_p \log u - \sum_{m\geq 1} s_m(p)/(mu^m)$, up to constants, with $s_m = A_1 u_1^m + A_2 u_2^m$ analytic in p and $s_3 \propto p$. Differentiating term by term in p at $p = 0$ diverges: along $u_k \approx -2/k$ the parameter sensitivity of the orbit grows like k while $F_0'(u_k)$ decays like $1/k$, and the s_3/u^3 term grows like k^3. So $\kappa^{(1)}$ is a regularised, resurgent quantity of the unfolding (compare the analytic dependence of the unfolded modulus on the canonical parameter in [18, 5]). Its closed-form evaluation is a natural question for that theory.

Remark 43 (On the definition and uniqueness in [21])**.** In [21], κ_b is defined by analytic continuation in b of Kneser's solution $\kappa_b(z_0)$, $\operatorname{Re} z_0 > -2$, from (η, ∞) (Prop. 3 there). That proposition asserts real-analytic dependence of Kneser's conformal uniformisation on the real base without a separate parameter-dependence proof; Proposition 44 supplies such a proof locally along the real exterior ray, and Corollary 70 gives uniform vertical tails there. Transporting those estimates through a complex corridor to the Shell–Thron boundary remains open. Off that boundary, Proposition 1 of that paper asserts two regular representations $\psi_1^{-1}(\rho_1)$ and $\psi_2^{-1}(\rho_2)$ from the limits at $\pm\mathrm{i}\infty$. We impose their degree-one form explicitly in Definition 37; Lemma 68 records the integer end label that a fixed-point limit alone leaves open. The continuation across the Shell–Thron boundary is argued in [21, §5] by the continuity of a numerical scheme; it is not proved there.

The uniqueness statement [21, Prop. 2] requires only analyticity on $\operatorname{Re} z > -2$, $F(0) = 1$ and the two limits. As stated, it is false, already at $b = \mathrm{e}$, by the following exact existence argument. Let $K = \operatorname{sexp}_{\mathrm{e}}$ be Kneser's real-normalised solution and $D_0 = \{|z| < \frac{1}{2}\}$. Since $K(0) = 1$ and K is nonconstant and holomorphic, the open-mapping theorem gives $\delta > 0$ such that $B(1, \delta) \subset K(D_0)$. Decrease δ so that $\mathrm{e}(1 - \mathrm{e}^{-\delta}) > \delta$. For every real t,

$$\exp(B(t, \delta)) \supset B\big(\mathrm{e}^t, \mathrm{e}^t(1 - \mathrm{e}^{-\delta})\big),$$

as follows by applying the principal logarithm and $|\log(1+v)| \leq -\log(1-|v|)$. Put $u_n = K(n)$; then $u_0 = 1$ and $u_{n+1} = \mathrm{e}^{u_n}$. Induction using the displayed inclusion gives $B(u_n, \delta) \subset K(D_0 + n)$ for every n, and indeed

$$B\big(u_{n+1}, u_{n+1}(1 - \mathrm{e}^{-\delta})\big) \subset K(D_0 + n + 1).$$

Choose n so large that the last radius exceeds $1 + \pi/2$, and let $k = \lceil \mathrm{e}^{u_{n+1}}/(2\pi) \rceil$. For large n, $0 \leq \log(2\pi k) - u_{n+1} < 1$. Hence $w = \log(2\pi k) + \mathrm{i}\pi/2$ lies in $K(D_0 + n + 1)$. Choose $z_* \in D_0 + n + 1$ with $K(z_*) = w$ and set $c = z_* + 2$. Then $\operatorname{Re} c > 0$ and, exactly,

$$K(c - 1) = \mathrm{e}^w = 2\pi\mathrm{i}k, \qquad K(c) = \mathrm{e}^{2\pi\mathrm{i}k} = 1.$$

Thus $F(z) = K(z + c)$ is analytic on $\operatorname{Re} z > -2$, obeys $F(z + 1) = \mathrm{e}^{F(z)}$, has $F(0) = 1$, and has the same two vertical limits as K. Yet $F(-1) = 2\pi\mathrm{i}k \neq 0 = K(-1)$. The construction works for every real $b > \eta$. Put $a = \log b > 1/\mathrm{e}$ and $u_n = K_b(n) \to \infty$, since $\mathrm{e}^{ax} > x$ for $x > 0$. Choose N with $au_{N+1} > 1$ and then $\delta > 0$ so that $B(u_N, \delta) \subset K_b(B(N, \frac{1}{2}))$ and $u_{N+1}(1 - \mathrm{e}^{-a\delta}) > \delta$. The inclusion

$$E_b(B(t, \delta)) \supset B\big(\mathrm{e}^{at}, \mathrm{e}^{at}(1 - \mathrm{e}^{-a\delta})\big)$$

propagates the fixed δ-disks and yields image disks of radii $u_{n+1}(1 - \mathrm{e}^{-a\delta}) \to \infty$. For large $n \geq N$ choose $k = \lceil a \exp(au_{n+1})/(2\pi) \rceil$. The target $w_k = a^{-1}(\log(2\pi k/a) + \mathrm{i}\pi/2)$ lies in the resulting image disk. Then $E_b^2(w_k) = 1$ and $E_b(w_k) = 2\pi\mathrm{i}k/a \neq K_b(-1) = 0$. Choose $z_k \in B(n + 1, \frac{1}{2})$ with $K_b(z_k) = w_k$, put $c_k = z_k + 2$, and set $F_k(z) = K_b(z + c_k)$. Since

$\operatorname{Re} c_k > 0$, each F_k meets the hypotheses of Proposition 2, but $F_k(-1) = 2\pi \mathrm{i} k/a$. The values k are unbounded, giving infinitely many distinct solutions. The gap in [21, Prop. 2] is the last step of its proof, which infers $C = 0$ from $F_1(C) = 1$. For real $b > \eta$ the uniqueness theorem of [20] is not affected: it also assumes analyticity on $\mathbb{C} \setminus (-\infty, -2]$ and $F(\bar{z}) = \overline{F(z)}$, both of which this F violates. Indeed c is nonreal, and K cannot extend holomorphically through -2, since $K(-1) = 0$ whereas an exponential never vanishes; hence F is singular at $-2 - c$ off the standard cut. For complex bases, some root-selection condition must be added to [21, Prop. 2]. The seam-height condition of Definition 37 is such a condition, and Theorem 38 shows that it suffices near $(1, \eta)$.

7.11 Identification with an independent Kneser-type implementation

The hypothesis of Theorem 39 can be tested against a completely independent computation of the base-continued family. Sheldon Levenstein's PARI/GP program `fatou.gp` builds the "merged" two-fixed-point solution for complex bases. At each base it constructs, independently, the solution with a θ-mapping at each of the two fixed points: the two regular representations of [21, Prop. 1 and Lemma 1]. Its author regards this as the continuation of Kneser's solution in the base. We use the unmodified program (sha256 `70559daf...`): published reference values with 100 verified digits at $b = 0.8 + 0.4\mathrm{i}$ and $b = 1 + \mathrm{i}$, and our own runs at `\p 40` at $b = 1.2 + 0.1\mathrm{i}$ and $b = 1.3 + 0.2\mathrm{i}$, near the interval $(1, \eta)$. In those runs $\operatorname{sexp}(0) = 1$ to 10^{-53}, and the functional equation holds to 10^{-54}.

We solved the sewing equation of Theorem 32 directly at these complex bases (script `weld_complex.py`). R is the regular solution at the attracting fixed point and S the one at the repelling fixed point; the seam lies in the band of holomorphy of $T = R^{-1} \circ S$, and the root is selected as in Definition 37. Table 5 shows the result. At the three bases where the band is not too narrow, the sewing solution agrees with the independent value *to the full working precision of* 40 *digits*. That is 27 to 35 orders of magnitude below the size of the separation $|\kappa - R|$ itself. At $1 + \mathrm{i}$ the band is narrowest and the agreement $(2 \cdot 10^{-25})$ is limited by the number of samples. The neighbouring roots $c_{\pm 1}$ reproduce R instead, and miss by the full separation. The θ-mapping builder used for the ladders does not converge at $0.8 + 0.4\mathrm{i}$, whereas the sewing iteration converges in a few dozen steps.

Table 5: Sewing solution versus the independent `fatou.gp` values of $\kappa_b(\frac{1}{2})$ (working precision 40 digits).

b	$\lvert\Lambda\rvert$	$\lvert\kappa_b(\frac{1}{2}) - R_b(\frac{1}{2})\rvert$	$\lvert K_b^W(\frac{1}{2}) - \kappa_b(\frac{1}{2})\rvert$	$\lvert\hat{c}_1\rvert$
$1.2 + 0.1\mathrm{i}$	1.91×10^{-11}	6.91×10^{-13}	4.6×10^{-40}	0.0869956
$1.3 + 0.2\mathrm{i}$	2.92×10^{-11}	1.81×10^{-12}	8.3×10^{-41}	0.0876095
$0.8 + 0.4\mathrm{i}$	1.05×10^{-3}	9.86×10^{-5}	5.5×10^{-39}	0.0856072
$1 + \mathrm{i}$	2.13×10^{-2}	4.41×10^{-3}	2.0×10^{-25}	0.0873850

The agreement identifies the sewing solution with an independent implementation of the two-sided regular representation, at four bases, two of them near $(1, \eta)$. It does *not* test the analytic continuation across the Shell–Thron boundary: both programs work at one base at a time. That continuation is the one remaining hypothesis (Section 7.13).

7.12 A pointwise test near the Shell–Thron boundary

The analytic continuation of Kneser's solution across the Shell–Thron boundary is argued in [21, §5] only through the continuity of a numerical scheme. In particular, continuity of each finite quadrature approximant does not imply continuity, let alone holomorphy, of its limit without locally uniform convergence in the base parameter. The transfer in [21,

Prop. 3] of fixed-point limits from real to complex bases likewise requires such uniformity. We tested whether the pointwise two-fixed-point constructions on the two sides are numerically compatible near the irrational boundary point $\lambda_c = e^{i\phi}$, $\phi = 2\pi(\sqrt{2}-1)/5$, where $b_c = \exp(\lambda_c e^{-\lambda_c}) \approx 1.5217 + 0.0148i$. Along the normal $\lambda = re^{i\phi}$ we computed the pointwise output $F_b^{\mathrm{GP}}(\frac{1}{2})$ of `fatou.gp` (`\p 24`). We used three bases inside ($r = 0.98, 0.99, 0.995$, one attracting fixed point) and three outside ($r = 1.005, 1.01, 1.02$, both fixed points repelling). So the two sides are computed with structurally different representations. A degree-4 polynomial in b fitted to five of the six values predicts the sixth to 10^{-12}–10^{-13}, uniformly on both sides. The quadratic through the three inside values extrapolates to the outside values with errors $1.3 \cdot 10^{-8}$, $3.2 \cdot 10^{-8}$ and $1.1 \cdot 10^{-7}$, growing with the distance as the cubic truncation term of a smooth function should. The reverse extrapolation gives the mirror-image errors. The errors are consistent with a smooth join of these pointwise constructions at the tested scales. Finite sampling cannot exclude a smaller jump, a higher-order nonanalytic term, or failure of continuation elsewhere on the boundary. Nor does the pointwise algorithm identify its outputs with the base-continued family κ. This is numerical evidence compatible with the continuation claimed in [21], not a proof.

We repeated the six-base calculation at five noninteger heights, using the unmodified `fatou.gp` at 50-digit precision (`boundary_multiheight.py`; raw values in `data/boundary-multiheight.json`). Table 6 reports the largest leave-one-out error when a degree-4 polynomial in b through five bases predicts the sixth. It also reports the difference at b_c between quadratics fitted separately to the three inside and three outside values. Recomputing the two closest bases at 70-digit precision changed all ten values by less than $5 \cdot 10^{-50}$. The five tested heights form only a finite set, and sampling at one boundary point cannot verify condition (P2).

Table 6: Finite two-sided boundary comparison at five noninteger heights. The quadratic gap is an extrapolation diagnostic, not a bound on a jump.

z	max. degree-4 leave-one-out error	two-sided quadratic gap at b_c
1/6	$1.21 \cdot 10^{-12}$	$4.25 \cdot 10^{-9}$
1/3	$1.62 \cdot 10^{-12}$	$6.64 \cdot 10^{-9}$
1/2	$1.45 \cdot 10^{-12}$	$7.02 \cdot 10^{-9}$
2/3	$9.68 \cdot 10^{-13}$	$5.57 \cdot 10^{-9}$
5/6	$4.28 \cdot 10^{-13}$	$2.89 \cdot 10^{-9}$

Proposition 44 (Local base analyticity on the real exterior ray)**.** *For every real $b_0 > \eta$, Kneser's real-normalised solution admits a jointly holomorphic extension in (b, z) to $\Delta \times W$, where Δ is a complex disc about b_0 and W is a complex neighbourhood of $[0, 1]$. These local extensions agree on overlap, so the classical exterior branch has a jointly holomorphic germ along the entire real ray (η, ∞). In particular, every compact subinterval of that ray has a common complex base neighbourhood and a common complex height neighbourhood of $[0, 1]$ on which this family is jointly holomorphic. The assertion supplies no corridor reaching a non-cusp Shell–Thron boundary arc.*

Proof. Write $a = \log b$ and choose the principal fixed-point pair $L_+(b)$ and $L_-(b)$, conjugate when $b > \eta$ and characterised there by $0 < a \operatorname{Im} L_+ < \pi$. They are simple fixed points and hence holomorphic in b near b_0. Set $u_b = (L_+ + L_-)/2$, $v_b = (L_+ - L_-)/(2\mathrm{i})$, and $\gamma_b(t) = u_b + \mathrm{i}v_b t$ for $-1 \le t \le 1$. At a real base, $u_b, v_b \in \mathbb{R}$, $v_b > 0$, and

$$E_b(\gamma_b(t)) = r_b \mathrm{e}^{\mathrm{i}\theta_b t}, \qquad r_b = \mathrm{e}^{a u_b} = |L_+|, \quad \theta_b = a v_b \in (0, \pi).$$

Since $v_b = r_b \sin\theta_b$, the multipliers satisfy $|E_b'(L_\pm)| = a r_b = \theta_b / \sin\theta_b > 1$. The straight chord γ_b and this circular arc meet only at $L_\pm$ and bound the initial region $H_b = \{\operatorname{Re} w \ge u_b,\ |w| \le r_b\} \setminus \{L_+, L_-\}$. This is Kneser's initial region as described in [27, §4, Lem. 4]. The arcs depend holomorphically on b through their real-analytic parameter t. They remain disjoint away from their endpoints after shrinking a disc Δ about b_0; at the endpoints their tangents remain distinct because the fixed-point multipliers are nonreal. The bounded Jordan interiors vary continuously, so their union in $\Delta \times \mathbb{C}$ is open; the ambient coordinates (b, w) make its projection a holomorphic family. Identify thin analytic collars of the two arcs by $(b, w) \mapsto (b, E_b(w))$. The map is injective on the collar after shrinking Δ, since it is injective on the compact chord at b_0 and its derivative never vanishes. This gives a Hausdorff holomorphic family of cylinders with one end at each $L_\pm(b)$.

The ends can be filled holomorphically. Indeed, parameter-dependent Koenigs coordinates $\sigma_{\pm,b}$, normalised by $\sigma_{\pm,b}(L_\pm) = 0$ and $\sigma'_{\pm,b}(L_\pm) = 1$, linearise E_b near the two repelling fixed points. If $\ell_{\pm,b} = \log E_b'(L_\pm(b))$ is chosen continuously, then $\exp(2\pi\mathrm{i} \log \sigma_{\pm,b}/\ell_{\pm,b})$ (or its reciprocal) is a puncture coordinate on the corresponding glued end. Filling both punctures yields a proper holomorphic submersion of compact genus-zero surfaces over Δ, just as in the collar construction of Proposition 16.

We also have a holomorphic marked section represented by $w = 1$. For real $b > \eta$, $E_b(x) > x$ for every real x: the difference is positive at $x \to \pm\infty$ and has no real zero. The adjacent real intervals between the chord and its image are (u_b, r_b), with $r_b = E_b(u_b) > u_b$. The forward endpoints $E_b^n(u_b)$ increase to $+\infty$, since a finite limit would be a real fixed point. For any $x > 0$ below u_b, forward iteration eventually enters (u_b, r_b) or one of its boundary seams; for $x > r_b$, repeated real logarithms eventually do the same. Indeed, a monotone orbit staying on either side would have a finite real fixed-point limit. Thus these dynamical translates and their seam collars cover the positive real axis. In particular, 1 is in the interior of one such interval or on a seam. A finite chain of holomorphic dynamical charts represents it near b_0; at a seam the analytic collar gives the same section. The entire compact real segment from 1 to $b = E_b(1)$ is covered by finitely many of these charts.

The two filled ends and this marked point are three disjoint holomorphic sections. Fischer–Grauert local triviality [8] and the fibrewise three-point Möbius normalisation give a jointly holomorphic coordinate q_b on the compactified cylinder, taking the upper end to 0, the lower end to ∞, and $w = 1$ to 1. The upper puncture coordinate $\exp(2\pi\mathrm{i} \log \sigma_{+,b}/\ell_{+,b})$ shows that the logarithmic lift $z = (2\pi\mathrm{i})^{-1} \log q_b$, with $z(1) = 0$, gains $+1$ under the edge identification by E_b. On the real slice, reflection exchanges the two ends and fixes the marked point, so $q_b(\bar{w}) = 1/\overline{q_b(w)}$ in the real dynamical charts. Thus z is real there; its local injectivity and $z(E_b(w)) = z(w) + 1$ make it strictly increasing along the positive axis. Inverting on the finite

chart chain covering the real segment from 1 to b gives a jointly holomorphic superfunction on $\Delta \times W$, after shrinking both.

For real b, the lift restricted to H_b is injective and its integer translates cover the universal-cover plane. Choosing any $d \in H_b$ and subtracting its value gives the initial-region criterion of [27, Thm. 1]; the injectivity and tiling assertions for the Kneser generalisation are proved in [27, Thm. 5]. We normalise at d here, since 1 need not itself belong to H_b. Their Abel coordinates differ only by a constant on H_b, and continuation along the real dynamical charts to $w = 1$ makes that constant zero. Thus the inverse just constructed is the classical real-normalised Kneser branch. The marking fixes the local family uniquely, and the identity theorem patches the parameter discs on overlaps. A finite subcover of any compact real base interval, followed by intersection of its finitely many height neighbourhoods, gives the final uniformity assertion. □

Proposition 45 (Persistence of the exterior initial curve near the cusp)**.** *Put, for small $\theta \neq 0$,*

$$r(\theta) = \mathrm{e}^{\theta \cot \theta}, \qquad a(\theta) = \frac{\theta \mathrm{e}^{-\theta \cot \theta}}{\sin \theta}, \qquad b(\theta) = \mathrm{e}^{a(\theta)}, \qquad L_{\pm}(\theta) = r(\theta)\mathrm{e}^{\pm \mathrm{i}\theta}.$$

For every fixed $M > 0$ there is $p_0 > 0$ such that, whenever $\theta = p + \mathrm{i}q$, $0 < p < p_0$ and $|q| \leq Mp$, the chord $\gamma_\theta(t) = r(\theta)(\cos\theta + \mathrm{i}t \sin\theta)$ and its image $E_{b(\theta)}(\gamma_\theta(t)) = r(\theta)\mathrm{e}^{\mathrm{i}\theta t}$ are individually injective and meet only at their common endpoints $L_{\pm}(\theta)$. Thus the exterior initial Jordan curve persists geometrically in this complex parameter cone.

There is moreover a real-analytic function $q_c(p) = p^2/6 + O(p^3)$ such that, for $0 < p < p_0$ and $0 \leq q < q_c(p)$, both fixed-point multipliers have modulus greater than one. The upper multiplier has modulus one at $q = q_c(p)$; this is a non-cusp Shell–Thron boundary arc. Therefore the unmarked two-end quotient surfaces of Proposition 44 exist locally throughout this exterior wedge, arbitrarily close to that boundary arc. The map $\theta \mapsto b(\theta)$ is injective on a sufficiently small such wedge, so its image is a genuine complex base corridor attached to the real exterior ray. This geometric assertion does not supply a holomorphic section represented by $w = 1$ on a common parameter corridor or a common regular height domain there.

Proof. The identities $ar \sin \theta = \theta$ and $\log r = \theta \cot \theta$ show that $L_{\pm}$ are fixed points and give the displayed image arc. The functions $a(\theta)$ and $b(\theta)$ are even and holomorphic at 0, with $b(\theta) = \eta(1 + \theta^2/(2\mathrm{e}) + O(\theta^4))$; in particular $b'(\theta) \neq 0$ for small $\theta \neq 0$. More strongly, $b(\theta) = B(\theta^2)$ with $B'(0) = \eta/(2\mathrm{e}) \neq 0$. The map B is locally injective, as is $\theta \mapsto \theta^2$ in $\operatorname{Re}\theta > 0$; hence b is injective on the small wedge.

An intersection at an interior arc point would require the number

$$s_\theta(t) = \frac{\mathrm{e}^{\mathrm{i}\theta t} - \cos\theta}{\mathrm{i}\sin\theta} \qquad (-1 < t < 1)$$

to be real. Taylor expansion with its endpoint zeros factored out gives, uniformly for $-1 \leq t \leq 1$,

$$\mathrm{e}^{\mathrm{i}\theta t} - \cos\theta - \mathrm{i}t\sin\theta = (1 - t^2)\theta^2\left(\tfrac{1}{2} + O(|\theta|)\right), \qquad \operatorname{Im} s_\theta(t) = -(1 - t^2)\left(\tfrac{p}{2} + O(|\theta|^2)\right).$$

The error is uniform even as $t \to \pm 1$: every term of order at least two in the power series has the factor $1 - t^2$, with the remaining polynomial bounded on $[-1, 1]$. In the stated cone, $|\theta|^2 \leq (1 + M^2)p^2$, so the last imaginary part is strictly negative for $-1 < t < 1$ and small p. The chord is injective since $\sin\theta \neq 0$, and the arc is injective since $\mathrm{e}^{\mathrm{i}\theta(t_1 - t_2)} = 1$ with $|t_1 - t_2| \leq 2$ and small $|\theta|$ forces $t_1 = t_2$. The same expansion gives distinct endpoint tangents, hence a Jordan curve.

The multipliers are $\lambda_{\pm}(\theta) = a(\theta)L_{\pm}(\theta) = \theta\cot\theta \pm \mathrm{i}\theta$. For $\theta = p + \mathrm{i}q$ near zero,

$$|\lambda_+|^2 = 1 + \frac{p^2}{3} - 2q + O(p^3 + p|q| + q^2), \qquad |\lambda_-|^2 = 1 + \frac{p^2}{3} + 2q + O(p^3 + p|q| + q^2).$$

At $q = 0$ both are repelling, while $\partial_q|\lambda_+|^2 = -2 + O(p + |q|)$. The real implicit function theorem gives a unique nearby zero $q_c(p) = p^2/6 + O(p^3)$, and the signs above show that both multipliers are repelling for $0 \le q < q_c(p)$. Since $a(\theta) = \lambda_+(\theta)\mathrm{e}^{-\lambda_+(\theta)}$, the locus $|\lambda_+| = 1$ is precisely the stated Shell–Thron boundary parameterisation. The collar gluing and repelling-end filling in Proposition 44 apply locally at every point of the open wedge. □

Proposition 46 (A certified marked exterior path)**.** *Set $\theta(q) = \frac{1}{10} + \mathrm{i}q$ and use $b(\theta)$, $L_\pm(\theta)$ and the initial region H_θ of Proposition 45. There is a unique $q_c \in (0, 1667/10^6)$ with $|\lambda_+(\theta(q_c))| = 1$. For every $0 \le q \le q_c$,*

$$E^{\circ 28}_{b(\theta(q))}(1) \in \operatorname{int} H_{\theta(q)}.$$

For $0 \le q < q_c$ both fixed points are repelling, and the exterior Kneser germ normalised by $F(0) = 1$ analytically continues along the embedded base path $q \mapsto b(\theta(q))$. The result concerns the germ at height $z = 0$; it does not give a single regular height domain containing $[0, 1]$ uniformly as $q \uparrow q_c$.

Proof. The following strict enclosures are certified with outward-rounded Arb complex ball arithmetic by the reproducible script `docs/certify_exterior_mark.py`. It uses only the rational inputs $p = 1/10$, $Q = 1667/10^6$, 28 iterates, and 1000 equal subintervals of $[0, Q]$; the recorded run used `python-flint` 0.9.0 with FLINT 3.6.0. For $0 \le q \le Q$ it checks that the arc coordinate $s_\theta(t) = (e^{\mathrm{i}\theta t} - \cos\theta)/(\mathrm{i}\sin\theta)$ has $\partial_t \operatorname{Re} s_\theta(t) > 0.98$ for $-1 \le t \le 1$, with $\operatorname{Re} s_\theta(-1/10) < -0.08$ and $\operatorname{Re} s_\theta(1/10) > 0.08$. The estimate behind the remaining whole-arc check is elementary: writing

$$e^{\mathrm{i}\theta t} - \cos\theta - \mathrm{i}t\sin\theta = (1 - t^2)\theta^2 H(\theta, t), \qquad |H(\theta, t) - \tfrac{1}{2}| \le \frac{e^R - 1 - R}{2R}, \quad R = 0.101,$$

the termwise interpolation remainder and $|\sin\theta/\theta - 1| \le \sinh(R)/R - 1$ imply $\operatorname{Im} s_\theta(t) < -0.04(1 - t^2)$ for $-1 < t < 1$. Indeed, after division by $1 - t^2$ the order-n interpolation polynomial is a sum of at most $n/2$ powers of t, each of modulus at most one; this gives the stated bound on H. Thus the image arc is a graph strictly below the chord in the s-plane. On the horizontal interval $[-0.01, 0.04]$ its height is below -0.0396.

Let $w_{28}(\theta) = E^{\circ 28}_{b(\theta)}(1)$ and $\zeta = (w_{28}/r - \cos\theta)/(\mathrm{i}\sin\theta)$. The same script encloses ζ on every parameter subinterval and verifies $-0.01 < \operatorname{Re}\zeta < 0.04$ and $-0.03 < \operatorname{Im}\zeta < -0.01$. Hence w_{28} lies strictly between the two arcs. It also verifies that $|\lambda_+|^2 - 1$ is positive at $q = 0$, negative at $q = Q$, and has derivative less than -1 throughout $[0, Q]$; that $|\lambda_-|^2 > 1.003$ there; and that $\operatorname{Im}(db(\theta(q))/dq) > 0.05$. These strict inequalities prove the root, repelling and embedded-path claims.

For $q < q_c$, the two repelling ends and the analytic collars give a holomorphic family of quotient spheres locally in q. The point $w_{28}(\theta)$ is a holomorphic section in the initial region, so it marks the physical point $w = 1$ through the finite iterate $E^{\circ 28}_b$; that iterate has nonzero derivative at 1. After three-point normalisation choose the logarithmic Abel lift with value 28 at w_{28}, and pull it back by the local inverse of $E^{\circ 28}_b$ to obtain value 0 at $w = 1$. Local inversion of this Abel coordinate gives a jointly holomorphic normalised superfunction germ at $z = 0$. At $q = 0$ it is the classical Kneser germ by Proposition 44; the embedded path continues that germ uniquely. □

Proposition 47 (A prescribed Brjuno endpoint for the marked path)**.** *Let*

$$\alpha = \frac{\sqrt{2} - 1}{26}, \qquad \lambda_c = \mathrm{e}^{2\pi\mathrm{i}\alpha}.$$

There is a unique solution $\theta_c = p_c + \mathrm{i}q_c$ of $\lambda_+(\theta_c) = \lambda_c$ in the coordinate square of radius 10^{-8} centred at

$$\theta_0 = 0.1000435842 + 0.0016682119\,\mathrm{i}.$$

The number α is Brjuno. Along the embedded path $\theta(q) = p_c + \mathrm{i}q$, $0 \le q < q_c$, the two fixed points remain repelling and $E^{\circ 28}_{b(\theta(q))}(1)$ remains strictly inside $H_{\theta(q)}$, also at $q = q_c$ in the latter geometric assertion. Consequently the classical real Kneser germ has a marked analytic continuation along an exterior path ending at a specified non-cusp Brjuno boundary point. No common regular z-domain containing $[0,1]$ is asserted.

Proof. The number α is a root of $676X^2 + 52X - 1$. For a rational m/n near it, that integer polynomial has nonzero value of modulus at least $1/n^2$ at m/n. Its derivative is bounded nearby, hence $|\alpha - m/n| \ge c/n^2$ for some $c > 0$. The denominators of its continued-fraction convergents therefore satisfy $n_{k+1} \le c^{-1} n_k$, while they grow at least as fast as Fibonacci numbers. Thus $\sum_k (\log n_{k+1})/n_k < \infty$, the Brjuno condition.

The remaining claims are certified with outward-rounded Arb complex balls by `docs/certify_brjuno_exterior_mark.py`. Put $f(\theta) = \theta \cot\theta + \mathrm{i}\theta - \lambda_c$ and $T(\theta) = \theta + \mathrm{i}f(\theta)$. On the stated closed square the script verifies

$$|f(\theta_0)| < 3 \cdot 10^{-11}, \qquad \sup |T'(\theta)| < 0.07, \qquad 3 \cdot 10^{-11} + 0.07\sqrt{2} \cdot 10^{-8} < 10^{-8}.$$

The contraction theorem gives the unique root θ_c in the square. In particular $0 < q_c < 0.00166823$.

The same script subdivides $[0, 0.00166823]$ into 1000 rational intervals and encloses the entire coordinate box $|p - 0.1000435842| \le 10^{-8}$. Uniformly there, it verifies the strict arc-graph and multiplier inequalities used in Proposition 46, with the whole-arc bound $\operatorname{Im} s_\theta(t) < -0.04(1 - t^2)$ obtained now using $|\theta| < 0.102$. It verifies for $\zeta = (E^{\circ 28}_{b(\theta)}(1)/r - \cos\theta)/(\mathrm{i}\sin\theta)$ that

$$-0.01 < \operatorname{Re}\zeta < 0.04, \qquad -0.03 < \operatorname{Im}\zeta < -0.01.$$

The arc is below -0.0396 on this horizontal range, so the marked iterate lies strictly inside H_θ. Finally the script checks $\partial_q |\lambda_+|^2 < -1$, $|\lambda_-|^2 > 1.003$, and $\operatorname{Im}(db(p + \mathrm{i}q)/dq) > 0.05$ throughout the enclosing box. Since $|\lambda_+(\theta_c)| = 1$, both multipliers are repelling for $q < q_c$ and the base path is embedded. The marked quotient construction and local inversion from Proposition 46 now apply along this path. □

Proposition 48 (A marked pullback cell along the Brjuno path)**.** *For every $0 \le q \le q_c$ on the path of Proposition 47, the inverse branch of $E^{\circ 28}_{b(q)}$ taking $w_{28}(q)$ to 1 extends conformally to a neighbourhood of $\overline{H_q}$. Denote it by Φ_q. It depends continuously on q up to q_c and holomorphically in the base parameter locally along the path. Moreover*

$$\Phi_q(E_{b(q)}(w)) = E_{b(q)}(\Phi_q(w)) \qquad (w \in \gamma_q([-1,1])).$$

Consequently $D_q = \Phi_q(H_q)$ is a Jordan cell containing $w = 1$; its two boundary arcs are $\Phi_q(\gamma_q)$ and $E_{b(q)}(\Phi_q(\gamma_q))$, and their endpoints are the original fixed points $L_\pm(q)$. This supplies a global physical pullback of the marked initial region, including at the Brjuno endpoint.

Proof. The certificate `docs/certify_brjuno_exterior_mark.py` now also encloses the singular orbit $E^{\circ k}_{b(q)}(0)$ for every $0 \le k \le 28$ on all 1000 parameter slabs. In the chord coordinate $\zeta = (w/r - \cos\theta)/(\mathrm{i}\sin\theta)$ it verifies

$$-0.1 < \operatorname{Re}\zeta < 0.2, \qquad \operatorname{Im}\zeta > 0.02 \quad \big(w = E^{\circ k}_{b(q)}(0),\ 0 \le k \le 28\big).$$

The closed cell $\overline{H_q}$ lies below the chord $\operatorname{Im}\zeta = 0$, so it avoids every one of these orbit points with a uniform margin. Choose a simply connected thin neighbourhood U_q of $\overline{H_q}$ which still avoids them.

Set $g_0(w) = w$. Inductively choose a logarithm on U_q so that

$$E_{b(q)}(g_j(w)) = g_{j-1}(w), \qquad g_j(w_{28}(q)) = E_{b(q)}^{\circ(28-j)}(1) \quad (1 \le j \le 28).$$

For $j = 1$, g_0 never vanishes on U_q. If g_{j-1} vanished at $w \in U_q$, then $w = E_{b(q)}^{\circ(j-1)}(0)$, contrary to the certificate. Thus each logarithm exists on the simply connected U_q. It is univalent, and the induction gives $E_{b(q)}^{\circ j} \circ g_j = \mathrm{id}$. In particular $\Phi_q = g_{28}$ is the asserted inverse branch. The uniform clearance and the analytic motion of the two boundary arcs allow the U_q and logarithm branches to be chosen locally uniformly in q, including at q_c. This proves the stated parameter dependence.

It remains to check the seam identity; it does not follow merely from the equation $E_b^{\circ 28} \circ \Phi_q = \mathrm{id}$. At the real base $q = 0$, put $u = \gamma_0(0)$ and $r = E_{b(0)}(u)$. The branch Φ_0 is the real inverse of the strictly increasing map $E_{b(0)}^{\circ 28}$ on the real interval through $w_{28}(0)$. Hence $\Phi_0(r) = E_{b(0)}(\Phi_0(u))$. For arbitrary (q, t) in the connected rectangle $[0, q_c] \times [-1, 1]$, both $\Phi_q(E_{b(q)}(\gamma_q(t)))$ and $E_{b(q)}(\Phi_q(\gamma_q(t)))$ are inverse images of $E_{b(q)}(\gamma_q(t))$ under $E_{b(q)}^{\circ 28}$. They vary continuously, and $E_b^{\circ 28}$ has nonzero derivative everywhere. Their equality set is therefore both open and closed in the rectangle. It contains $(0, 0)$, so it is the whole rectangle.

The conformal image $D_q = \Phi_q(H_q)$ is Jordan and contains $\Phi_q(w_{28}(q)) = 1$. The seam identity identifies its second boundary arc with the image under E_b of its first. At either endpoint $L_\pm$, it says that $\Phi_q(L_\pm)$ is fixed by E_b; since $E_b^{\circ 28}(\Phi_q(L_\pm)) = L_\pm$, it follows that $\Phi_q(L_\pm) = L_\pm$. These are the claimed endpoints. □

Proposition 49 (A quantitative flat-cylinder model)**.** *For every parameter on the marked Brjuno path, including its boundary endpoint, the sewn two-end cylinder has a quasiconformal parameterisation from the standard cylinder $\mathbb{C}/\mathbb{Z}$ with maximal dilatation less than $11/9$. In the chord coordinate of Proposition 45, it is induced by*

$$z = x + 20\mathrm{i}y, \quad t = \tanh y, \quad \zeta_q(z) = t + x(s_{\theta(q)}(t) - t), \qquad 0 \le x \le 1, \quad y \in \mathbb{R},$$

where the sides $x = 0$ and $x = 1$ are identified by $E_{b(q)}$. The bound is uniform as $q \uparrow q_c$ and controls the entire punctured cylinder, not only a compact middle part. It does not by itself assert that the image of the flat seam meets a normalised height circle only once.

Proof. The chord and image arc are disjoint Jordan edges, and the latter is an increasing graph in the chord coordinate by the certified inequalities of Proposition 47. Write

$$s_\theta(t) - t = (1 - t^2)D_\theta(t), \qquad D_\theta(t) = \frac{\theta^2 H(\theta, t)}{\mathrm{i}\sin\theta},$$

where the entire interpolation remainder H is defined by

$$e^{\mathrm{i}\theta t} - \cos\theta - \mathrm{i}t\sin\theta = (1 - t^2)\theta^2 H(\theta, t).$$

Set $A_\theta(t, x) = 1 + x(s'_\theta(t) - 1)$. Since $dt/dy = 1 - t^2$, differentiation in $z = x + 20\mathrm{i}y$ gives the Beltrami coefficient

$$\mu_\theta(t, x) = \frac{D_\theta(t) + \mathrm{i}A_\theta(t, x)/20}{D_\theta(t) - \mathrm{i}A_\theta(t, x)/20}.$$

The affine change from ζ to the physical w-coordinate is conformal and leaves this coefficient unchanged.

The reproducible outward-rounded Arb calculation in `certify_exterior_flat_cylinder.py` encloses this formula on the whole box $|\operatorname{Re}\theta - 0.1000435842| \le 10^{-8}$, $0 \le \operatorname{Im}\theta \le 0.00166823$, $-1 \le t \le 1$, $0 \le x \le 1$, and proves $|\mu_\theta| < 1/10$. To avoid interval division by $1 - t^2$ at the

ends, it evaluates the Taylor terms of orders 2 through 11 and bounds the remaining modulus by

$$\frac{R^{10}}{2\,11!\,(1-R/12)}, \qquad R = 0.102.$$

Thus the interpolation has positive Jacobian in its interior. Its boundary maps once around the certified Jordan curve; the degree formula then makes it a homeomorphism onto H_q and a local diffeomorphism inside. The two vertical sides match under E_b, so it descends to a quasiconformal homeomorphism of the sewn punctured cylinder. The standard inequality $K \le (1+\|\mu\|_\infty)/(1-\|\mu\|_\infty)$ gives $K < 11/9$. The uniform bound also makes the two isolated ends removable for the pulled-back complex structure. □

Proposition 50 (Marked exterior germ limits on a boundary subarc)**.** *Let* $I = [0.1000435842 - 10^{-8}, 0.1000435842 + 10^{-8}]$. *For every* $p \in I$ *there is a unique* $q_c(p) \in (0, 0.00166823)$ *with* $|\lambda_+(p + \mathrm{i}q_c(p))| = 1$. *On the exterior side* $0 \le q < q_c(p)$, *the real-normalised Kneser germ transported from* $q = 0$ *has a common complex height disc* V *about* $z = 0$, *independent of* (p, q) *in this parameter region. As* $q \uparrow q_c(p)$, *these germs converge locally uniformly on* V *to nonconstant holomorphic boundary germs, uniformly in* $p \in I$. *Every limit takes the value* 1 *and has nonzero derivative at* 0. *In particular this holds at the prescribed Brjuno multiplier of Proposition 47; the existence of these local boundary germs does not require a Brjuno hypothesis. No common domain containing* $[0, 1]$ *or matching with an inner branch is asserted.*

Proof. The interval certificate of Proposition 47 verifies $|\lambda_+(p)| > 1$ throughout I and $\partial_q|\lambda_+(p+\mathrm{i}q)|^2 < -1$ on the whole enclosing box. It also subdivides I into 100 rational intervals and checks $|\lambda_+(p + 0.00166823\,\mathrm{i})| < 1$ on each one. The intermediate-value theorem and strict derivative give the unique $q_c(p)$, which depends real-analytically on p. The same certificate puts the marked iterate in $H_{p+\mathrm{i}q}$ with a uniform strict margin for the entire closed region $K = \{(p, q) : p \in I,\ 0 \le q \le q_c(p)\}$.

Use the standard punctured sphere coordinate $\xi = \exp(2\pi\mathrm{i}z)$ on the flat cylinder of Proposition 49. Pulling back the physical complex structure of the sewn cylinder gives a Beltrami coefficient $\mu_{p,q}$ on this fixed sphere, with $\|\mu_{p,q}\|_\infty < 1/10$. The explicit formula in that proposition, including its factored endpoint limits, shows that $(p, q) \mapsto \mu_{p,q}$ is continuous in the L^∞ norm on K. The normalised solutions of the Beltrami equation therefore vary continuously on the sphere by [17, Thm. 8]. Their source-coordinate marked points also vary continuously, because $w_{28}(p, q)$ stays uniformly inside the Jordan cell and the flat-cylinder parameterisations are homeomorphisms varying continuously. Three-point normalisation at the two punctures and w_{28} consequently gives conformal coordinates $Q_{p,q}$ that vary locally uniformly in the physical chart near the marked point. The boundary coordinates are defined by this same uniformisation, even when the upper multiplier is neutral. For $q < q_c(p)$ they coincide with the repelling-end quotient coordinates used to transport the classical exterior germ, by uniqueness of the three-point normalisation.

The certified interior margin supplies a fixed physical disc about $w_{28}(p, q)$ in the moving cells. On this compact family of discs, $Q_{p,q}$ and all its w-derivatives depend continuously on (p, q) by Cauchy's formula. Their first derivatives at the marked points never vanish, since each $Q_{p,q}$ is conformal. Compactness of K and the inverse function theorem give a single small disc V on which the inverses of the normalised Abel charts

$$A_{p,q}(w) = \frac{1}{2\pi\mathrm{i}} \log Q_{p,q}(E^{\circ 28}_{b(p+\mathrm{i}q)}(w)), \qquad A_{p,q}(1) = 0,$$

exist and depend continuously on (p, q). For $q < q_c(p)$ their inverses are precisely the transported Kneser germs. Uniform continuity on the compact set K proves the asserted boundary convergence and nondegeneracy, uniformly in p. □

Proposition 51 (An exterior analytic base corridor for the germ)**.** *There are a convex complex θ-neighbourhood U of the closed rectangle*

$$\widehat{K} = \{\theta : |\operatorname{Re}\theta - 0.1000435842| \le 10^{-8},\ |\operatorname{Im}\theta| \le 10^{-2}\}$$

and a disc V about $z = 0$ such that the real-normalised exterior Kneser germ extends jointly holomorphically to $U \times V$. The map $\theta \mapsto b(\theta)$ is injective on U, so its image is a genuine complex base corridor containing the real exterior seed segment and crossing a non-cusp Shell–Thron boundary subarc. The assertion is about the germ on V; it does not give a common regular neighbourhood of $[0, 1]$ or identify the continuation with the inner sewn branch.

Proof. The outward-rounded certificate `certify_exterior_wide_corridor.py` checks on $\widehat{K}$ that the image arc remains an increasing graph below the chord, $E_b^{\circ 28}(1)$ remains inside the Jordan cell, and the singular orbit $E_b^{\circ k}(0)$ lies above its chord for $0 \le k \le 28$. It also gives $|\mu_\theta| < 1/10$ on the entire flat cylinder and $\operatorname{Re} b'(\theta) > 0.05$. All inequalities are strict, so they persist on a slightly larger convex open rectangle U. In the fixed flat-cylinder coordinate $\xi = e^{2\pi i z}$ of Proposition 49, the pulled-back Beltrami coefficient μ_θ depends holomorphically on θ as an L^∞-valued function: its explicit formula is holomorphic in θ pointwise, uniformly bounded away from modulus one, and has uniform Taylor bounds on a slightly larger parameter rectangle. The Ahlfors–Bers holomorphic-parameter theorem [17, Thm. 11] supplies a solution $h_\theta(\xi)$ of the Beltrami equation, normalised at the fixed source points $0, 1, \infty$, that is holomorphic in θ for each fixed ξ.

One must account for the fact that the inverse of the explicit quasiconformal map Ψ_θ from the flat cylinder to the physical Jordan cell is not holomorphic in the parameter at fixed physical w. Put $\xi_\theta(w) = \Psi_\theta^{-1}(w)$ near the marked point. The maps h_θ and Ψ_θ have the same Beltrami coefficient in ξ, while $\Psi_\theta(\xi)$ is holomorphic in θ for fixed ξ. Differentiating $\Psi_\theta(\xi_\theta(w)) = w$ at fixed w gives

$$\partial_{\bar\theta}\xi_\theta(w) + \mu_\theta(\xi_\theta(w))\partial_{\bar\theta}\overline{\xi_\theta(w)} = 0.$$

The same identity and $\partial_{\bar\xi} h_\theta = \mu_\theta \partial_\xi h_\theta$ imply $\partial_{\bar\theta}[h_\theta(\xi_\theta(w))] = 0$. Differentiating instead at fixed θ with respect to $\bar w$ proves that this composition is holomorphic in w. These calculations take place in a smooth interior chart, where all derivatives are classical; hence the composition is jointly holomorphic. They apply also at the holomorphic marked section $w_{28}(\theta) = E_{b(\theta)}^{\circ 28}(1)$.

Divide this composition by its nonzero value at $w_{28}(\theta)$. The resulting coordinate $Q_\theta(w)$ takes the two punctures to $0, \infty$ and the marked point to 1, and is jointly holomorphic in (θ, w) near the marked section. Composing with $E_{b(\theta)}^{\circ 28}$, taking the logarithm with value zero at $w = 1$, and applying the parameter-dependent inverse function theorem yields a jointly holomorphic normalised superfunction on $U \times V$, after shrinking U, V if necessary. At $q = 0$ it is the classical Kneser germ, so uniqueness of analytic continuation identifies it throughout the connected corridor.

Finally the wide certificate checks $\operatorname{Im}(db(p+iq)/dq) > 0.05$ on $\widehat{K}$. Since $db/dq = ib'(\theta)$, this is $\operatorname{Re} b'(\theta) > 0.05$ there. Shrink the convex U so that $\operatorname{Re} b' > 0$ on it. For distinct $\theta_1, \theta_2 \in U$, the integral of b' along their joining segment has positive real part, so $(b(\theta_2) - b(\theta_1))/(\theta_2 - \theta_1) \neq 0$. Thus b is injective on U as claimed. □

Proposition 52 (The real seam anchor and an elliptic crossing test)**.** *For every real $p \in I$ of Proposition 50, the normalised seam function*

$$g_{p,q}(t) = \log |Q_{p,q}(\gamma_{p+\mathrm{i}q}(t))|, \qquad -1 < t < 1,$$

has exactly one zero at $q = 0$, namely $t = 0$, and that zero is simple with $\partial_t g_{p,0}(0) < 0$. For fixed p, a change in the number of seam crossings on $0 \le q \le q_c(p)$ requires a pair (q, t) with

$$g_{p,q}(t) = \partial_t g_{p,q}(t) = 0.$$

On the symmetric wide parameter box, the seam functions also obey $g_{p,-q}(-t) = -g_{p,q}(t)$. In particular, for $p = p_c$, exclusion of such pairs on this compact parameter path proves the hypothesis of Proposition 63 at its prescribed endpoint.

There is an equivalent scalar elliptic test on the fixed flat cylinder. Write $Y = 20 \operatorname{artanh} t$, let $\Psi_{p,q}$ denote the parameterisation of Proposition 49 in the physical w-coordinate, and set

$$v_{p,q}(x, Y) = -\frac{1}{2\pi} \log |Q_{p,q}(\Psi_{p,q}(x + \mathrm{i}Y))|.$$

With $\mu = \mu_{p+\mathrm{i}q}$ from that proposition, v is periodic in x and satisfies, weakly across the seam,

$$\nabla \cdot (\mathsf{A}_\mu \nabla v) = 0, \qquad \mathsf{A}_\mu = \frac{1}{1 - |\mu|^2} \begin{pmatrix} |1-\mu|^2 & -2 \operatorname{Im} \mu \\ -2 \operatorname{Im} \mu & |1+\mu|^2 \end{pmatrix}.$$

The two equations in the displayed crossing condition are exactly $v_{p,q}(0, Y) = \partial_Y v_{p,q}(0, Y) = 0$.

Proof. At a real parameter, complex conjugation preserves the initial region and commutes with E_b. It descends to an antiholomorphic involution of the sewn sphere that exchanges its two punctures and fixes the real marked point. In the three-point coordinate it is $Q \mapsto 1/\overline{Q}$; its fixed locus is $|Q| = 1$. Conjugation sends the seam point labelled t to the one labelled $-t$. The quotient seam is embedded, so a point on it is fixed only when $t = 0$. Near this point the real fixed locus has the horizontal physical tangent, while the chord seam has a nonzero vertical tangent. Conformality of the quotient chart makes their intersection transverse. Since g tends from $+\infty$ to $-\infty$ as t increases, its derivative at the unique zero is negative. For complex-conjugate parameters, the same involution interchanges the two punctures and preserves the marked normalisation, giving the displayed symmetry for g.

The uniform quasiconformal bound and compactness of the parameter region give uniform puncture tails for the normalised $Q_{p,q}$. Thus all zeros of $g_{p,q}$ lie in one compact subinterval of $(-1, 1)$. On that subinterval, the physical seam charts and the normalised uniformisation vary continuously in C^1 with q. If no double equation in the statement occurs, the implicit function theorem continues every simple zero locally in q; compactness and the uniform tails keep their number constant.

For the elliptic formula, choose a local lift $h = (2\pi\mathrm{i})^{-1} \log(Q \circ \Psi)$ on the flat cylinder. Its imaginary part is v, and it satisfies $h_{\bar{z}} = \mu h_z$. Writing $h = u + \mathrm{i}v$ and separating real and imaginary parts gives $\mathsf{A}_\mu \nabla v = (-u_Y, u_x)$. Taking the divergence proves the weak equation, including the periodic seam matching. Finally, $g_{p,q}(t) = -2\pi v_{p,q}(0, 20 \operatorname{artanh} t)$ and $dY/dt = 20/(1 - t^2)$, which gives the stated equivalence. □

Lemma 53 (Uniform positive slopes at the two punctures)**.** *On the wide rectangle $\widehat{K}$ of Proposition 51, let v_θ be the elliptic height function of Proposition 52. Put $\lambda_\pm = \theta\cot\theta \pm \mathrm{i}\theta$ and take the principal logarithms $\ell_\pm = \operatorname{Log}\lambda_\pm$. Uniformly for $\theta \in \widehat{K}$,*

$$\partial_Y v_\theta(0,Y) \longrightarrow \operatorname{Im}\!\left(-\frac{1}{10\ell_+}\right) > 0.98 \quad (Y \to +\infty),$$
$$\partial_Y v_\theta(0,Y) \longrightarrow \operatorname{Im}\!\left(\frac{1}{10\ell_-}\right) > 0.98 \qquad (Y \to -\infty).$$

Thus some finite Y_0 works uniformly on $\widehat{K}$: $\partial_Y v_\theta(0,Y) > 0$ for $|Y| > Y_0$. Every tangency in Proposition 52 therefore lies in $|Y| \le Y_0$.

Proof. The wide interval certificate also verifies $\operatorname{Re}\lambda_\pm > 0.9$, $\operatorname{Im}\lambda_+ > 0.09$, and $\operatorname{Im}\lambda_- < -0.09$ throughout $\widehat{K}$, as well as the two 0.98 slope bounds above. In particular the stated logarithms vary continuously, have imaginary parts of the indicated signs, and never vanish. The factored formula for $\mu_\theta(x,Y)$ is analytic in $t = \tanh(Y/20)$ at $t = \pm 1$. It therefore tends, uniformly in θ and piecewise smoothly in x, to periodic coefficients $\mu_{\theta,\pm}(x)$ as $Y \to \pm\infty$.

Choose the uniformising lift h_θ on the universal cover of the cylinder so that $h_\theta(z+1) = h_\theta(z)+1$ and $h_\theta(0) = 0$. It is a quasiconformal homeomorphism of the plane with coefficient μ_θ and dilatation below $11/9$; its imaginary part differs from v_θ by a constant. For any $\theta_j \to \theta_\infty$ in $\widehat{K}$ and $Y_j \to +\infty$, the maps

$$h_{\theta_j}(z + \mathrm{i}Y_j) - h_{\theta_j}(\mathrm{i}Y_j)$$

fix $0, 1, \infty$ and have a locally uniformly convergent subsequence by quasiconformal compactness. The limit is the unique normalised solution for $\mu_{\theta_\infty,+}$.

This limit is explicit. Since $1 - \tanh(Y/20)$ is asymptotic to $2e^{-Y/10}$, the physical interpolation near L_+, after a nonzero conformal rescaling, tends to $e^{-Y/10}[1 + x(\lambda_+ - 1)]$. The normalised straight-sector solution for its coefficient is

$$h_{\theta,+}(x + \mathrm{i}Y) = \frac{\log[1 + x(\lambda_+ - 1)] - Y/10}{\ell_+}, \qquad 0 \le x \le 1,$$

extended by $h_{\theta,+}(z+1) = h_{\theta,+}(z)+1$. The logarithm along the segment from 1 to λ_+ is chosen to run from 0 to ℓ_+. The segment remains in the right half-plane, and its argument increases strictly; this formula is the normalised quasiconformal homeomorphism with coefficient $\mu_{\theta,+}$. Uniqueness identifies it with every translated limit. At the lower end, $1+\tanh(Y/20) \sim 2e^{Y/10}$ and the same argument gives

$$h_{\theta,-}(x + \mathrm{i}Y) = \frac{\log[1 + x(\lambda_- - 1)] + Y/10}{\ell_-}.$$

The preceding convergence also holds for the tangential Y-derivatives at the seam. Indeed, after dividing the physical coordinate $w - L_\pm$ by the corresponding factor $e^{\mp Y_j/10}$, the analytic gluing maps converge on fixed seam disks to multiplication by $\lambda_\pm$, uniformly in θ_j. In these rescaled physical charts the lifted maps are holomorphic. Local uniform convergence followed by Cauchy's formula gives convergence of their physical derivatives, and the explicit rescaled interpolation gives convergence of their Y-derivatives. Hence the limits of $\partial_Y v$ are the imaginary parts of $\partial_Y h_{\theta,\pm}$ displayed above. Compactness of $\widehat{K}$ makes the convergence uniform. The two positive continuous limiting slopes have a positive minimum there, proving the last assertion. □

Proposition 54 (A certified frozen Abel coordinate)**.** *Use the chord data $D_\theta(t)$ and $A_\theta(t,x)$ of Proposition 49, with $t=\tanh(Y/20)$. Set*

$$L_\theta(Y)=\operatorname{Log}\left(\frac{\theta}{\sin\theta}\right)+\mathrm{i}\theta t,\quad B_\theta=e^{L_\theta}=s_\theta'(t),\quad P_\theta(x,Y)=\frac{A_\theta(t,x)}{20D_\theta(t)}.$$

On the full marked path, including its boundary endpoint, $|P_\theta-\mathrm{i}|<0.09$. On the wider rectangle $\widehat{K}$, the explicit frozen vertical slope

$$c_\theta(Y)=\frac{e^{L_\theta(Y)}-1}{20D_\theta(t)L_\theta(Y)}$$

has $\operatorname{Im}c_\theta(Y)>0.98$ for every real Y. The quotient at $L_\theta=0$ is interpreted by analytic continuation. Define

$$F_\theta(x,Y)=\frac{\operatorname{Log}[1+x(e^{L_\theta(Y)}-1)]}{L_\theta(Y)},\qquad h_\theta^0(x+\mathrm{i}Y)=F_\theta(x,Y)+\int_0^Y c_\theta(s)\,ds.$$

Extending h_θ^0 by $h_\theta^0(z+1)=h_\theta^0(z)+1$ gives an explicit approximate Abel lift. Its exact residual in the pulled-back Beltrami equation is

$$\partial_Y h_\theta^0-P_\theta\partial_x h_\theta^0=\partial_Y F_\theta,\qquad |\partial_Y F_\theta(x,Y)|<10^{-3}\operatorname{sech}^2(Y/20).$$

Its residual has L^2 norm below 0.005 on the marked path. At the seam the approximate height derivative is greater than 0.98. The residual and norm bounds do not yet give pointwise control of the exact uniformising derivative.

Proof. The Beltrami formula of Proposition 49 rearranges exactly to $h_Y=P_\theta h_x$. Since $A_\theta=1+x(B_\theta-1)$, differentiation gives

$$\partial_x F_\theta=\frac{B_\theta-1}{L_\theta A_\theta},\qquad P_\theta\partial_x F_\theta=c_\theta.$$

The displayed residual follows. The factor F_θ equals 0 at $x=0$ and 1 at $x=1$, so the periodic extension is continuous and its seam height derivative is c_θ.

For a bound without cancellation at $L=0$, write

$$F(L,x)=\int_0^1\frac{xe^{sL}}{1+x(e^{sL}-1)}\,ds.$$

Its L-derivative has integrand

$$\frac{sx(1-x)e^{sL}}{[1+x(e^{sL}-1)]^2}.$$

The accompanying Arb script `certify_exterior_frozen_abel.py` verifies $|L|<0.12$ on $\widehat{K}$. Consequently,

$$|F_L|\le\frac{e^{0.12}}{8(2-e^{0.12})^2},\qquad |L_Y|\le\frac{0.102}{20}\operatorname{sech}^2(Y/20).$$

Their product is less than $0.000945\operatorname{sech}^2(Y/20)$. Squaring and integrating uses

$$\int_{\mathbb{R}}\operatorname{sech}^4(Y/20)\,dY=80/3$$

and gives the stated L^2 bound. The same outward-rounded script encloses P_θ on the marked path and c_θ on the wide box; it evaluates $(e^L-1)/L$ by a Taylor polynomial with a rigorous tail, so intervals containing $L=0$ cause no division problem. □

Proposition 55 (An L^4 correction to the frozen coordinate)**.** *On the full marked path let h_θ be the exact uniformising lift with $h_\theta(0) = 0$ and $h_\theta(z+1) = h_\theta(z) + 1$. The coordinate h^0_θ of Proposition 54 satisfies, on the infinite flat cylinder,*

$$\|\partial_x(h_\theta - h^0_\theta)\|_{L^4} < 0.032, \qquad \|\partial_Y(h_\theta - h^0_\theta) - \mathrm{i}\partial_x(h_\theta - h^0_\theta)\|_{L^4} < 0.005.$$

In particular $\|\partial_Y(h_\theta - h^0_\theta)\|_{L^4} < 0.037$. These are global integral bounds; they do not assert a pointwise sign for the exact seam derivative.

Proof. Write $\mathcal{L} = \partial_Y - \mathrm{i}\partial_x$ and $f = \partial_Y F_\theta$. The preceding proposition gives

$$\|P_\theta - \mathrm{i}\|_\infty < 0.09, \qquad \|f\|_4 < 0.000945\left(20\int_{\mathbb{R}} \operatorname{sech}^8 s\, ds\right)^{1/4} = 0.000945(640/35)^{1/4} < 0.002.$$

An exact correction e must solve $\mathcal{L}e = (P_\theta - \mathrm{i})e_x - f$. On the cylinder let B_{per} be the periodised Beurling transform. The planar bound of [1, Thm. 1.2] at exponent 4 gives $\|B\|_4 \le 4.725 < 12$. The same bound $\|B_{\mathrm{per}}\|_4 < 12$ follows by cutting a smooth periodic input of compact vertical support to M horizontal periods, applying the planar estimate, and averaging the output over the interior $(1-\varepsilon)M$ periods. On each fixed-height output strip the omitted kernel tails tend uniformly to zero, since the kernel is $O(|x|^{-2})$ away from the input; first let $M \to \infty$, then $\varepsilon \downarrow 0$, then release the output-height cutoff. Density gives the bound on cylinder L^4.

Since $\mathcal{L} = -2\mathrm{i}\partial_{\bar z}$, the cylinder Cauchy operator has

$$T := \partial_x\mathcal{L}^{-1} = -\frac{I + B_{\mathrm{per}}}{2\mathrm{i}}, \qquad \|T\|_4 < 6.5.$$

Solve for u as the unique fixed point in cylinder L^4 of $u = T[(P_\theta - \mathrm{i})u - f]$, because $6.5(0.09) = 0.585 < 1$. It obeys

$$\|u\|_4 < \frac{6.5(0.002)}{1 - 0.585} < 0.032, \qquad \|g\|_4 \le 0.09\|u\|_4 + \|f\|_4 < 0.005.$$

For completeness, construct e from this fixed point by applying the periodic Cauchy potential to $g = (P_\theta - \mathrm{i})u - f$. For compactly supported smooth inputs, its gradient is the usual Cauchy and Beurling pair. Approximation in L^4, the bounds above, and local Morrey compactness give a periodic $e \in W^{1,4}_{\mathrm{loc}}$ with $\nabla e \in L^4$ and $e(0) = 0$. The identity $e_x = u$ passes to the limit; uniqueness modulo constants follows because a periodic entire holomorphic function with derivative in L^4 is constant.

Set $H = h^0_\theta + e$. The exact equation $H_Y = P_\theta H_x$ says that H has the same Beltrami coefficient as h_θ, of modulus below $1/10$. The positive bound $\operatorname{Im} c_\theta > 0.98$ and boundedness of F_θ show that $\operatorname{Im} h^0_\theta$ tends to $+\infty$ and $-\infty$ at the corresponding cylinder ends. The L^4 gradient bound makes $e = o(|Y|)$ uniformly in $0 \le x \le 1$: its horizontal mean grows at most $O(|Y|^{3/4})$ by Hölder's inequality, while the oscillation on each unit-height strip is uniformly bounded by Morrey's inequality. Hence H is proper. The homotopy $(1-s)z + sH(z)$ is proper for $0 \le s \le 1$ and has degree one. The weak quasiregular factorisation (or the quasiconformal chain rule followed by Weyl's lemma) makes $H \circ h_\theta^{-1}$ an entire holomorphic map. It is proper of degree one, so affine; the period relation forces its slope to be 1, and the value at 0 removes the constant. Therefore $H = h_\theta$, proving the claimed bounds for the exact lift. □

Proposition 56 (Certified pointwise seam monotonicity)**.** *On the closed parameter rectangle $|\operatorname{Re}\theta - 0.1000435842| \le 10^{-8}$, $|\operatorname{Im}\theta| \le 0.01$, including both sides of its non-cusp boundary subarc, the exact uniformising lift satisfies*

$$\sup_{z\in\mathbb{C}} |\partial_Y(h_\theta - h^0_\theta)(z)| < 0.041, \qquad \partial_Y \operatorname{Im} h_\theta(\mathrm{i}Y) > 0.93 \quad (Y \in \mathbb{R}).$$

Consequently the image of the chord seam under the three-point-normalised coordinate Q_θ crosses $|Q| = 1$ exactly once and transversely, for every parameter in this rectangle.

Proof. Let $e = h_\theta - h_\theta^0$ and $f = \partial_Y F_\theta$. The same outward-rounded interval computation as in Proposition 54, now differentiating the factored interpolation remainder in t, gives throughout the wider rectangle $\|P_\theta - \mathrm{i}\|_\infty < 0.18$ and $\|\partial_Y P_\theta\|_\infty < 0.004$. Here the derivative of its order-n polynomial has modulus at most $n^2/4$ on $[-1,1]$; starting at $n = 12$, its Taylor tail is bounded by $36(0.102)^{10}/[12!(1 - 0.102/10)]$. For the residual, put $E = e^{0.12}$ and $a = 2 - E$. Differentiating the integral formula for F_L in the preceding proof gives

$$|F_L| \le \frac{E}{8a^2}, \qquad |F_{LL}| \le \frac{E}{12a^2} + \frac{E^2}{6a^3}.$$

Since $|L_Y| \le (0.102/20)\operatorname{sech}^2(Y/20)$ and $|L_{YY}| \le (0.102/200)\operatorname{sech}^2(Y/20)$, the interval evaluation yields

$$|f_Y| < 0.00012\operatorname{sech}^2(Y/20), \qquad \|f_Y\|_4 < 0.00025.$$

The planar Beurling bound $\|B\|_4 \le 4.725 < 6$ quoted in Proposition 55 transfers to the periodic cylinder by the same cutoff-and-averaging argument used there. Hence the cylinder operator $T = \partial_x \mathcal{L}^{-1}$ has norm below 3.5. The fixed-point construction of that proposition now applies on the wider rectangle because $3.5(0.18) = 0.63 < 1$. It gives

$$\|e_x\|_4 < \frac{3.5(0.002)}{1 - 0.63} < 0.019, \quad \|\mathcal{L}e\|_4 < 0.18(0.019) + 0.002 < 0.0055, \quad \|e_Y\|_4 < 0.025.$$

The same properness and degree-one argument identifies this correction with the exact uniformising lift.

Take a vertical difference quotient $e^\delta(x,Y) = [e(x,Y+\delta) - e(x,Y)]/\delta$. Its x derivative belongs to cylinder L^4 for each $\delta \neq 0$, and it satisfies

$$\mathcal{L}e^\delta = (P_\theta - \mathrm{i})(e^\delta)_x + g^\delta, \quad g^\delta = P_\theta^\delta e_x(x, Y+\delta) - f^\delta.$$

The bounds above give $\|g^\delta\|_4 < 0.004(0.019) + 0.00025 < 0.00033$. The same cylinder operator T as before therefore gives

$$\|(e^\delta)_x\|_4 < \frac{3.5(0.00033)}{1 - 0.63} < 0.0032, \qquad \|(e^\delta)_Y\|_4 < 1.18(0.0032) + 0.00033 < 0.0042.$$

Passing to weak limits of the difference quotients proves that $v = e_Y \in W^{1,4}(\mathbb{C}/\mathbb{Z})$ and $\|v_x\|_4 < 0.0032$, $\|v_Y\|_4 < 0.0042$; the preceding bound gives $\|v\|_4 < 0.025$.

For clarity, a disk of radius $r < 1/2$ on the flat cylinder obeys the following pointwise Sobolev estimate, obtained by averaging along radial segments and applying Hölder's inequality:

$$|v(z_0)| \le (\pi r^2)^{-1/4}\|v\|_{L^4(B_r(z_0))} + (3/2)^{3/4}(2\pi)^{-1/4} r^{1/2}\|\nabla v\|_{L^4(B_r(z_0))}.$$

At $r = 1/4$ its two coefficients are below 1.51 and 0.43. Hence $|e_Y(z_0)| < 1.51(0.025) + 0.43(0.0032 + 0.0042) < 0.041$ for every z_0. The $W^{1,4}$ representative is continuous; on the seam it equals the classical tangential derivative supplied by the analytic physical chart. Since $h_Y^0(\mathrm{i}Y) = c_\theta(Y)$ and $\operatorname{Im} c_\theta(Y) > 0.98$, the stated strict derivative bound follows.

The seam height $\operatorname{Im} h_\theta(\mathrm{i}Y)$ tends to the opposite infinite limits at the two ends by the proof of Proposition 55. It is therefore strictly increasing through every real level. The three-point-normalised Q_θ differs from $\exp(2\pi\mathrm{i}h_\theta)$ by a nonzero constant factor, so $|Q_\theta| = 1$ is one such height level. It is crossed exactly once, with nonzero derivative. □

Proposition 57 (An attracting overlap of the exterior chord)**.** *At the exact interior parameter* $\theta_0 = 0.1000435842 + 0.01\mathrm{i}$*, the upper fixed point* $L_+(\theta_0)$ *is attracting and the lower fixed point is repelling. The entire physical chord subarc* $\gamma_{\theta_0}([-13/100, 1])$ *lies in the immediate attracting basin of* $L_+(\theta_0)$*. This supplies a common physical chart region for the exterior quotient and the upper attracting dynamics; it does not identify the exterior solution with the inner sewn solution. Moreover, the unique point of the chord mapped by the marked quotient coordinate to* $|Q_{\theta_0}| = 1$ *lies on this certified attracting subarc, at a parameter* $t_* > 0$*. Consequently there is a noninteger real height* $z_* \in (0, 1)$ *and product neighbourhoods of* (θ_0, z_*) *on which the continued exterior superfunction takes values in the attracting basin and admits a jointly holomorphic upper Koenigs coordinate.*

Proof. The outward-rounded Arb certificate `certify_interior_chord_overlap.py` uses exact rational coordinates for θ_0 and partitions the indicated real t-interval into 113 cells of width $1/100$. Put $a = \log b(\theta_0)$, $L = L_+(\theta_0)$, and $\delta = 1/100$. It verifies $|aL|e^{|a|\delta} < 1$ and $|aL_-| > 1$. For a cell centred at t_j, let $c_0 = \gamma_{\theta_0}(t_j)$ and $\rho_0 = |\mathrm{i}r \sin\theta_0|/200$. The rigorous recurrence

$$c_{n+1} = e^{ac_n}, \qquad \rho_{n+1} = |c_{n+1}|(e^{|a|\rho_n} - 1)$$

encloses the whole cell image under $E_b^{\circ n}$ in $\overline{B(c_n, \rho_n)}$. For each of the 113 cells the certificate finds an $n \le 505$ with $|c_n - L| + \rho_n < \delta$. The disk $B(L, \delta)$ is strictly forward invariant by the first verified inequality, because $|E_b(L+u) - L| \le |aL|e^{|a|\delta}|u|$. Thus the whole subarc belongs to the attracting basin. It is connected and contains L, so it lies in the immediate component.

For the final assertion, write the marked iterate in the flat interpolation coordinate as $\zeta_{28} = t_m + x_m(s_{\theta_0}(t_m) - t_m)$, $0 < x_m < 1$, and put $Y_m = 20\,\mathrm{artanh}\, t_m$. The same Arb certificate checks $\mathrm{Re}\,\zeta_{28} > 0.15$ and $\mathrm{Re}(s_{\theta_0}(t) - t) < 0.01$ for every $-1 \le t \le 1$; hence $t_m > 0.14$ and $Y_m > 2.8$. The formula for F_x in Proposition 54, together with $|L_{\theta_0}(Y)| < 0.12$, gives

$$|F_x - 1| \le \frac{(e^{0.12} - 1 - 0.12)/0.12 + (e^{0.12} - 1)}{2 - e^{0.12}} < 0.23.$$

The exact equation $e_Y = P_\theta e_x - f$, the wide-box bounds $|P_\theta - \mathrm{i}| < 0.18$, $|f| < 0.001$, and $|e_Y| < 0.041$ give $|e_x| < 0.052$. Thus $|h_{\theta_0,x} - 1| < 0.3$ on the flat cylinder. Since $h_{\theta_0}(0) = 0$ and its seam-height slope exceeds 0.93, we have

$$\mathrm{Im}\, h_{\theta_0}(x_m + \mathrm{i}Y_m) > 0.93Y_m - 0.3 > 0.$$

Three-point normalisation at the mark divides $\exp(2\pi\mathrm{i}h_{\theta_0})$ by its value there. Its unit-circle level on the seam therefore has positive height. Strict seam monotonicity makes its unique crossing satisfy $Y_* > 0$, hence $t_* = \tanh(Y_*/20) > 0$. The seam point represents a noninteger height $z_* \in (0, 1)$, because the marked point is in the interior of the Jordan cell. The developed value at z_* is a preimage under $E_b^{\circ 28}$ of either $\gamma_{\theta_0}(t_*)$ or its forward image. Both lie in the immediate attracting basin, so that value lies in the full attracting basin. Finite entry into the invariant disk persists on product neighbourhoods of (θ_0, z_*); Proposition 78 then gives the stated joint Koenigs coordinate. □

Proposition 58 (A certified interior sewing circle)**.** *At $\theta_0 = 0.1000435842 + 0.01\mathrm{i}$, the entire lower regular curve $S_{\theta_0}([0,1])$ lies in the full attracting basin of $L_+(\theta_0)$ and avoids every backward preimage of L_+. Let σ_+ be the upper Koenigs coordinate, normalised by $\sigma_+'(L_+) = 1$, and let $\ell_+ = \operatorname{Log}\lambda_+$ be the branch continued from the inner real axis. The upper time*

$$T(x) = \frac{1}{\ell_+}\log\frac{\sigma_+(S_{\theta_0}(x))}{\sigma_+(1)}$$

has a branch along $[0,1]$ for which

$$0.94 < \operatorname{Re} T'(x) < 1.06, \qquad |\operatorname{Im} T'(x)| < 0.056, \qquad T(1) - T(0) = 1.$$

Consequently $T([0,1])/\mathbb{Z}$ is an embedded analytic Jordan circle in the upper cylinder. With $\varpi = 2\pi\mathrm{i}/\ell_+$, its shifted image $(T+\varpi)([0,1])$ lies in $\operatorname{Im} w < -2.5$. It therefore defines an analytic two-chart sewing near this strictly interior parameter, with the normalising upper point $w = 0$ above the seam. The resulting locally normalised sewn solution is jointly holomorphic in the base and height on its regular domains. This local family has not been identified with the continuation of the real inner sewn family or with the exterior family.

Proof. The accompanying outward-rounded Arb certificates use the exact rational coordinates of θ_0. Write $a = \log b$, $L_- = L_-(\theta_0)$, $\mu = aL_-$, $M = |\mu|$, $A = |a|$, and $f(u) = L_-(e^{au} - 1)$. They verify $1.01 < M < 1.02$, $A < 0.38$, and, with $\rho = 1/3000$, $\exp(4A\rho/(M-1)) < 2$. The normalised lower Poincaré function is the limit of

$$P_N(t) = L_- + f^{\circ N}(t\mu^{-N}).$$

For $|t| \le \rho$, elementary exponential bounds give $|f^{\circ j}(v)| \le 4\rho M^{j-N}$ and $|(f^{\circ N})'(v)| \le 2M^N$ on the line segment joining $t\mu^{-N}$ to $f(t\mu^{-(N+1)})$. Indeed the latter endpoint differs from the former by at most $\frac{1}{2}A\rho^2 M^{-2N-1}e^{A\rho}$, and both have modulus below $2\rho M^{-N}$. Summing the resulting telescoping differences gives

$$\sup_{|t|\le\rho}|P(t) - P_N(t)| \le \frac{A\rho^2 e^{A\rho}}{M-1}M^{-N} < 39\rho^2(100/101)^N.$$

At $N = 1000$ this last bound is below $2.1 \cdot 10^{-10}$.

Put $m = 700$ and $t(z) = -\exp((z-m)\operatorname{Log}\mu)$. The certificates check $|t(z)| < \rho$ on $0 \le z \le 1$ and use $S(z) = E_b^{\circ m}(P(t(z)))$. They partition $[0,1]$ into 32 rational cells. On each cell the initial t-disk has radius $|t(z_j)|(e^{|\operatorname{Log}\mu|/64} - 1)$. For a disk of radius r centred at u under f, the new radius is at most $|L_-e^{au}|(e^{Ar} - 1)$; under E_b the analogous radius is $|E_b(u)|(e^{Ar} - 1)$. Adding the above uniform Poincaré tail after 1000 steps, the certificates find that every cell enters $B(L_+, 1/100)$ within a further 427 iterates after the 700 shifts. The disk is strictly invariant since $|aL_+|e^{A/100} < 1$. At entry each enclosure also excludes L_+; since $|a|/100 < 2\pi$, no later point of the invariant disk can map exactly to L_+. Thus $\sigma_+(S(x)) \ne 0$ throughout the interval.

For the slope, the second certificate propagates derivative disks on a finer 64-cell partition, using Cauchy's estimate for the Poincaré tail on $|t| < \rho$. Alongside a value disk $B(c,r)$ and a derivative disk $B(d,s)$, one exponential step uses

$$c_{\mathrm{new}} = e^{ac}, \qquad d_{\mathrm{new}} = ac_{\mathrm{new}}d, \qquad s_{\mathrm{new}} = |ac_{\mathrm{new}}|((e^{Ar} - 1)|d| + e^{Ar}s).$$

The lower Poincaré steps have the analogous formula with $c_{\mathrm{new}} = f(c)$. If $u_n = E_b^{\circ n}(S(z)) - L_+$ and $d_n = (E_b^{\circ n} \circ S)'(z)$, then

$$D_n = \frac{d_n}{u_n\ell_+} \longrightarrow T'(z), \qquad \frac{D_{n+1}}{D_n} = \frac{au_ne^{au_n}}{e^{au_n} - 1}.$$

On $|u| < 1/100$, the last factor differs from 1 by at most $A|u|$; subsequent displacements contract by $q = |aL_+|e^{A/100} < 1$. Thus the multiplicative tail after any certified iterate is bounded by $\exp(A|u_n|/(1-q)) - 1$. The outward-rounded derivative and tail disks after 1200 upper iterates give the stated slope inequalities.

Since $S(1) = E_b(S(0))$, logarithm continuation gives $T(1) - T(0) = 1 + k\varpi$ for some integer k. The certified bound $\operatorname{Re}\varpi > 50$ and the integral of the slope over $[0,1]$ force $k = 0$. For the seam location, compare the same-depth ratio $\mathcal{R}_n = (E_b^{\circ n}(S(0)) - L_+)/(E_b^{\circ n}(1) - L_+)$. On the invariant disk, the factor $h(u) = (e^{au} - 1)/(au)$ gives the ratio-tail bound

$$\left|\frac{\mathcal{R}_\infty}{\mathcal{R}_n} - 1\right| \le \exp\left(\frac{A(|u_n^{(0)}| + |u_n^{(1)}|)}{1-q}\right) - 1.$$

The certificate encloses the principal logarithm of this ratio, including its tail, and locates $\operatorname{Im}(T(0) + \varpi) < -2.5 - 0.056$. The imaginary-slope bound then places the entire shifted circle below -2.5. Positive real slope and degree one make it an embedded circle. A thin analytic collar persists for parameters near θ_0, and the three-section compactification and normalisation argument of Proposition 16 produces the stated local holomorphic sewn family. □

Proposition 59 (A transverse crossing on the straight interior path)**.** *Set* $\theta_\dagger = 23541/2560000 + 0.01\mathrm{i}$ *and let* $S = S_{\theta_\dagger}$ *be the lower regular superfunction. There are distinct real heights*

$$x \in 0.04563349481384156132545513 + [-10^{-4}, 10^{-4}],$$
$$y \in 0.27675003438644006456213 32 + [-10^{-4}, 10^{-4}]$$

for which $S(x) = S(y)$*. This crossing is transverse. The physical subarc* $S([0.04, 0.28])$ *lies in a simply connected disc contained in the upper attracting chart and avoiding* L_+*. Hence the same two heights have equal upper Koenigs times on one logarithmic branch. Any upper-time sewing curve containing this subarc therefore self-intersects at* $\theta_\dagger$*. Moreover,* $\operatorname{Re} T'(0) < -0.1$*. In particular, the straight path* $\theta = p + 0.01\mathrm{i}$ *cannot transport the real inner sewing by maintaining this lower regular circle as an embedded, positive-slope seam at every parameter. This does not rule out continuation by a different seam or parameter path.*

Proof. Both accompanying certificates use the exact rational parameter and the lower Poincaré limit from the preceding proposition. For the crossing, `certify_interior_self_intersection.py` takes $\rho = 1/2000$, Poincaré depth 4500, and 800 further exponential iterates. Its uniform Poincaré value tail is bounded by

$$\frac{A\rho^2 e^{A\rho}}{M-1} M^{-4500}, \qquad A = |\log b|, \quad M = |E_b'(L_-)|,$$

and Cauchy's estimate bounds the derivative tail. It propagates value and derivative discs over the two stated height intervals. For $F(x,y) = S(x) - S(y)$, a rational preconditioner of the real 2×2 Jacobian gives a Krawczyk image within $[-1.3 \cdot 10^{-6}, 1.3 \cdot 10^{-6}] \times [-10^{-6}, 10^{-6}]$ relative to the box centre. The two Jacobian defect row sums are < 0.1. Thus the Krawczyk map contracts the box strictly into itself; its unique zero is transverse, and the two height intervals are disjoint.

Eight more rational height cells enclose $S([0.04, 0.28])$ in $B(2.69185 + 0.019515\mathrm{i}, 0.001)$; the maximum certified distance from its centre is $< 1.1 \cdot 10^{-6}$. This disc lies in $B(L_+, 0.02)$ and avoids L_+. The latter disc is invariant since $|E_b'(L_+)|e^{A/50} < 1$; the upper Koenigs coordinate has no zero there except L_+, by the local linearisation, the absence of other preimages in this small disc, and Hurwitz's theorem. Its logarithm therefore has a single branch on the smaller

disc containing the whole subarc. Because $S(x) = S(y)$, that branch gives $T(x) = T(y)$; the nonzero conformal multiplier from S' to T' preserves transversality.

Independently, `certify_interior_path_fold.py` propagates the value and derivative discs at height 0 to the invariant upper disc and bounds the remaining Koenigs product. It gives $\operatorname{Re} T'(0) < -0.11272 < -0.1$. □

Proposition 60 (The lower sewing circle in the exterior quotient)**.** *At $\theta_0 = 0.1000435842 + 0.01\mathrm{i}$, let H_{θ_0} be the exterior initial Jordan cell and set $w_n(x) = E_{b(\theta_0)}^{\circ n}(S_{\theta_0}(x))$. There is a unique $x_* \in (0,1)$ such that $w_{76}(x_*)$ lies on the chord boundary of H_{θ_0}. For $0 \le x < x_*$ one has $w_{77}(x) \in H_{\theta_0}$, and for $x_* < x \le 1$ one has $w_{76}(x) \in H_{\theta_0}$. At x_* these two representatives are the boundary pair $w_{77}(x_*) = E_b(w_{76}(x_*))$; at the other join, $w_{77}(0) = w_{76}(1)$. The resulting closed curve is an embedded essential analytic circle in the exterior quotient cylinder. Its annular physical chart extends jointly holomorphically to parameters near θ_0. Thus the local lower sewing circle and the continued exterior quotient have a common annular physical chart. This does not identify their global uniformising coordinates.*

Proof. Use the chord coordinate $\zeta(w) = (w/r - \cos\theta_0)/(\mathrm{i}\sin\theta_0)$. The chord is $[-1,1]$, its exponential image is $s([-1,1])$, and direct substitution gives the exact identity

$$\zeta(E_b(w)) = s(\zeta(w)), \qquad s(t) = \frac{e^{\mathrm{i}\theta_0 t} - \cos\theta_0}{\mathrm{i}\sin\theta_0}.$$

On the central rectangle $\mathcal{R} = [0.45, 0.55] + \mathrm{i}[-0.05, 0.10]$, the outward-rounded certificate verifies $\operatorname{Re} s' > 0.98$ and $\operatorname{Im} s(t) < -0.03$ for real $t \in [0.45, 0.55]$, while $\operatorname{Re} s(0.45) < 0.48 < 0.52 < \operatorname{Re} s(0.55)$. Thus s is injective on this convex rectangle, its real arc lies below the chord, and for $\zeta \in \mathcal{R} \cap s(\mathcal{R})$

$$\zeta \in \zeta(H_{\theta_0}) \quad \Longleftrightarrow \quad \operatorname{Im}\zeta < 0 < \operatorname{Im} s^{-1}(\zeta).$$

The inverse here is the branch on $s(\mathcal{R})$.

The same certificate uses the Poincaré approximation and explicit tail from Proposition 58, followed by 75–77 exponential iterates. It partitions $[0,1]$ into 64 rational cells and propagates both value and derivative disks. Throughout the interval it verifies that $\zeta(w_{75}), \zeta(w_{76}), \zeta(w_{77})$ belong to $\mathcal{R}$, all with real parts in $(0.48, 0.52)$, and

$$\operatorname{Im}\zeta(w_{75}) > 0.03, \qquad \operatorname{Im}\zeta(w_{77}) < -0.005, \qquad \frac{d}{dx}\operatorname{Im}\zeta(w_{76}) < -0.03.$$

Its endpoint disks put $\operatorname{Im}\zeta(w_{76}(0)) > 0$ and $\operatorname{Im}\zeta(w_{76}(1)) < 0$, so the middle curve crosses the chord exactly once and transversely. The criterion above, applied to $w_{77} = E_b(w_{76})$ before the crossing and to $w_{76} = E_b(w_{75})$ afterwards, proves the asserted membership in H_{θ_0}.

For both $n = 76, 77$ the certificate also gives $d\operatorname{Re}\zeta(w_n(x))/dx > 10^{-4}$. Since $w_{77}(0) = w_{76}(1)$, the two interior arcs have disjoint real-coordinate ranges except at this common endpoint. The only other join is the single identified boundary pair at x_*. Hence the quotient curve is simple and analytic, and crossing the glued boundary once makes it essential. All inequalities are strict and all arcs avoid the two fixed endpoints, so compactness gives a jointly holomorphic annular collar for nearby parameters. □

Corollary 61 (Two distinct return circles in the exterior quotient)**.** *At the same parameter θ_0, the lower regular circle has a second embedded essential return circle in the exterior quotient, represented by $w_{15}(x)$ before a unique chord crossing of $w_{14}(x)$ and by $w_{14}(x)$ after it. This circle is disjoint from the return circle of Proposition 60. In particular, a forward return of the lower circle to the exterior initial cell does not by itself specify a canonical comparison map between the two quotient cylinders.*

Proof. The independent outward-rounded certificate `certify_interior_early_return.py` uses the same Poincaré tail and 64 rational cells as the preceding proposition. On $[0.25, 0.33] + \mathrm{i}[-0.10, 0.08]$ it verifies $\operatorname{Re} s' > 0.98$ and $\operatorname{Im} s(t) < -0.03$ for real $t \in [0.25, 0.33]$. All three curves $\zeta(w_{13})$, $\zeta(w_{14})$, and $\zeta(w_{15})$ lie in this rectangle, with real parts in $(0.26, 0.31)$, and the certificate gives

$$\operatorname{Im}\zeta(w_{13}) > 0.005, \qquad \operatorname{Im}\zeta(w_{15}) < -0.03, \qquad \frac{d}{dx}\operatorname{Im}\zeta(w_{14}) < -0.03.$$

The endpoint signs of $\operatorname{Im}\zeta(w_{14})$ are positive and negative. Thus w_{14} crosses the chord once, and the same initial-cell criterion as above puts w_{15} before the crossing and w_{14} afterwards inside H_{θ_0}. For $n = 14, 15$, $d\operatorname{Re}\zeta(w_n)/dx > 10^{-4}$; the endpoint identity $w_{15}(0) = w_{14}(1)$ and the boundary gluing make a simple essential circle. Its real-coordinate strip $(0.26, 0.31)$ is disjoint from the strip $(0.48, 0.52)$ certified for the earlier proposition, even after boundary identification. □

Corollary 62 (Degree-one regular germs at the exterior ends)**.** *Let Q_{θ_0} be the three-point-normalised coordinate of the exterior quotient at θ_0. On the corresponding quotient end germs, the lower regular superfunction S and the upper regular superfunction R satisfy*

$$Q_{\theta_0}(S(z))^{-1} = C_- e^{-2\pi \mathrm{i} z}(1 + O(e^{-2\pi \mathrm{i} z})) \quad (\operatorname{Im} z \to -\infty),$$

and

$$Q_{\theta_0}(R(z)) = C_+ e^{2\pi \mathrm{i} z}(1 + O(e^{2\pi \mathrm{i} z})) \quad (\operatorname{Im} z \to +\infty),$$

where $C_\pm \neq 0$. These are degree-one local puncture maps. Together with a chosen return circle from Proposition 60 or Corollary 61, this locates the remaining chart-identification problem in the compact middle bands between that circle and the two puncture germs. It does not assert that the two local identifications continue through those bands or agree with one another.

Proof. Let τ_- be the Koenigs coordinate at the repelling lower fixed point, normalised by $\tau_-'(L_-) = 1$, and let $\mu = E_b'(L_-)$. On a deep lower time half-cylinder, $\tau_-(S(z)) = -\mu^z$ exactly. The lower exterior puncture coordinate is, on its logarithmic sector quotient,

$$\xi_- = \exp\left(-\frac{2\pi \mathrm{i}}{\log \mu}\log(-\tau_-(w))\right).$$

The choice of logarithmic branch is fixed by the lower regular curve; multiplication by μ changes its exponent by an integer multiple of $2\pi\mathrm{i}$, so ξ_- descends to the glued end. Consequently $\xi_-(S(z)) = e^{-2\pi \mathrm{i} z}$. The exterior quotient was compactified by this simple puncture coordinate, so its global coordinate has the expansion $Q_{\theta_0}^{-1} = C_-\xi_-(1 + O(\xi_-))$ with $C_- \neq 0$.

At the upper end let σ_+ be the attracting Koenigs coordinate and $\lambda_+ = E_b'(L_+)$. On a deep upper time half-cylinder, $\sigma_+(R(z)) = \sigma_+(1)\lambda_+^z$. The upper exterior puncture coordinate

$$\xi_+ = \exp\left(\frac{2\pi \mathrm{i}}{\log \lambda_+}\log\frac{\sigma_+(w)}{\sigma_+(1)}\right)$$

therefore obeys $\xi_+(R(z)) = e^{2\pi \mathrm{i} z}$. The global coordinate has a simple zero there, giving $Q_{\theta_0} = C_+\xi_+(1 + O(\xi_+))$ with $C_+ \neq 0$. The local end degrees are one. The statements concern germs in the respective quotient end charts; they do not supply analytic continuation from the middle circle to those germs. □

Proposition 63 (A single-crossing criterion for a full height segment)**.** *Fix the prescribed parameter p_c of Proposition 47 and abbreviate $Q_q = Q_{p_c,q}$ for the three-point-normalised coordinates of Proposition 50. Suppose the curve $t \mapsto Q_{q_c}(\gamma_{q_c}(t))$, $-1 < t < 1$, meets the unit circle exactly once, and the meeting is transverse; equivalently,*

$$\log |Q_{q_c}(\gamma_{q_c}(t))| = 0$$

has exactly one zero t_* and its derivative there is nonzero. Then there are a complex parameter neighbourhood U_* of $\theta_c = p_c + \mathrm{i}q_c$ and a connected open height neighbourhood $W \supset [0,1]$ such that the exterior germ of Proposition 51 extends jointly holomorphically to $U_* \times W$. In particular, it crosses a non-cusp boundary subarc and takes the values $F_\theta(0) = 1$, $F_\theta(1) = b(\theta)$ there. The single-crossing hypothesis is verified by Proposition 56.

Proof. The normalised coordinates Q_θ and their inverses vary continuously in C^1 on compact collars away from the two punctures by Propositions 50 and 51. In puncture coordinates, Q_θ tends uniformly to 0 at the upper end and to ∞ at the lower end. Thus the unique transverse intersection with $|Q_{\theta_c}| = 1$ persists, and remains the only such intersection, throughout a sufficiently small complex neighbourhood U_* of θ_c.

The marked point $Q_\theta = 1$ and the seam point split $|Q_\theta| = 1$ into two compact subarcs. On each subarc the inverse of Q_θ has a representative in H_θ. Starting at the interior marked point $w_{28}(\theta)$, pull the first representative back by the conformal branch Φ_θ of Proposition 48. At the unique seam crossing the representative passes from the image arc to the chord; this is the orientation of the positive Abel deck generator. Continue on the second subarc using $E_{b(\theta)} \circ \Phi_\theta$. The seam identity in that proposition makes the two physical charts agree on their common analytic collar. They therefore define a holomorphic developing map at every height $z \in [0,1]$ with $Q_\theta = e^{2\pi \mathrm{i} z}$, taking the values 1 and $E_{b(\theta)}(1) = b(\theta)$ at its endpoints.

The path is compact and avoids both punctures. Finitely many of the collars just described cover it. Their domains, transition maps and physical developing maps vary jointly holomorphically in the parameter by Proposition 51 and the finite-log construction of Proposition 48. The seam identity of that proposition extends to the complex parameter neighbourhood by the identity theorem, since both sides are holomorphic in θ there and agree along the real q path. After shrinking U_* and these finitely many charts, they cover one connected complex neighbourhood W of $[0,1]$ for every $\theta \in U_*$. The analytic overlap identities provide a jointly holomorphic family on $U_* \times W$. Near $z = 0$ it is the previously transported normalised Kneser germ, so the identity theorem identifies the family throughout W. □

Corollary 64 (A full exterior height corridor)**.** *At the prescribed non-cusp Brjuno parameter θ_c there are a complex parameter neighbourhood U_* and a connected open height neighbourhood $W \supset [0,1]$ on which the exterior Kneser family continued from real $b > \eta$ is jointly holomorphic and satisfies $F_\theta(0) = 1$ and $F_\theta(1) = b(\theta)$. The parameter neighbourhood crosses the Shell–Thron boundary. No comparison with the inner sewn family K^W is asserted.*

Proof. Proposition 56 verifies the unique transverse seam crossing at θ_c. Apply Proposition 63. □

Corollary 65 (Full exterior height along an interior-reaching path)**.** *Put $p_0 = 0.1000435842$ and $J = \{p_0 + \mathrm{i}q : 0 \le q \le 0.01\}$. There are a connected complex parameter neighbourhood $\Omega \supset J$ and a connected height neighbourhood $W_J \supset [0,1]$ such that the exterior Kneser branch continued from $q = 0$ is jointly holomorphic on $\Omega \times W_J$, the map $\theta \mapsto b(\theta)$ is injective on Ω, and the family satisfies $F_\theta(0) = 1$, $F_\theta(1) = b(\theta)$. In particular its full-height branch is defined at the strictly interior parameter $\theta_0 = p_0 + 0.01\mathrm{i}$ of Proposition 57.*

Proof. The wide-box certificate of Proposition 51 supplies the marked pullback cell, its analytic seam identity, and a jointly holomorphic normalised germ throughout a neighbourhood of J. Proposition 56 supplies a unique transverse seam crossing at every point of J. The compact-arc development in the proof of Proposition 63 therefore applies locally at each such parameter; it uses the Jordan cell and its puncture tails, not the sign of $|\lambda_+| - 1$. Finitely many resulting parameter charts cover J. Their full-height families agree on overlaps because all extend the same jointly holomorphic germ at $z = 0$; the identity theorem then extends the agreement along $[0, 1]$. Choose a thin connected parameter neighbourhood Ω of J inside their union and inside the injectivity domain U of Proposition 51, and a common thin connected tube W_J about $[0, 1]$ inside their finitely many height domains. Patching gives the assertion. □

Corollary 66 (The exterior zero label at the boundary)**.** *After shrinking U_* in Corollary 64, the exterior family extends jointly holomorphically to $U_* \times W_-$ for a connected height neighbourhood $W_- \supset [-1, 1]$. Throughout this parameter neighbourhood,*

$$F_\theta(-1) = 0, \qquad \partial_z F_\theta(-1) = \frac{\partial_z F_\theta(0)}{\log b(\theta)} \neq 0.$$

Its next local backward branch at $z = -2$ has the logarithmic monodromy $2\pi\mathrm{i}/\log b(\theta)$ of Proposition 84. Thus the exterior boundary germ has the same zero backward label as the real inner sewn family; this label alone does not identify the two families.

Proof. The proof of Proposition 63 applies locally at every point of the compact marked path, since Proposition 56 gives the same strict crossing throughout it. The resulting full-height charts agree on overlaps: they contain the transported germ at $z = 0$, and the identity theorem extends that agreement along $[0, 1]$. They therefore give a continuous family of height paths $F_\theta([0, 1])$ from the real seed to θ_c.

These paths avoid 0. Indeed, each developing representative lies either in $\Phi_\theta(H_\theta)$ or in $E_{b(\theta)}(\Phi_\theta(H_\theta))$. The second set omits 0 because the exponential map does. If $\Phi_\theta(w) = 0$ for $w \in \overline{H_\theta}$, then $w = E_{b(\theta)}^{\circ 28}(0)$, contrary to the certified singular-orbit clearance of Proposition 48. The compact height paths and the strict clearance allow a common small simply connected tube $V \supset [0, 1]$, after restricting the parameter to a neighbourhood of θ_c, on which F_θ has no zeros.

Choose its jointly holomorphic logarithm $L_\theta(z)$ on $U_* \times V$ by $L_\theta(0) = 0$, and choose the local holomorphic branch $\ell(\theta) = \log b(\theta)$ from the real exterior seed. Since $F_\theta(1) = b(\theta)$, $[L_\theta(1) - \ell(\theta)]/(2\pi\mathrm{i})$ is an integer, constant along the marked path. It is zero at the real seed, where $F_\theta([0, 1])$ is positive. Hence $L_\theta(1) = \ell(\theta)$ near θ_c.

For $z \in V - 1$ put $\widetilde{F}_\theta(z) = L_\theta(z + 1)/\ell(\theta)$. On the connected overlap with V, the Abel equation and the choice at $z = 0$ give $\widetilde{F}_\theta = F_\theta$; shrink the tube if needed so this overlap is connected. They glue on $W_- = V \cup (V - 1)$ and give $F_\theta(-1) = L_\theta(0)/\ell(\theta) = 0$. Differentiation gives the displayed nonzero derivative because the exterior height chart is locally conformal at $z = 0$. The final assertion is Proposition 84. □

Corollary 67 (Exterior root isolation and local normality)**.** *There are a parameter disc $U_0 \Subset U_*$ about θ_c, a radius $r > 0$, and constants $0 < c < C$ such that, for every $\theta \in U_0$ and $|z| \leq r$,*

$$c|z| \leq |F_\theta(z) - 1| \leq C|z|, \qquad c \leq |F_\theta(z)| \leq C.$$

After reducing r, the same constants can be chosen so that $c|z + 1| \leq |F_\theta(z)| \leq C|z + 1|$ for $|z + 1| \leq r$. Joint holomorphy also gives the exterior half of condition (P1) on every $V \Subset W_-$ across this boundary subarc, while the displayed estimates isolate its normalised root uniformly. No corresponding inner estimate is asserted.

Proof. The jointly holomorphic quotient $A(\theta, z) = (F_\theta(z) - 1)/z$ extends across $z = 0$ with $A(\theta_c, 0) = \partial_z F_{\theta_c}(0) \neq 0$. Shrink U_0 and r so that A is bounded above and away from zero on $\overline{U_0} \times \overline{B(0,r)}$. The same compactness argument applies to F_θ near $F_{\theta_c}(0) = 1$ and to $F_\theta(z)/(z+1)$ near $z = -1$, using Corollary 66 for its nonzero value there. Enlarge C and decrease c to cover all three inequalities. □

Lemma 68 (The integer label at either vertical end)**.** *Let $E_b(w) = b^w$ have a fixed point L admitting a local Koenigs coordinate σ, with multiplier λ. Choose $\varepsilon \in \{+1, -1\}$ and $\ell = \log\lambda$ so that $0 < \varepsilon \operatorname{Im}\ell < 2\pi$. Suppose F is nonconstant and holomorphic on a sufficiently high ε-half-plane, satisfies $F(z+1) = E_b(F(z))$, and $F(x + \varepsilon\mathrm{i}t) \to L$ uniformly for $0 \leq x \leq 1$ as $t \to +\infty$. Then, deep in that end and in a fundamental vertical strip, there are an integer $m \geq 0$ and a holomorphic function h near $q = 0$, with $h(0) \neq 0$, such that*

$$\sigma(F(z)) = \mathrm{e}^{\ell z} q^m h(q), \qquad q = \mathrm{e}^{\varepsilon 2\pi\mathrm{i}z}.$$

The Koenigs logarithm consequently has the form

$$\frac{\log\sigma(F(z))}{\ell} = \left(1 + \frac{\varepsilon 2\pi\mathrm{i}m}{\ell}\right) z + \frac{\log h(q)}{\ell}.$$

The degree-one form $z+$ a bounded holomorphic 1-periodic correction at this end is equivalent to $m = 0$. For a jointly holomorphic parameter family with a common such end and holomorphically chosen L, σ and ℓ, the constant Laurent coefficient $a_0(b)$ is holomorphic in b (and equals $h_b(0)$ when $m = 0$). On a connected parameter domain, if nonzero at one base, its zeros are isolated.

Proof. For large t, the values $F(x + \varepsilon\mathrm{i}t)$, $0 \leq x \leq 1$, lie in the Koenigs chart. There $g(z) = \mathrm{e}^{-\ell z}\sigma(F(z))$ is 1-periodic, so it descends to a holomorphic function on a punctured q-disk. Its Laurent coefficient at index n is

$$a_n = \int_0^1 \mathrm{e}^{-\ell(x+\varepsilon\mathrm{i}t)} \sigma(F(x + \varepsilon\mathrm{i}t)) \mathrm{e}^{-\varepsilon 2\pi\mathrm{i}n(x+\varepsilon\mathrm{i}t)}\, dx.$$

For $n < 0$ its modulus is bounded by a constant times $\sup_{0\leq x\leq 1} |\sigma(F(x + \varepsilon\mathrm{i}t))| \mathrm{e}^{(\varepsilon \operatorname{Im}\ell + 2\pi n)t}$, which tends to zero. Thus every negative Laurent coefficient vanishes and g extends holomorphically to $q = 0$. It is not identically zero, so it has finite order $m \geq 0$ there and $g = q^m h(q)$ with $h(0) \neq 0$. Taking a logarithm on the deep end gives the displayed formula. For the parameter assertion, use the same coefficient integral on a fixed sufficiently deep line locally in b; it is holomorphic in b. The identity theorem makes its zeros isolated unless it vanishes identically, which the nonzero seed excludes. □

A fixed-point limit at either end therefore does not, by itself, select the degree-one time used in Definition 37. This is an actual ambiguity for a repelling fixed point: the inverse Koenigs map P extends to an entire Poincaré function, and for every $m \geq 0$ and $c \neq 0$ the entire solution $F_m(z) = P(c\mathrm{e}^{(\ell+\varepsilon 2\pi\mathrm{i}m)z})$ satisfies the functional equation and tends to L uniformly on $0 \leq \operatorname{Re} z \leq 1$ as $\varepsilon \operatorname{Im} z \to +\infty$, with exactly the label m. These examples impose one end condition; they do not assert the normalisation or the limit at the other end. The two integer labels and their nonvanishing leading coefficients are branch data for a two-regular-chart identification.

Corollary 69 (The exterior branch has degree-one time at both ends)**.** *At every real $b > \eta$, Kneser's classical solution has $m = 0$ in Lemma 68 at both conjugate fixed-point ends. Its two leading coefficients are nonzero and vary holomorphically and without zeros in the local parameter families of Proposition 44.*

Proof. In the compactified cylinder of that proposition, the local puncture coordinates are $\xi_+ = \exp(2\pi\mathrm{i}\log\sigma_+/\ell_+)$ and $\xi_- = \exp(-2\pi\mathrm{i}\log\sigma_-/\ell_-)$. The global sphere coordinate q_b has a simple zero at the upper end and a simple pole at the lower end. Consequently $q_b = c_+(b)\xi_+(1+O(\xi_+))$ and $q_b^{-1} = c_-(b)\xi_-(1+O(\xi_-))$, with $c_\pm(b)$ holomorphic and nonzero. Taking logarithms and using $z = (2\pi\mathrm{i})^{-1}\log q_b$ shows at either end that $\log\sigma_\pm/\ell_\pm = z + C_\pm(b) + O(e^{\varepsilon 2\pi\mathrm{i}z})$, where $\varepsilon = +1$ at the upper end and -1 at the lower end. Thus the coefficient of z is one, so the label is zero. The same local expansions give nonvanishing holomorphic leading coefficients. □

Corollary 70 (Parameter-uniform exterior tails)**.** *Fix $b_0 > \eta$. There are a complex disc Δ about b_0, $\delta, T, \rho > 0$, and functions $h_\pm \in \mathcal{O}(\Delta \times B(0,\rho))$ which are nowhere zero after shrinking the disc and radius, such that the local exterior family satisfies*

$$\sigma_{\pm,b}(\kappa_b(z)) = \mathrm{e}^{\ell_{\pm,b}z} h_\pm(b, \mathrm{e}^{\pm 2\pi\mathrm{i}z}) \quad \text{for } -\delta < \operatorname{Re} z < 1+\delta, \quad \pm\operatorname{Im} z > T.$$

For every smaller closed base disc $\Delta' \Subset \Delta$ and every $0 < c < \inf_{b\in\Delta'}\{\operatorname{Im}\ell_{+,b}, -\operatorname{Im}\ell_{-,b}\}$, a constant C gives

$$\sup_{\substack{b\in\Delta'\\ 0\le x\le 1}} |\kappa_b(x \pm \mathrm{i}t) - L_\pm(b)| \le C\mathrm{e}^{-ct} \qquad (t > T).$$

The same estimate holds uniformly over a complex neighbourhood of any compact subinterval of (η,∞), with a possibly smaller common $c > 0$.

Proof. In the proof of Corollary 69, the global coordinate and the upper puncture coordinate satisfy $q_b = \phi_+(b,\xi_+)$, where $\phi_+(b,0) = 0$ and $\partial_\xi\phi_+(b,0) \neq 0$. The joint inverse function theorem therefore gives $\xi_+ = q_b\psi_+(b,q_b)$ with ψ_+ holomorphic and nonzero near $q_b = 0$. Since $q_b = \mathrm{e}^{2\pi\mathrm{i}z}$, taking a holomorphic logarithm of ψ_+ and then exponentiating the relation between ξ_+ and $\sigma_{+,b}$ yields the asserted h_+; the constant integer choice of the lift is absorbed into it. At the lower end apply the same argument to $q_b^{-1} = \phi_-(b,\xi_-)$, with $\xi_- = \mathrm{e}^{-2\pi\mathrm{i}\log\sigma_{-,b}/\ell_{-,b}}$. Choose ρ and T uniformly after shrinking Δ. On Δ', both $h_\pm$ and their reciprocals are bounded on a smaller closed q-disc, and $|\mathrm{e}^{\ell_{\pm,b}(x\pm\mathrm{i}t)}| \le C_0\mathrm{e}^{-c_0 t}$ for $0 \le x \le 1$ and some $c_0 > c$. The inverse Koenigs maps satisfy $\sigma_{\pm,b}^{-1}(\zeta) = L_\pm(b) + \zeta + O(\zeta^2)$ uniformly there, giving the estimate. A finite subcover of a compact real base interval gives common parameters and a positive minimum decay rate. □

Proposition 71 (A normal-family criterion for base continuation)**.** *Let Ω and W be connected open sets in the base and height planes, respectively, and suppose $I = \Omega \cap (\eta,\infty)$ contains a nonempty interval. Let $D \subset W$ have an accumulation point in W. Suppose $A_N \in \mathcal{O}(\Omega \times W)$ is locally uniformly bounded in N and, for every $(b,z) \in I \times D$, $A_N(b,z) \to \kappa_b(z)$, where κ_b is Kneser's classical solution on W. Then A_N converges locally uniformly on $\Omega \times W$ to a jointly holomorphic family A, and $A_b = \kappa_b$ on W for every $b \in I$. If, in addition, $U_0 \subset \Omega$ is a nonempty open set where the sewn family K^W is jointly holomorphic on $U_0 \times W$, and $A_N(b,z) \to K_b^W(z)$ for $(b,z) \in U_0 \times D$, then $A = K^W$ on $U_0 \times W$.*

Proof. Local boundedness makes the sequence normal on $\Omega \times W$. Any two subsequential limits agree on $I \times D$ by the prescribed pointwise convergence. For each $b \in I$ the identity theorem in z makes them agree on $\{b\} \times W$; for each $z \in W$ the identity theorem in b then makes them agree throughout $\Omega \times \{z\}$. Hence the normal sequence has a unique cluster limit and converges locally uniformly to it. Applying the identity theorem in z once more identifies A_b with κ_b for $b \in I$ and, under the additional assumption, with K_b^W for $b \in U_0$. □

This gives an alternative to checking (P0)–(P2) one boundary point at a time: a single locally bounded sequence of jointly holomorphic approximants on a corridor crossing the

boundary would provide the continuation and, if its inner limit were verified, the branch identification. At each fixed finite stage, the quadrature formulas are locally holomorphic in b on a cell where all logarithm and inverse-chart branches are fixed and the normalising denominator is nonzero. A common branch-safe corridor for every stage, and a bound uniform in quadrature order and iteration count, have not been established for the approximants of [21, §5]. Real-base convergence by itself cannot supply the bound: for real b_0 the entire functions $a_N(b) = \exp[-N - N^3(b - b_0)^2]$ tend to zero at every real b, but are unbounded along every vertical line $b = b_0 + \mathrm{i}y$ with $y \neq 0$.

7.13 Reduction of the continuation to Brjuno boundary points

We record how the one remaining hypothesis reduces to statements at boundary points with Brjuno rotation number. Let Γ be an open arc of the Shell–Thron boundary in the upper half-plane. Parametrise it by $\lambda_c = \mathrm{e}^{2\pi\mathrm{i}\alpha}$, where λ_c is the multiplier of the fixed point L_1. Let $U = U_{\mathrm{out}} \cup \Gamma \cup U_{\mathrm{in}}$ be a neighbourhood of Γ split by it.

Lemma 72 (Radial small divisors)**.** *For $\lambda = r\lambda_c$ with $r > 0$ and every $m \geq 1$, $|\lambda^m - 1| \geq \min\bigl(\frac{1}{\sqrt{2}}|\lambda_c^m - 1|,\ \frac{1}{2}\bigr)$.*

Proof. Write $\lambda_c^m = \mathrm{e}^{\mathrm{i}\theta}$ with $|\theta| \leq \pi$. The distance from 1 to the ray $\{t\mathrm{e}^{\mathrm{i}\theta} : t > 0\}$ equals $|\sin\theta|$ if $|\theta| \leq \pi/2$ and 1 otherwise. In the first case, $|\sin\theta| = |\mathrm{e}^{\mathrm{i}\theta} - 1|\cos(\theta/2) \geq |\mathrm{e}^{\mathrm{i}\theta} - 1|/\sqrt{2}$. In the second, $|\mathrm{e}^{\mathrm{i}\theta} - 1| \leq 2$. □

The local germ has a useful exact form. Since $b = \exp(\lambda\mathrm{e}^{-\lambda})$ and $L = \mathrm{e}^{\lambda}$, the coordinate $v = (\log b)(w - L)$ transforms $w \mapsto b^w$ into

$$f_\lambda(v) = \lambda(\mathrm{e}^v - 1). \tag{29}$$

Lemma 73 (Radial convergence of local charts)**.** *Let $\lambda_c = \mathrm{e}^{2\pi\mathrm{i}\alpha}$ with α Brjuno, and put $\lambda = r\lambda_c$. The normalised inverse Schröder charts H_λ, defined by $f_\lambda \circ H_\lambda = H_\lambda \circ (\lambda\,\cdot)$ and $H_\lambda(0) = 0$, $H_\lambda'(0) = 1$, are holomorphic and uniformly bounded on one fixed disk $|v| < \rho$ for r sufficiently close to 1 from either side. They converge locally uniformly there to the Siegel chart H_{λ_c}. On a possibly smaller fixed disk, their inverses have the same convergence property.*

Proof. The map $v \mapsto \mathrm{e}^v - 1$ is univalent on $|v| < 1$. Thus the cone-uniform Brjuno majorant estimate of [16] applies directly to (29); the radial paths lie in such a cone. Lemma 72 gives the requisite uniform lower bounds for the small divisors. Each Taylor coefficient of H_λ is rational in λ with poles only at roots of unity, hence converges to its value at the irrational λ_c. The uniform majorant makes this coefficientwise convergence locally uniform on a fixed disk. Since $H_\lambda'(0) = 1$, Cauchy estimates on that disk give a smaller disk on which the H_λ are uniformly close to the identity and univalent; their inverses then converge locally uniformly as well. □

This lemma concerns germs at L. Neither the Siegel disk nor its finite-forward-pullback domain supplies a complete sewing line (Proposition 85).

Lemma 74 (The other fixed points are repelling)**.** *For $b = \exp(\lambda\mathrm{e}^{-\lambda})$ with $0 < |\lambda| \leq 1$ and $\lambda \neq 1$, every fixed point of E_b other than $L_1 = \mathrm{e}^{\lambda}$ is simple and has a multiplier of modulus greater than 1.*

Proof. Put $a = \log b = \lambda\mathrm{e}^{-\lambda}$ and $g(\mu) = \mu\mathrm{e}^{-\mu}$. Fixed points $w = E_b(w)$ correspond to $g(\mu) = a$ through $\mu = aw$, $w = \mathrm{e}^{\mu}$; μ is their multiplier. The chosen logarithm is indeed a, since $|a| \leq \mathrm{e} < \pi$. We show that g is injective on the closed unit disk. In polar coordinates,

$$\arg g(r\mathrm{e}^{\mathrm{i}\theta}) = \theta - r\sin\theta, \qquad |g(r\mathrm{e}^{\mathrm{i}\theta})| = r\mathrm{e}^{-r\cos\theta}.$$

For each $0 < r \le 1$, the first expression is strictly increasing in θ modulo 2π. Holding its value fixed and varying $r < 1$, implicit differentiation gives

$$\frac{d}{dr}\log|g(re^{i\theta(r)})| = \frac{|1 - re^{i\theta(r)}|^2}{r(1 - r\cos\theta(r))} > 0.$$

Thus different radii cannot have the same image, including by continuity at $r = 1$. Since λ is already a preimage of a in the closed unit disk, every other preimage has modulus greater than 1. Finally $g'(\mu) = e^{-\mu}(1-\mu)$ vanishes only at $\mu = 1$, which cannot be another preimage when $\lambda \neq 1$. □

Proposition 75 (Transport of the lower fixed point)**.** *Fix $\lambda_0 \in (0,1)$ and its real second multiplier $\mu_0 > 1$, where $\mu_0 e^{-\mu_0} = \lambda_0 e^{-\lambda_0}$. Let $\gamma : [0,1] \to \overline{\mathbb{D}} \setminus \{0,1\}$ be a piecewise C^1 path with $\gamma(0) = \lambda_0$, $\gamma([0,1)) \subset \mathbb{D}$, and $\gamma(1) = \lambda_c \in \partial\mathbb{D} \setminus \{1\}$. Then μ_0 continues analytically along γ to a multiplier μ_c with $|\mu_c| > 1$. The associated lower fixed point and its regular superfunction continue along a narrow parameter corridor and extend locally jointly holomorphically in the base and z through $b_c = \exp(\lambda_c e^{-\lambda_c})$, with a transported branch of $\log\mu$.*

Proof. Put $g(t) = te^{-t}$. At every point of the path, the implicit equation $g(\mu) = g(\lambda)$ has a locally unique continuation of the second root as long as $\mu \neq 1$. Lemma 74 keeps it outside the closed unit disk, so it cannot meet the root λ. On the compact trace $K = \gamma([0,1])$, which avoids 1, there is a $\delta > 0$ such that every such second root obeys $|\mu - 1| \ge \delta$: otherwise $\mu_j \to 1$ and $\lambda_j \to \lambda_* \in K$ would give $g(\lambda_*) = g(1)$, contradicting injectivity of g on the closed unit disk. Along each smooth piece of γ,

$$\frac{d\mu}{d\lambda} = \frac{g'(\lambda)}{g'(\mu)} = \frac{1-\lambda}{\lambda}\,\frac{\mu}{1-\mu}.$$

Here $|\lambda|$ is bounded below on K, and $|\mu/(1-\mu)| \le 1 + 1/\delta$. The derivative is therefore bounded by a constant times the path speed. At any hypothetical first stopping point, finite path length gives a finite limit $\mu \neq 1$; the implicit-function theorem would extend the branch. Thus it reaches λ_c. It remains outside the unit disk by Lemma 74: it cannot merge with λ_c, because $g'(\lambda_c) \neq 0$ would make the local root uniquely equal to λ near λ_c, contrary to $|\mu| > 1$ inside the disk. The lower Koenigs construction and the proof of Lemma 81 apply in neighbourhoods along the path and at its endpoint. Their normalisation and the continued branch of $\log\mu$ patch these local superfunctions. Finally $d\exp(g(\lambda))/d\lambda \neq 0$ at $\lambda_c \neq 1$, so the endpoint chart may be expressed in the base b. □

This transports the lower regular chart only. It does not transport the sewing correction or the solution K^W to the boundary arc.

Corollary 76 (A global lower regular family in the inner parameter disk)**.** *Let $\Omega = \mathbb{D} \setminus (-1, 0]$. The real second multiplier $\mu_2 > 1$ extends to a single-valued holomorphic function on Ω with $|\mu_2(\lambda)| > 1$. It has a holomorphic logarithm fixed by the positive real value on $(0,1)$. With this choice, the lower regular superfunction $S_\lambda(z)$ is jointly holomorphic on $\Omega \times \mathbb{C}$ and extends locally across every point of $\partial\mathbb{D} \setminus \{1, -1\}$.*

Proof. The continuation argument of Proposition 75 works along every finite path in Ω. Since Ω is simply connected, the monodromy theorem makes the continuation single-valued. Lemma 74 gives $|\mu_2| > 1$, hence a global holomorphic logarithm. At each parameter, the Koenigs coordinate at $L_2 = e^{\mu_2}$, normalised by derivative 1, is unique and varies locally holomorphically; so the local constructions of Lemma 81 patch. Its forward-iterate formula makes S_λ entire in z and jointly holomorphic locally in (λ, z). Proposition 75 gives the stated local boundary extensions. □

For $0 < |\lambda| < 1$ the normalised attracting Koenigs germ also varies locally holomorphically in λ, since its coefficient equations have no interior resonances. Thus both local fixed-point charts can be transported through Ω; persistence of the overlap needed for their sewing remains unproved.

Proposition 77 (Infinite monodromy around $b = 1$)**.** *The map $\lambda \mapsto b = \exp(\lambda e^{-\lambda})$ is injective on $\mathbb{D}$. On the universal cover of $0 < |\lambda| < 1$, the lower multiplier is single-valued and holomorphic, and its regular superfunction is jointly holomorphic in the parameter and height. The lower multiplier branch of Corollary 76 has infinite-order monodromy around $b = 1$: continuing it around any positively oriented loop that winds once around $b = 1$ in the image of $0 < |\lambda| < 1$ changes its value, and its values after different integer winding numbers are distinct. Consequently neither the lower fixed point nor its normalised lower regular superfunction is single-valued on that punctured base domain.*

Proof. Injectivity of $g(\lambda) = \lambda e^{-\lambda}$ on $\mathbb{D}$ was proved in Lemma 74. Since $|g(\lambda)| < e < \pi$, the exponential is injective on $g(\mathbb{D})$ as well. Moreover $b'(\lambda) = b(\lambda)e^{-\lambda}(1 - \lambda) \neq 0$ in $\mathbb{D}$, so this is a conformal parameterisation. The continuation proof of Proposition 75 applies to every finite path in the punctured disk. Monodromy on its simply connected universal cover gives a holomorphic root μ. Since $|\mu| > 1$, it has a global holomorphic logarithm there. The local construction of Lemma 81 then patches to give the asserted superfunction.

Choose $0 < \varepsilon < 1$ so small that $R := -\log \varepsilon > \varepsilon$, and continue the real lower root $\mu_0 > 1$ along $\lambda(t) = \varepsilon e^{it}$, with t running from 0 to $2\pi n$ for any integer n. The path-continuation proof of Proposition 75 applies to every finite number of turns. All continued roots obey $|\mu(t)| > 1$. Their common modulus equation gives

$$\operatorname{Re} \mu(t) = R + \operatorname{Re} \lambda(t) + \log |\mu(t)| > R - \varepsilon > 0.$$

Thus the principal $\operatorname{Log} \mu(t)$ remains defined throughout. Taking continuous logarithms in $g(\mu(t)) = g(\lambda(t))$ and using the real equation at $t = 0$ yields

$$\mu(t) - \operatorname{Log} \mu(t) = R - it + \varepsilon e^{it}.$$

At the end of n turns, the right side is $R + \varepsilon - 2\pi i n$. Since the same single-valued function $\mu \mapsto \mu - \operatorname{Log} \mu$ is used at every endpoint, roots obtained after different winding numbers cannot coincide. A radial path from any real $\lambda_0 \in (0, 1)$ to ε conjugates this monodromy to the loop based at λ_0.

Distinct endpoint roots give distinct fixed points $L_2 = e^{\mu}$, because $\mu = (\log b)L_2$, and distinct regular superfunctions, because $S(x) \to L_2$ as $x \to -\infty$. Homotopy invariance extends the conclusion to every loop of the same winding number. □

Proposition 78 (The upper chart at the normalising orbit)**.** *Let $0 < |\lambda| < 1$, $a = \lambda e^{-\lambda}$, $b = e^a$, and $L_1 = e^{\lambda}$. Then $E_b^{\circ n}(1) \to L_1$ locally uniformly in λ. The attracting Koenigs coordinate σ_λ, normalised by $\sigma_\lambda(L_1) = 0$ and $\sigma'_\lambda(L_1) = 1$, extends to the attracting basin of L_1 and depends locally jointly holomorphically on (λ, w) there. In particular $c(\lambda) = \sigma_\lambda(1)$ is holomorphic and nonzero on the punctured unit disk, and $\sigma'_\lambda(1) \neq 0$. It extends across $\lambda = 0$ with*

$$c(\lambda) = -\lambda - \tfrac{1}{2}\lambda^2 + O(\lambda^3).$$

Consequently there is a unique holomorphic function h on $|\lambda| < 1$ with $h(0) = 0$ and $c(\lambda) = -\lambda \exp h(\lambda)$; in particular $h(\lambda) = \lambda/2 + O(\lambda^2)$.

On the universal cover of $0 < |\lambda| < 1$, the normalised upper regular superfunction germ

$$R_\lambda(z) = \sigma_\lambda^{-1}(c(\lambda) \exp(z \log \lambda)), \qquad R_\lambda(0) = 1,$$

is jointly holomorphic near each parameter and $z = 0$. Along every compact parameter path avoiding 0 it has a common positive height radius on a sufficiently narrow corridor. On a possibly smaller common disk, $R_\lambda(z) = 1$ only at $z = 0$, and R_λ and $1/R_\lambda$ are uniformly bounded along the path. The deck transformation around $\lambda = 0$ acts with infinite order on these normalised upper germs.

Proof. Under the change of coordinate $v = aw$, the map E_b becomes $v \mapsto ae^v$, whose attracting fixed point is $v = \lambda$ and whose only finite singular value is 0. The singular-value theorem for exponential maps places 0 in the immediate attracting basin of this fixed point [25, §2]. Since $0 \mapsto a$ in the v coordinate, $w = 1$ belongs to the same basin. Attraction is locally uniform in the parameter: at any fixed λ_0, a finite iterate of 1 enters a small invariant contraction disk about $L_1(\lambda_0)$, and this persists for nearby parameters.

The local Koenigs coordinate varies holomorphically with λ. For a basin point w, choose N with $E_b^{\circ N}(w)$ in its local domain and define

$$\sigma_\lambda(w) = \lambda^{-N}\sigma_\lambda(E_b^{\circ N}(w)).$$

This formula is independent of larger N and locally jointly holomorphic. Its w derivative never vanishes, since E_b has no critical points and the local Koenigs coordinate is univalent. The local coordinate has only the zero L_1, so $c(\lambda) = 0$ exactly when a finite iterate of 1 lands on L_1. In that case the singular orbit of $v \mapsto ae^v$ would be finite. For a postsingularly finite exponential map, a preperiodic dynamic ray lands at its singular value [14, Thm. 3.1]; its points escape, so it cannot land at a point in the open attracting basin. This contradicts the membership of 0 in that basin proved above. Thus c never vanishes.

To analyse the removable endpoint $\lambda = 0$, write $u = w - L_1$ and

$$G_\lambda(u) = E_b(L_1 + u) - L_1 = e^\lambda(\exp(\lambda e^{-\lambda}u) - 1) = \lambda u + O(\lambda^2 u^2).$$

On a fixed small u-disk and for sufficiently small $|\lambda|$, $|G_\lambda(u)| \le C|\lambda||u|$ with $C^2|\lambda| < 1$. Put $u_n = G_\lambda^{\circ n}(u)$. Each $\lambda^{-n}u_n$ has a removable singularity at $\lambda = 0$, and

$$\left|\lambda^{-n-1}u_{n+1} - \lambda^{-n}u_n\right| \le C'|\lambda|(C^2|\lambda|)^n|u|^2.$$

The telescoping limit is the normalised Koenigs coordinate and extends jointly holomorphically through $\lambda = 0$ with $\sigma_\lambda(L_1+u) = u+O(\lambda u^2)$. Since $1-L_1 = -\lambda-\lambda^2/2+O(\lambda^3)$, this gives the stated expansion of c. The function $-c(\lambda)/\lambda$ is now holomorphic and zero-free on the whole unit disk, with value 1 at the origin. It therefore has the asserted normalised holomorphic logarithm.

Choose a holomorphic branch of $\log\lambda$ on the universal cover. The inverse-function theorem at $w = 1$, applied to $\sigma'_\lambda(1) \neq 0$, gives the displayed germ and its joint holomorphy. A finite cover of a compact lifted parameter path gives one positive height radius after narrowing its corridor. Since $R'_\lambda(0) = c(\lambda)\log\lambda/\sigma'_\lambda(1) \neq 0$, the same compactness argument gives a smaller common root-isolation disk. Shrink it once more to avoid zeros of R_λ and obtain the stated bounds. On one positive turn around 0, $c(\lambda)$ and $\sigma'_\lambda(1)$ return unchanged, whereas $\log\lambda$ gains $2\pi\mathrm{i}$. After n turns, $R'_\lambda(0)$ therefore changes by $2\pi\mathrm{i}nc(\lambda)/\sigma'_\lambda(1) \neq 0$ for $n \neq 0$. □

Corollary 79 (Diverging chart-entry depth at a Brjuno boundary point)**.** *Let $\lambda_c = e^{2\pi\mathrm{i}\alpha} \neq 1$ with α Brjuno, $\lambda = r\lambda_c$, and $b(r) = \exp(\lambda e^{-\lambda})$. Choose $\rho > 0$ so that the charts of Lemma 73 are univalent near $\{|\zeta| \le \rho\}$ for r close to 1 from either side, and let $0 < s < \rho$. Put*

$$\mathcal{D}_{r,s} = L_1(b(r)) + (\log b(r))^{-1}H_{r\lambda_c}(B(0,s)).$$

For every fixed $N \ge 0$, the point $E_{b(r)}^{\circ N}(1)$ lies outside $\mathcal{D}_{r,s}$ for all $r \neq 1$ sufficiently close to 1. For $r < 1$ the first-entry time

$$m_s(r) = \min\{n \ge 0 : E_{b(r)}^{\circ n}(1) \in \mathcal{D}_{r,s}\}$$

is finite and tends to infinity as $r \uparrow 1$.

Proof. For the boundary map $f_{\lambda_c}(v) = \lambda_c(\mathrm{e}^v - 1)$, the singular value $-\lambda_c$ belongs to the Julia set whenever the map has a Siegel disk [23, Introduction, remark after Thm. 2]. Under the coordinate $v = (\log b_c)(w - L_1(b_c))$, this singular value is $w = 0$. The Julia set is forward invariant, and $E_{b_c}(0) = 1$; hence every $E_{b_c}^{\circ N}(1)$ belongs to the Julia set and lies outside the boundary Siegel chart $\mathcal{D}_{1,\rho}$.

If the first assertion failed, there would be $r_j \to 1$, $r_j \neq 1$, and $\zeta_j \in B(0,s)$ with

$$E_{b(r_j)}^{\circ N}(1) = L_1(b(r_j)) + (\log b(r_j))^{-1} H_{r_j \lambda_c}(\zeta_j).$$

After taking a subsequence, $\zeta_j \to \zeta$ with $|\zeta| \leq s < \rho$. Lemma 73 and finite-iterate continuity put $E_{b_c}^{\circ N}(1)$ in $\mathcal{D}_{1,\rho}$, a contradiction. For each $r < 1$, Proposition 78 says the orbit of 1 converges to $L_1(b(r))$, so it eventually enters $\mathcal{D}_{r,s}$. The first assertion excludes every bounded entry time as $r \uparrow 1$. □

This concerns finite pullbacks of a fixed local upper chart at the normalising orbit. It does not preclude continuation of the sewn solution by a different mechanism.

Proposition 80 (Collapse of the attracting basin at a Brjuno boundary point)**.** *Let $\lambda_c = \mathrm{e}^{2\pi\mathrm{i}\alpha} \neq 1$ with α Brjuno, $b(\lambda) = \exp(\lambda \mathrm{e}^{-\lambda})$, and let $\mathcal{A}_\lambda$ be the immediate attracting basin of $L_1(\lambda) = \mathrm{e}^\lambda$ for $0 < |\lambda| < 1$. Then $1 \in \mathcal{A}_\lambda$ and*

$$d(\lambda) := \operatorname{dist}(1, \mathbb{C} \setminus \mathcal{A}_\lambda) \longrightarrow 0 \qquad (\lambda \to \lambda_c,\ |\lambda| < 1).$$

The Poincaré distance within the immediate basin also satisfies

$$\operatorname{dist}^{\mathrm{hyp}}_{\mathcal{A}_\lambda}(1, b(\lambda)) \longrightarrow \infty \qquad (\lambda \to \lambda_c,\ |\lambda| < 1).$$

Moreover, let $\lambda_j \to \lambda_c$ from the disk and let V be a connected height domain containing 0. If holomorphic maps $G_j : V \to \mathcal{A}_{\lambda_j}$ satisfy $G_j(0) = 1$, then $G_j \to 1$ locally uniformly on V, without a boundedness assumption. Consequently, if $1 \in V$, there is a neighbourhood U of λ_c such that no $\lambda \in U \cap \{0 < |\lambda| < 1\}$ admits a holomorphic $G : V \to \mathcal{A}_\lambda$ with $G(0) = 1$ and $G(1) = b(\lambda)$.

Proof. The proof of Proposition 78 puts 0 in the immediate attracting basin for every inner parameter; since $E_{b(\lambda)}(0) = 1$, this basin also contains 1. The proof of Corollary 79 puts 1 in the Julia set of the boundary map $E_{b(\lambda_c)}$. Repelling periodic points are dense in this Julia set [2, Thm. 4]. Given $\epsilon > 0$, choose such a point p_c with $|p_c - 1| < \epsilon/2$. If its period is m, then $|(E_{b(\lambda_c)}^{\circ m})'(p_c)| > 1$, so the implicit-function theorem continues it to repelling periodic points p_λ for all λ near λ_c. They lie outside $\mathcal{A}_\lambda$ and satisfy $|p_\lambda - 1| < \epsilon$. Thus $d(\lambda) < \epsilon$ near λ_c.

For the second assertion, first note that $b(\lambda_c) \neq 1$: indeed $0 < |\lambda_c \mathrm{e}^{-\lambda_c}| \leq \mathrm{e} < 2\pi$, so its exponential cannot equal 1. Both 1 and $b(\lambda_c) = E_{b(\lambda_c)}(1)$ belong to the Julia set. Choose a repelling periodic point $q_c \neq 1$ near $b(\lambda_c)$; it continues to repelling periodic points $q_j \to q_c$ of $E_{b(\lambda_j)}$. By the first assertion, choose $p_j \in \mathbb{C} \setminus \mathcal{A}_{\lambda_j}$ with $p_j \to 1$. For all large j the functions

$$H_j(z) = \frac{G_j(z) - p_j}{q_j - p_j}$$

omit both 0 and 1, while $H_j(0) \to 0$. Montel's theorem gives spherical normality of this family. Every subsequential limit takes the value 0 at the origin; Hurwitz's theorem for the zero-free H_j forces that limit to be identically zero. Hence $H_j \to 0$ and $G_j \to 1$ locally uniformly on V. Since $\mathcal{A}_{\lambda_j}$ omits the two distinct points p_j, q_j, it is a hyperbolic domain.

If the asserted Poincaré distances stayed bounded along a sequence, take universal covering maps $\phi_j : \mathbb{D} \to \mathcal{A}_{\lambda_j}$ with $\phi_j(0) = 1$. The lifts of $b(\lambda_j)$ can then be chosen as points t_j in a fixed compact subdisk of $\mathbb{D}$. Applying the preceding collapse to ϕ_j gives $b(\lambda_j) = \phi_j(t_j) \to 1$, contrary to $b(\lambda_c) \neq 1$. If $1 \in V$ and $G_j(1) = b(\lambda_j)$, this gives $b(\lambda_c) = 1$, a contradiction. The neighbourhood assertion follows by choosing a contrary sequence $\lambda_j \to \lambda_c$. □

This collapse rules out a fixed basin-valued upper chart carrying both normalising heights 0 and 1 near the boundary. If a normalised holomorphic solution exists on such a fixed connected height domain, its image must meet $\partial\mathcal{A}_\lambda \subset J(E_{b(\lambda)})$ for all sufficiently close inner parameters. This does not decide whether the sewn family reaches such a fixed height domain near λ_c; continuation through another chart remains possible.

Lemma 81 (Parameter stability of the lower chart)**.** *Let b_c be a point of the boundary arc at which the chosen lower fixed point $L_2(b_c)$ is simple and its multiplier $\mu_c = E'_{b_c}(L_2(b_c))$ satisfies $|\mu_c| > 1$. Choose the local branches of $\log b$ and $\log \mu$ used to define E_b and the lower regular superfunction. Near b_c, the normalised Koenigs coordinate τ_b at $L_2(b)$ and its local inverse depend jointly holomorphically on the base and their arguments on fixed disks. Moreover the entire superfunction*

$$S_b(z) = E_b^{\circ m}\Big(\tau_b^{-1}\big(-\exp((z-m)\log\mu(b))\big)\Big)$$

is independent of every sufficiently large integer m, locally jointly holomorphic in (b, z), and converges with all z-derivatives locally uniformly to S_{b_c} as $b \to b_c$ from either side.

Proof. The implicit-function theorem gives holomorphic $L_2(b)$ and $\mu(b)$ near b_c. In the coordinate $u = w - L_2(b)$, set $G_b(u) = E_b(L_2(b) + u) - L_2(b)$ and let h_b be its local inverse. Then $h_b'(0) = \mu(b)^{-1}$. Choose $|\mu_c|^{-1} < q < |\mu_c|^{-1/2}$. After shrinking the parameter neighbourhood and a fixed disk $|u| < r$, we have $|h_b(u)| \le q|u|$, $|\mu(b)h_b(u) - u| \le C|u|^2$, and $Mq^2 < 1$, where $M = \sup_b |\mu(b)|$. Hence

$$\tau_b(L_2(b) + u) := \lim_{n\to\infty} \mu(b)^n h_b^{\circ n}(u)$$

converges normally in (b, u): successive terms differ by at most $C(Mq^2)^n|u|^2$. It has derivative 1 at the fixed point and obeys $\tau_b(E_b(w)) = \mu(b)\tau_b(w)$ wherever both sides are local. The inverse-function theorem, uniformly after one more shrinking, gives jointly holomorphic inverses on a fixed disk about 0.

Since $\operatorname{Re}\log\mu(b) = \log|\mu(b)|$ stays positive, for any compact set of (b, z) one integer m puts $-\exp((z-m)\log\mu(b))$ in that inverse disk. The local conjugacy makes the displayed formula independent of increasing m. These formulas therefore define a locally jointly holomorphic function for all $z \in \mathbb{C}$. Holomorphy and Cauchy's formula give the asserted locally uniform convergence, including all z-derivatives. □

Proposition 82 (Backward-value label of the lower regular family)**.** *In the parameter neighbourhood of Lemma 81, put $a(b) = \log b \neq 0$. For every $k \in \mathbb{Z} \setminus \{0\}$ there is a simple root c of $S_{b_c}(c) = 1$ with $S_{b_c}(c-1) = 2\pi\mathrm{i}k/a(b_c)$. Each such root continues locally holomorphically as $c(b)$, and $G_b(z) := S_b(z + c(b))$ is a jointly holomorphic, entire-in-z solution with $G_b(0) = 1$ and*

$$G_b(-1) = \frac{2\pi\mathrm{i}k}{\log b}.$$

Conversely every normalised translate of S_b has one of these nonzero integer labels, which is constant along local parameter continuation. In fact every entire-in-z solution with $F(0) = 1$ has a nonzero backward-value label. For real $1 < b < \eta$, the geometric sewn solution satisfies $K_b^W(-1) = 0$; this identity persists on its local complex parameter discs. Consequently a normalised lower regular translate cannot agree with the sewn solution on a common connected regular z-domain containing -1.

Proof. The functional equation gives $S_b(z) = E_b(S_b(z-1)) \neq 0$, so the nonconstant entire function S_b omits 0. Picard's theorem makes it attain every nonzero value, including $2\pi\mathrm{i}k/a(b_c)$. Choose t with that value and put $c = t + 1$; then $S_{b_c}(c) = 1$. The formula in Lemma 81, differentiated after choosing a sufficiently large m, shows $S_b'(z) \neq 0$: the derivatives of $E_b^{\circ m}$ and of the local inverse τ_b^{-1}, and the exponential factor, are all nonzero. Thus the root c is simple and the implicit-function theorem continues it holomorphically in b. The displayed value at -1 persists by the functional equation and continuity. Conversely $S_b(c) = 1$ forces $a(b)S_b(c-1) \in 2\pi\mathrm{i}\mathbb{Z}$; omission of 0 excludes the zero integer. The integer is locally constant because it depends holomorphically on b. For an arbitrary entire solution F, the same argument applies: $F(z) = E_b(F(z-1))$ makes F zero-free, so $F(0) = 1$ forces $F(-1) = 2\pi\mathrm{i}k/a(b)$ with $k \neq 0$.

For real $b \in (1, \eta)$, the marked upper point $w = 0$ of Proposition 15 corresponds to $z = 0$, and its deck translate $w = -1$ corresponds to $z = -1$. The real attracting regular branch has $R_b(-1) = 0$, since $E_b(0) = 1 = R_b(0)$ and this inverse branch is real. Hence $K_b^W(-1) = 0$. Proposition 16 and the identity theorem preserve the equality on each local parameter disc. The two values at -1 exclude the asserted agreement. □

Corollary 83 (Rigidity of the backward branch)**.** *Let U be a connected base domain on which $a(b) = \log b$ is holomorphic and nonzero. Suppose $F_b(z)$ is jointly holomorphic in (b, z) near $U \times \{-1, 0\}$, satisfies $F_b(0) = 1$, and obeys the tetration equation near $z = -1$. Then*

$$k(b) := \frac{a(b)F_b(-1)}{2\pi\mathrm{i}} \in \mathbb{Z}$$

is constant on U. If $k = 0$, none of the F_b can extend holomorphically through $z = -2$ while preserving the tetration equation; in particular they cannot be entire solutions. The inner sewn family has $k = 0$ on its real-axis parameter neighbourhood, as does any continuation of Kneser's real-normalised family from $b > \eta$ that remains regular at $z = -1$. Thus their values and all base derivatives at $z = -1$ agree automatically wherever both branches are defined.

Proof. The equation at $z = -1$ gives $\exp(a(b)F_b(-1)) = 1$. The displayed function is holomorphic and integer-valued, hence constant. If $k = 0$ and F_b extended regularly through -2, the equation would give $E_b(F_b(-2)) = F_b(-1) = 0$, impossible because an exponential never vanishes. Proposition 82 gives $k = 0$ for the sewn family. Kneser's family has $F_b(-1) = 0$ for real $b > \eta$ by its real normalisation; the identity theorem transports this value on any stated parameter continuation. □

Proposition 84 (Logarithmic monodromy at the first backward singularity)**.** *Let U be a simply connected parameter disc on which $a(b) = \log b \neq 0$. Suppose $F_b(z)$ is jointly holomorphic near $U \times \{-1\}$, $F_b(-1) = 0$, and $F_b'(-1) \neq 0$. Locally in (b, t) write $F_b(-1 + t) = t\,u_b(t)$ with $u_b(0) \neq 0$. On any simply connected sector $0 < |t| < r$, all holomorphic backward branches satisfying $\exp(a(b)G_b(-2 + t)) = F_b(-1 + t)$ have the form*

$$G_b(-2 + t) = \frac{\operatorname{Log} t + \log u_b(t) + 2\pi\mathrm{i}k}{a(b)}, \qquad k \in \mathbb{Z}.$$

Continuation once around $t = 0$ adds $2\pi\mathrm{i}/a(b)$. Hence no single-valued holomorphic or meromorphic backward branch satisfying the equation exists on a full punctured disc about $z = -2$. For every real $b \in (1, \eta)$, the sewn solution K_b^W has a simple zero at $z = -1$, so this description applies to it and to its local holomorphic parameter family.

Proof. The quotient $u_b(t)$ extends jointly holomorphically through $t = 0$. After shrinking U and r, it has a joint holomorphic logarithm. Two logarithms of $tu_b(t)$ on a connected sector differ by a constant multiple of $2\pi\mathrm{i}$, giving the formula. A turn about $t = 0$ changes $\operatorname{Log} t$

by $2\pi\mathrm{i}$. If a single-valued backward branch existed on the punctured disc, integrating the logarithmic derivative of $\exp(aG) = tu$ around a small circle would give both 0 and $2\pi\mathrm{i}$. A meromorphic branch has no pole away from the centre, since its exponential would have an essential singularity there; it is therefore subject to the same contradiction.

For the sewn family, the upper chart of Proposition 15 contains $w = -1$ and its inverse uniformising coordinate $w(z)$ is conformal, with $w(-1) = -1$. Proposition 78 gives $R_b'(0) \neq 0$, and differentiating the functional equation at $z = -1$ gives $R_b'(-1) = R_b'(0)/a(b) \neq 0$. Thus $(K_b^W)'(-1) = R_b'(-1)w'(-1) \neq 0$. Proposition 16 preserves this simple zero on a sufficiently small complex parameter disc about each real base. □

This monodromy is shared by every simple-zero branch with $F(-1) = 0$; it does not by itself identify the two families in (P2).

The families G_b show that the functional equation and the normalisation $F(0) = 1$ allow entire solutions jointly holomorphic across a non-cusp boundary point. The integer at $z = -1$ distinguishes these families from the inner sewn branch. Thus (P0) still requires continuation of the *specified* branches with backward value 0.

Proposition 85 (Siegel pullbacks have no complete sewing band)**.** *Let $\lambda_c = \mathrm{e}^{2\pi\mathrm{i}\alpha}$ be Brjuno and different from 1, and let $b_c = \exp(\lambda_c \mathrm{e}^{-\lambda_c})$. Take $\rho > 0$ small enough that the chart of Lemma 73 is univalent on a neighbourhood of $\{|\zeta| \le \rho\}$ and its image lies in that lemma's common inverse-chart disk, and put*

$$\mathcal{D}_\rho = L_1(b_c) + (\log b_c)^{-1} H_{\lambda_c}(\{|\zeta| < \rho\}), \qquad \chi_c(w) = H_{\lambda_c}^{-1}((\log b_c)(w - L_1(b_c))).$$

Then $E_{b_c}(\mathcal{D}_\rho) = \mathcal{D}_\rho$ and $\chi_c \circ E_{b_c} = \lambda_c \chi_c$ there. Define

$$\Omega_\rho = \{z \in \mathbb{C} : \ S_{b_c}(z+n) \in \mathcal{D}_\rho \text{ for some } n \in \mathbb{Z}_{\ge 0}\}, \qquad V_c(z) = \lambda_c^{-n} \chi_c(S_{b_c}(z+n)).$$

The following assertions hold.

1. *Ω_ρ is nonempty, open and invariant under $z \mapsto z+1$ and $z \mapsto z-1$. The formula for V_c is independent of the admissible integer n and defines a holomorphic function with $V_c(z+1) = \lambda_c V_c(z)$. It has zeros, while $S_{b_c}^{-1}(1) \neq \varnothing$ and $S_{b_c}^{-1}(1) \cap \Omega_\rho = \varnothing$.*

2. *Ω_ρ contains no nonempty path-connected set C with $C + m = C$ for any nonzero integer m. In particular it contains no complete horizontal line, horizontal strip, or lift of an essential closed curve in $\Omega_\rho/\mathbb{Z}$. The same statement holds if $\mathcal{D}_\rho$ is replaced by the maximal Siegel disk.*

3. *For every compact $K \subset \Omega_\rho$ there is one integer N such that the pulled-back upper coordinates*
$$V_{b,N}(z) = \lambda(b)^{-N} H_{\lambda(b)}^{-1}((\log b)(S_b(z+N) - L_1(b)))$$
are defined near K for $b = \exp(\lambda \mathrm{e}^{-\lambda})$, $\lambda = r\lambda_c$, and r sufficiently close to 1 from either side. They converge to V_c locally uniformly near K, together with all z-derivatives.

Proof. The Schröder identity and $|\lambda_c| = 1$ make $\mathcal{D}_\rho$ invariant in both directions. The admissible sets $\{z : S_{b_c}(z+n) \in \mathcal{D}_\rho\}$ increase with n. If $m \ge n$ are both admissible, the Schröder identity gives $\chi_c(S_{b_c}(z+m)) = \lambda_c^{m-n} \chi_c(S_{b_c}(z+n))$. This proves independence, holomorphy and the translation identity; admissibility after either integer translation follows by increasing the pullback depth when needed. The nonconstant entire function S_{b_c} omits 0, since its defining formula has at least one exponential iterate. Little Picard's theorem therefore makes it attain $L_1(b_c) \neq 0$; any such preimage lies in Ω_ρ and is a zero of V_c. The same argument shows that S_{b_c} attains 1. By Corollary 79, 1 belongs to the Julia set of E_{b_c}. Every point

in its forward orbit remains in that set, whereas $\mathcal{D}_\rho$ lies in the Fatou set. Hence no z with $S_{b_c}(z) = 1$ can satisfy the defining condition of Ω_ρ.

Put $a = \log b_c$ and $\mathcal{B} = \bigcup_{n\geq 0} E_{b_c}^{-n}(\mathcal{D}_\rho)$. Because E_{b_c} maps $\mathcal{D}_\rho$ bijectively onto itself,

$$E_{b_c}^{-1}(\mathcal{D}_\rho) = \bigcup_{k\in\mathbb{Z}} (\mathcal{D}_\rho + 2\pi \mathrm{i}k/a),$$

a disjoint union. Thus $\mathcal{D}_\rho$ is a component of its first preimage. Inductively it is a component of every $E_{b_c}^{-n}(\mathcal{D}_\rho)$: a path at level n starting in $\mathcal{D}_\rho$ maps to a path at level $n-1$ starting there, so the induction hypothesis and the first-preimage description keep it in $\mathcal{D}_\rho$. It is also a component of $\mathcal{B}$, since any path in the increasing union has compact image and is contained in one finite level.

Suppose $C \subset \Omega_\rho$ were nonempty, path-connected and $C+m = C$ for some $m > 0$. The set $S_{b_c}(C)$ is path-connected in $\mathcal{B}$ and meets $\mathcal{D}_\rho$: for any $z \in C$, choose an admissible depth n and then a multiple N of m with $N \geq n$; $z+N \in C$ and $S_{b_c}(z+N) \in \mathcal{D}_\rho$. Hence $S_{b_c}(C) \subset \mathcal{D}_\rho$. But for a fixed $z \in C$, $S_{b_c}(z - jm) \to L_2(b_c)$ as $j \to \infty$, whereas $\chi_c(S_{b_c}(z-jm)) = \lambda_c^{-jm}\chi_c(S_{b_c}(z))$ lies on a compact circle inside the chart disk. Its limit would be a fixed point inside $\mathcal{D}_\rho$, necessarily $L_1(b_c)$, not $L_2(b_c)$. The same argument uses the maximal Siegel disk in place of $\mathcal{D}_\rho$.

Compactness of K and the increasing admissible sets give one N with $S_{b_c}(K+N) \Subset \mathcal{D}_\rho$. Lemma 73 and Lemma 81 give the convergence in (3); Cauchy's formula gives all z-derivatives. □

Proposition 86 (No bounded periodic sewing collar in the attracting basin)**.** *Let $\lambda_c = \mathrm{e}^{2\pi\mathrm{i}\alpha} \neq 1$ with α Brjuno, and continue the chosen lower regular family S_λ to a neighbourhood of λ_c as in Lemma 81. For $0 < |\lambda| < 1$ let $\mathcal{A}_\lambda^{\mathrm{all}} = \bigcup_{n\geq 0} E_{b(\lambda)}^{-n}(\mathcal{A}_\lambda)$ be the full attracting basin of $L_1(\lambda)$. Let $V \subset \mathbb{C}$ be any nonempty connected open set with $V+1 = V$, and let $\lambda_j \to \lambda_c$ from $0 < |\lambda| < 1$ on the selected lower branch. There is no locally bounded sequence of holomorphic 1-periodic functions Q_j on V such that*

$$S_{\lambda_j}(z + Q_j(z)) \in \mathcal{A}_{\lambda_j}^{\mathrm{all}} \qquad (z \in V,\ j \geq 1).$$

In particular, taking $Q_j = 0$ excludes any fixed lower-time collar whose entire image stays in the attracting basin up to this boundary point.

Proof. Suppose the sequence existed. Montel's theorem gives a subsequence $Q_j \to Q$ locally uniformly on V. Set $\phi_j(z) = z + Q_j(z)$ and $\phi(z) = z + Q(z)$. The identity $\phi(z+1) = \phi(z)+1$ makes ϕ nonconstant and $W = \phi(V)$ a nonempty connected open set with $W + 1 = W$. Lemma 81 gives $H_j := S_{\lambda_j} \circ \phi_j \to H := S_{\lambda_c} \circ \phi$ locally uniformly on V.

We claim $H(V)$ avoids the Julia set. Otherwise its open image contains a repelling periodic point p_c, by the density theorem [2, Thm. 4]. Continue it to repelling periodic points p_j of the nearby maps. Choose a closed disk $\overline{D} \Subset V$ containing a zero of $H - p_c$ and none on its boundary. Rouché's theorem gives a zero of $H_j - p_j$ in D for large j, contrary to $H_j(V) \subset \mathcal{A}_{\lambda_j}^{\mathrm{all}}$.

Thus the connected set $H(V) = S_{\lambda_c}(W)$ lies in one Fatou component U. The identity $W+1 = W$ and the functional equation make U invariant. The Fatou-component classification for exponential maps [22, §1, Remarks on notation] says that, in the presence of this Siegel disk, every Fatou component is an iterated preimage of it. It applies here because, with $a = \log b(\lambda_c) \neq 0$, the affine map $w \mapsto aw + \log a$ conjugates $E_{b(\lambda_c)}$ to $u \mapsto \mathrm{e}^u + \log a$. Hence U is the maximal Siegel disk. Thus W lies in the eventual maximal-Siegel domain of Proposition 85(2), whose maximal-disk version forbids a nonempty connected translation-invariant set. □

Corollary 87 (No uniformly controlled boundary sewing)**.** *Keep the Brjuno parameter and disk $\mathcal{D}_\rho$ of Proposition 85. Let $r_j \to 1$ with $r_j \neq 1$, $\lambda_j = r_j\lambda_c$, and $b_j = \exp(\lambda_j e^{-\lambda_j})$. For $0 < s \le \rho$ put*

$$\mathcal{D}_{j,s} = L_1(b_j) + (\log b_j)^{-1} H_{\lambda_j}(B(0,s)).$$

On a complete horizontal strip Σ, there is no locally bounded sequence of holomorphic 1-periodic functions Q_j with the following property: for every compact $K \subset \Sigma$ there are an integer $N_K \ge 0$ and a radius $s_K < \rho$ such that, for all large j,

$$S_{b_j}(z + Q_j(z) + N_K) \in \mathcal{D}_{j,s_K} \qquad (z \in K).$$

Proof. By Montel's theorem a subsequence of Q_j converges locally uniformly on Σ to a holomorphic 1-periodic Q. The lower-chart lemma and the radial-chart lemma imply, for each compact K, that $S_{b_c}(z+Q(z)+N_K)$ belongs to the image of the closed disk $\{|\zeta| \le s_K\}$ under the boundary chart. This image is contained in $\mathcal{D}_\rho$, since $s_K < \rho$. Hence $z + Q(z) \in \Omega_\rho$ for every $z \in \Sigma$. The image $(\mathrm{id} + Q)(\Sigma)$ is nonempty and path-connected, and periodicity gives $(\mathrm{id} + Q)(\Sigma) + 1 = (\mathrm{id} + Q)(\Sigma)$. This contradicts Proposition 85(2). □

In particular, a boundary solution cannot have overlapping upper-Siegel and lower-repelling regular representations on a complete horizontal line with 1-periodic time corrections: the lower times $z + Q(z)$ would form a path-connected, translation-invariant subset of the corresponding eventual Siegel domain, contradicting part (2).

Proposition 88 (The local chart has a natural parameter boundary)**.** *The family H_λ for (29) has no jointly holomorphic extension in (λ, v) from either side of the unit circle through any root of unity, on any fixed nonzero disk in v. Consequently it has no such extension across any open arc of the unit circle.*

Proof. If an extension existed at a primitive qth root λ_0, the identity $f_\lambda \circ H_\lambda = H_\lambda \circ (\lambda\,\cdot)$ and $H'_\lambda(0) = 1$ would persist at λ_0. Iterating q times would give $f_{\lambda_0}^{\circ q} \circ H_{\lambda_0} = H_{\lambda_0}$. Local invertibility of H_{λ_0} would make $f_{\lambda_0}^{\circ q}$ the identity near 0, hence everywhere by the identity theorem for entire functions. This would make f_{λ_0} injective, contrary to its 2πi-periodicity. Roots of unity are dense on the unit circle. □

This obstruction concerns the chart, not the sewn Kneser-type function: parameter singularities in the separate charts might cancel in the sewing. The monogenic parameter regularity of [16] is therefore the relevant comparison for the local step.

Lemma 89 (Root isolation implies normality)**.** *Let $\mathcal{F} \subset \mathcal{O}(B(0,R))$ satisfy $F(0) = 1$, $F(z) \neq 0$ throughout $B(0,R)$, and $F^{-1}(1) \cap B(0,R) = \{0\}$ for every $F \in \mathcal{F}$. Then the normalised logarithms g_F with $g_F(0) = 0$, the functions F, and their reciprocals $1/F$ are uniformly bounded on every $|z| \le r < R$, with bounds depending only on r/R. In particular $|F'(0)| \le C/R$ for an absolute constant C. If $F(-1) = 0$ and $F(z) = \exp(aF(z-1))$ on $B(0,R)$, then $g_F(z) = aF(z-1)$; when $|a|$ is bounded below, the family is also uniformly bounded on every smaller disk $B(-1,r)$. For these solutions the zero-free hypothesis is automatic.*

Proof. Choose the holomorphic logarithm g_F with $g_F(0) = 0$. Root isolation makes $h_F = \frac{1}{2} + g_F/(4\pi\mathrm{i})$ omit 0 and 1, while $h_F(0) = \frac{1}{2}$. Schottky's theorem bounds h_F, hence g_F, $F = \exp g_F$, and $1/F$, uniformly on compact subdisks. Cauchy's estimate at radius $R/2$ gives the derivative bound. The functional equation makes $aF(z-1)$ a logarithm of $F(z)$ normalised at 0, so it equals g_F. □

For a zero-free $F \in \mathcal{O}(B(0,R_0))$ with $F(0) = 1$ and $F \not\equiv 1$, let d_F be the distance from 0 to the nearest other preimage of 1, or ∞ if there is none. Applying the lemma on every disk of radius $s < \min(R_0, d_F)$ gives $|F'(0)| \min(R_0, d_F) \le C$. Thus derivative blow-up on a fixed disk forces another preimage of 1 to approach 0.

Proposition 90 (Reduction)**.** *Let W be a nonempty open subset of a common regular z-domain for both families, with $\operatorname{Re} z > -2$ on W. Suppose the following hold.*

(P0) The branch κ from $b > \eta$ has a jointly holomorphic continuation on a connected exterior parameter corridor reaching $U_{\text{out}} \times W$. The sewing family of Theorem 35 has a jointly holomorphic continuation on a connected interior parameter corridor reaching $U_{\text{in}} \times W$. Both continuations use the stated branches and have W in their common regular z-domain.

(P1) The base-continued family $\kappa_b(z)$ on $U_{\text{out}} \times W$ and the two-sided regular family $K_b^W(z)$ on $U_{\text{in}} \times W$ are locally uniformly bounded up to Γ: for every compact subarc $\Gamma' \subset \Gamma$ and $V \Subset W$, they are bounded on a neighbourhood of Γ' on their respective sides, uniformly for $z \in V$.

(P2) Let $D \subset W$ be a countable set with an accumulation point in W. For each $z \in D$, the two one-sided radial limits exist and are equal at almost every point of Γ (in arc length).

Then κ continues jointly holomorphically in (b, z) across Γ on W and coincides with the continued K^W on $U_{\text{in}} \times W$. The equality extends to any common connected regular z-domain by the identity theorem.

Proof. Intersect the full-measure subsets in (P2) over the countable set D. At a point b_c in the intersection, (P1) makes each radial family normal as a family of holomorphic functions on W. Every subsequential limit has the values prescribed on D; the identity theorem makes it unique. Thus the two radial families converge locally uniformly on W and have the same limit. Fix $z \in W$. On each side, bounded holomorphic functions have non-tangential limits almost everywhere (Fatou); the radial limits just obtained agree with these. The H^∞ gluing theorem across an analytic arc therefore gives a holomorphic function of b through Γ. The bounds in (P1) are locally uniform in z. At a point of Γ, approach from U_{in}: the functions of z form a locally bounded family and converge pointwise to the glued values. Montel's theorem shows that the limit is holomorphic in z. The glued function is thus separately holomorphic and locally bounded, so Hartogs' theorem makes it jointly holomorphic. It equals κ on the outer side and K^W on the inner side; the identity theorem extends the equality in z wherever both are defined on a common connected domain. □

Proposition 91 (H^1 alternative to (P1))**.** *In Proposition 90, (P1) may be replaced by the following condition. In every local conformal coordinate $\xi = s + \mathrm{i}t$ straightening Γ, for each compact interval I on the straightened arc and each $V \Subset W$, there are $\epsilon > 0$ and $C < \infty$ such that on both sides*

$$\sup_{0<|t|<\epsilon} \int_I \sup_{z \in V} |F_\pm(s + \mathrm{i}t, z)|\, ds \le C, \tag{30}$$

where F_+ and F_- denote the respective families from (P0). The conclusion of Proposition 90 still holds.

Proof. Shrink I slightly to avoid the ends of the coordinate rectangle. By Fubini and (30), the two vertical sides of a smaller rectangle can be chosen with integrable $\sup_{z \in V} |F_\pm|$; its opposite horizontal edge is away from the arc. The local H^1 boundary theorem (or the disk theorem after conformally mapping this rectangle) [6] applied for each fixed z gives $L^1(I)$ traces $f_\pm(\cdot, z)$, with convergence in L^1 on the smaller interval as $t \to 0^\pm$. The radial parameter paths in (P2) meet the analytic arc transversely, so their limits agree almost everywhere with these Hardy traces. The bound (30) and Cauchy's formula in z show that $z \mapsto f_\pm(\cdot, z)$ are holomorphic as $L^1(I)$-valued maps: on each compact z-circle the L^1 norms are uniformly bounded, so Bochner dominated convergence passes the boundary limit through the Cauchy integral. By

(P2), their difference vanishes in $L^1(I)$ at every $z \in D$. The Banach-valued identity theorem makes the difference zero on W.

The two piecewise defined families are locally integrable in (s, t, z) by (30). Equality of their boundary traces removes the jump term in the distributional $\bar\partial_\xi$ derivative. The distributional $\bar\partial_z$ derivative also vanishes, since the families are jointly holomorphic away from the measure-zero arc. Weyl's lemma for $\bar\partial$ therefore extends the glued family jointly holomorphically across the arc. The identity theorem gives the same conclusion on the common connected regular z-domain as before. □

Remark 92. The exponent 1 is sharp for this abstract boundary-gluing argument. Indeed, on opposite sides of the real ξ-axis let $F_\pm(\xi, z) = 1 + z/\xi$. The two boundary values agree almost everywhere and $F_\pm(\xi, 0) = 1$. For every $0 < p < 1$ and compact I, V, the integrals $\int_I \sup_{z\in V} |F_\pm(s + \mathrm{i}t, z)|^p \, ds$ stay uniformly bounded as $t \to 0^\pm$, but the common function has a pole at $\xi = 0$ for $z \neq 0$. Thus replacing (30) by a bare H^p bound with $p < 1$ would not prove the gluing statement; this example does not address additional constraints of the tetration equation.

Lemma 89 gives a geometric sufficient condition for (P1) when the common z-domain contains $B(0, R) \cup B(-1, R)$: if both families have no preimage of 1 in $B(0, R)$ other than 0 for every base in the two collars, then (P1) holds on every smaller disk $W = B(0, r)$. Along a connected parameter corridor, the argument principle transports this root count from one base where 0 is the only, simple root, provided F_b is defined near the closed disk $\overline{B(0, R)}$ and $F_b(z) \neq 1$ on the fixed circle $|z| = R$ throughout the corridor. This contour exclusion near Γ has not been proved. Corollary 34 supplies a starting base and a fixed root-isolation disk for the inner sewing branch near the real cusp; transport to Γ still requires the contour exclusion.

The open set W in this proposition is essential. At the single point $z_0 = 0$, both families equal 1 by normalisation. More generally, at every $n \in \mathbb{Z}_{\geq 0}$ they equal $E_b^{\circ n}(1)$, including all derivatives with respect to b wherever the families are holomorphic. These integer heights therefore cannot test (P2). Corollary 83 adds $z = -1$ and all its base derivatives when both specified branches retain their zero backward label; that point distinguishes them from entire lower regular translates, but not from each other. An accumulating set of noninteger heights is needed. The monodromy in Proposition 78 gives an explicit illustration: after any nonzero number of parameter turns, the upper regular germs still take the same values $E_b^{\circ n}(1)$ at every nonnegative integer n, although their derivatives at $z = 0$ differ. A pointwise version of (P1)–(P2) would be vacuous and could not establish continuation of the family. Even finitely many z-derivatives at every integer do not remove this ambiguity. For any $m \geq 0$ and any solution F regular near the integers, the local solution $F_\epsilon(z) = F(z + \epsilon(\mathrm{e}^{2\pi\mathrm{i}z} - 1)^{m+1})$ obeys the same functional equation and has the same z-jets through order m at each integer where both are defined. For small nonzero ϵ it generally differs at noninteger heights; this observation makes no claim that F_ϵ remains in the two-chart sewing class. Condition (P0) is also substantive: Proposition 16 constructs K^W on discs along the whole real segment $(1, \eta)$, and Theorem 35 adds quantitative bounds near η; neither reaches an arbitrary Shell–Thron boundary arc. Proposition 44 supplies a jointly holomorphic exterior germ along the entire real ray $b > \eta$. Proposition 45 gives an explicit *geometric* base corridor on which its unmarked initial curve and two repelling ends persist up to a non-cusp boundary arc. Proposition 46 certifies transport of the normalising point $w = 1$ along one path in that corridor, hence of the exterior germ at $z = 0$; Proposition 47 places its endpoint at a prescribed Brjuno parameter. Proposition 48 pulls the whole initial region back through 28 iterates to a Jordan cell containing $w = 1$, even at that endpoint. Proposition 49 gives a uniform quasiconformal model of these quotient cylinders, and Proposition 50 gives one-sided limits on a fixed small height disk about $z = 0$ along a short boundary subarc, without an arithmetic restriction. Proposition 51 upgrades these limits to a jointly holomorphic germ across the subarc in a

genuine complex base corridor, certified on a rectangle with $|\operatorname{Im}\theta| \le 0.01$. Proposition 52 proves one transverse seam crossing at the real seed and reduces its possible loss along the path to a scalar elliptic tangency test. Lemma 53 confines any such tangency to a finite-height cylinder band, without yet giving its explicit width. Propositions 54 and 55 give an explicit positive-slope approximate Abel coordinate and a global L^4 bound on its correction to the exact uniformisation. Proposition 56 differentiates the equation vertically and upgrades this to a pointwise derivative bound, excluding all seam tangencies on a wider box that also contains interior bases. Proposition 57 certifies at one such base that a nontrivial subarc of the exterior chord lies in the immediate attracting basin. This physical overlap does not identify the two quotient coordinates. The single-crossing criterion of Proposition 63 is therefore met. Corollary 64 supplies a common regular height neighbourhood of $[0,1]$ across a non-cusp boundary subarc for the exterior family. This proves the exterior full-height substep of (P0); the inner sewn family has not been transported to the same subarc, so (P0) as a whole remains open. Corollary 65 carries that same full-height branch along the certified path to a strictly interior base where Proposition 57 supplies a physical attracting overlap. Proposition 58 separately certifies an embedded lower-to-upper sewing circle and a locally holomorphic sewn family at this interior base. Its parameter neighbourhood has not been connected to the real inner sewn family, and the two local sewing constructions have not been identified. Proposition 59 certifies a transverse self-intersection of that lower regular curve at another interior base on the proposed straight parameter path. Thus the same circle cannot remain an embedded positive-slope seam throughout that path; a parameter detour or a change of seam remains possible. Proposition 60 now places the entire lower sewing circle, after 76–77 forward iterates, as one embedded essential loop in the exterior quotient at that same base. This supplies a common annular physical chart, but does not yet identify the compactified quotient surfaces or their normalised coordinates. Corollary 61 certifies another, disjoint return loop after 14–15 iterates, so the physical return alone does not select a unique quotient comparison. Corollary 62 also matches both regular charts to degree-one puncture germs of the exterior quotient. For either chosen return loop, the unproved part is its analytic connection through the two compact middle bands to those germs. Corollary 66 continues this exterior family through height $z = -1$, where it has a simple zero and the same backward label as the inner sewn family. That label does not by itself identify the two branches. Corollary 67 gives a fixed disc of uniform root isolation and the exterior local bounds required by (P1) on this boundary subarc. The inner bounds remain open. This exterior substep is not established by the parameter argument of [21, Prop. 3 and §5]. Proposition 78 transports the upper normalised germ along every compact inner path avoiding $\lambda = 0$, but gives neither a boundary-uniform height disk nor the sewing correction. Corollary 79 shows that no fixed finite pullback depth at the normalising orbit can supply such a boundary chart: the required depth diverges at every Brjuno point. Proposition 80 further shows that any chart on a fixed connected height domain whose image stays in the immediate attracting basin and whose value at $z = 0$ is 1 must collapse to the constant 1 there, even without a prior bound. Such a chart cannot carry the normalised value at $z = 1$ near a Brjuno boundary point. Proposition 86 also rules out a locally bounded periodic lower-time correction on a fixed translation-invariant collar whose image remains in the full attracting basin up to that point.

Non-Brjuno numbers have measure zero, so (P2) may be verified at Brjuno points only. There, Lemma 73 makes the upper charts converge locally at L_1 from both sides to the Siegel chart. Lemmas 74 and 81 give locally uniform convergence of the entire lower superfunction at every point of Γ away from the cusp. Proposition 85(2) rules out a complete sewing band obtained from the Siegel chart even after arbitrary finite forward pullbacks. Corollary 87 further shows that, on a fixed complete strip, locally bounded periodic lower times cannot have locally uniform finite-depth entry into a fixed upper chart disk. The chart convergence in part (3) is only local on Ω_ρ and cannot furnish an essential sewing curve. Thus (P2) needs

control of the sewn solutions that does not assume a boundary transition map on such a band. The pointwise `fatou.gp` values in Section 7.12 are compatible with a smooth join at the Diophantine point $\alpha = (\sqrt{2}-1)/5$, but are not identified with the two families in (P2). We have not proved (P0), (P1), its Hardy-norm alternative (30), or (P2); in particular the proposed boundary-band argument cannot establish (P2).

8 What is claimed

Proved.

- Elementary facts: Lemma 1, the conditional first-mode expansion of Proposition 2, the scale Λ and constant C_η of Proposition 6, the formal identities (16), (20) and (21), and translation invariance of every κ_n.
- The Kneser-free reformulation: Lemma 12, the welding identity (Theorem 13), and the sewing construction (Proposition 15).
- The parabolic limit: the confluence statement (C) (Theorem 28). It gives $\tau_n(b) \to B_n e^{2\pi i n a}$ for every n and $B_0 = 0$; Proposition 8 gives $B_1 \neq 0$, hence the ratio limit and limiting phase (Corollary 29). The supporting lemmas are 25–27, 30 and 31.
- The Kneser side: the decay estimate with explicit constants (Theorem 32), and the characterisation $\mathcal{C}_b(Y) = \{K_b^W\}$ (Theorem 38). Hence $\hat{c}_n \to B_n e^{2\pi i n a}$ for every n holds for the unique solution; the ratio limit of Conjecture 9 follows. The solution has an attracting-regular representation above, a repelling-regular representation below, and a seam half a period below the real axis (Corollary 33).
- Local and uniform statements in the base: Theorems 35, 39 and 40.

These proofs were checked by two independent adversarial reviews and a second-round review of the repairs.

Conditional. The same conclusions for the base-continued Kneser family K_b follow once that family belongs to $\mathcal{C}_{b'}$ for b' in some open set of bases near (b_1, η) (Theorem 40), or once it satisfies the ladder hypothesis of Theorem 39. This is a qualitative statement about the definition of the family in [21] (Remark 42). The complex-base construction computes κ in exactly this form, and the ladder data are consistent with it.

Numerical, without certified error bounds. The horn coefficients (15) and (17)–(18), the constant a, and all ladder values. The deficit law (28), i.e. analyticity of $\log \tau_n$ in the unfolding parameter p with $C = 0.0101990063$. Agreement across settings is not a bound on total error.

Not proved. A closed form for the first-order coefficients $\kappa^{(n)}$. The absence of every negative mode for the ladder solutions, which is a statement about κ rather than K^W (for K^W it holds by construction). Large-order growth of the c_n beyond the bounds of Theorem 32. Global uniqueness of Kneser-type solutions across the Shell–Thron boundary.

Prior art. The qualitative statement $K_b \neq R_b$ is Paulsen's [21, Prop. 4], whose quantifier is over open subintervals; the pointwise measurements here are complementary, not a reduplication. We have not found the scale Λ, its mechanism, or the quantity $\hat{c}_1$ discussed in the tetration literature; the search term "$4\pi^2$" does not occur in [21]. We have *not* carried out

a systematic search of the exponentially-small-splitting literature for an equivalent statement about the exponential family, and regard the attribution of Section 6 as open.

9 Reproducibility

The computation of K uses the θ-mapping construction with two fixed points, seeded along ladders $b+\mathrm{i}y$ with $y \downarrow 0$, at requested precisions up to 52 digits; the regular solution R is computed independently from the Schröder series. Both are part of the `kneser` Python library. The analyses use stored ladder data and separate parabolic computations. The relevant scripts include: `first_mode_check.py` (Table 1), `c1_invariant.py` (Table 2), `theta_spectrum.py` (Table 3, Observation 5), `lambda_law_analysis.py` (Lemma 1, Proposition 6), `parabolic_horn.py` (Section 6.1; this one is independent of all the ladder data), `parabolic_horn_inverse.py`, `kappa_ladder.py`, `kappa_gauge_check.py`, `deficit_law.py`, and, for Section 7, `transition_map.py`, `eta_approach.py`, `eta_sweep.py`, `confluence_check.py` and `weld_solve.py` (output in `docs/data/eta-sweep.txt`). The mathematical interpretation of the last two follows the corrected gauge distinction and sign calculation above, not their earlier explanatory comments. Reproducing historical fit values also requires the recorded data selection, correction order, and external-point phase convention. The additional `deficit_audit.py` records those choices and matched-order fits, omission checks, and archived-ladder sensitivity in `docs/data/deficit-audit.json`. The external values of K are the published `fatou.gp` reference set of the tetration forum and the number (4) from [21].

One instrument-level caveat is worth recording because it invalidated an earlier version of these measurements: in the complex-base builder the fixed points were computed at a hard-coded precision of 30 digits, and since the repelling-side super-function is evaluated as $E^d(L + \lambda^{z-d})$ with depth d proportional to the requested precision, that error was amplified by $|\lambda|^d$ — so that asking for more digits made the residual *worse*. After the fix, residuals fall to $\sim 5 \cdot 10^{-22}$ at 20 digits and the signal-to-noise ratio of D reaches 10^{10}–10^{13}.

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